arXiv:2510.03342v3 [cs.RO] 28 Nov 2025

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Gemini Robotics 1.5: Pushing the Frontier of Generalist Robots with Advanced Embodied Reasoning, Thinking, and Motion Transfer

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General-purpose robots need a deep understanding of the physical world, advanced reasoning, and general and dexterous control. This report introduces the latest generation of the Gemini Robotics model family: Gemini Robotics 1.5, a multi-embodiment Vision-Language-Action (VLA) model, and Gemini Robotics-ER 1.5, a state-of-the-art Embodied Reasoning (ER) model. We are bringing together three major innovations. First, Gemini Robotics 1.5 features a novel architecture and a Motion Transfer (MT) mechanism, which enables it to learn from heterogeneous, multi-embodiment robot data and makes the VLA more general. Second, Gemini Robotics 1.5 interleaves actions with a multi-level internal reasoning process in natural language. This enables the robot to "think before acting" and notably improves its ability to decompose and execute complex, multi-step tasks, and also makes the robot's behavior more interpretable to the user. Third, Gemini Robotics-ER 1.5 establishes a new state-of-the-art for embodied reasoning, i.e., for reasoning capabilities that are critical for robots, such as visual and spatial understanding, task planning, and progress estimation. Together, this family of models takes us a step towards an era of physical agents—enabling robots to perceive, think and then act so they can solve complex multi-step tasks.

1. Introduction

Truly general robots will require a deep understanding of the physical world. Our previous work, Gemini Robotics (Gemini-Robotics-Team et al., 2025), established a strong foundation by leveraging Gemini's rich world knowledge to create a Vision-Language-Action (VLA) model that exhibits impressive interactivity, generality, and dexterity in direct robot control. We now introduce the **Gemini Robotics 1.5** (GR 1.5) family of robot foundation models, built on the latest generation of Gemini (Comanici et al., 2025). The new model family significantly enhances the capabilities of Gemini Robotics and brings Gemini's advanced thinking and agentic paradigm to the physical world. It includes Gemini Robotics 1.5, a multi-embodiment VLA model (Bjorck et al., 2025; Intelligence et al., 2025; Wen et al., 2025; Zitkovich et al., 2023) with strong reasoning and generalization, and Gemini Robotics-ER 1.5, a generalist Vision-Language Model (VLM) that achieves a new state-of-the-art across embodied reasoning benchmarks. We combine these two models into an agentic system that enables robots to solve complex problems by orchestrating user dialogue, high-level reasoning and planning, agentic tool use and low-level action.

Gemini Robotics 1.5 advances the frontier of Vision-Language-Action (VLA) pre-training by integrating two core breakthroughs. Firstly, a novel architecture and a Motion Transfer (MT) mechanism enable the model to learn from diverse robot data sources, forming a unified understanding of motion and physics. This multi-embodiment pre-training allows GR 1.5 to control multiple robots, including the ALOHA, Bi-arm Franka, and Apollo humanoid robots, without any robot-specific post-training, and it also enables zero-shot skill transfer from one robot to another. Secondly, GR 1.5 is a Thinking VLA

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Gemini Robotics 1.5：以先进具身推理、思考与运动迁移推动通用机器人前沿

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通用机器人需要对物理世界有深入理解，并具备高级推理能力以及通用且灵巧的控制能力。本报告介绍了 Gemini Robotics 模型系列的最新一代成果：Gemini Robotics 1.5，一种多具身形态的视觉-语言-动作（VLA）模型，以及 Gemini Robotics-ER 1.5，一种最先进的具身推理（ER）模型。我们汇聚了三项重大创新。首先，Gemini Robotics 1.5 采用新颖架构和运动迁移（MT）机制，使其能够从异构、多具身形态的机器人数据中学习，并让该 VLA 模型更具通用性。其次，Gemini Robotics 1.5 将动作与自然语言形式的多层级内部推理过程交织在一起。这使机器人能够"先思考，后行动"，显著提升其分解并执行复杂多步骤任务的能力，也让机器人的行为对用户而言更易解释。第三，Gemini Robotics-ER 1.5 在具身推理方面确立了新的最先进水平，即针对机器人至关重要的推理能力，例如视觉与空间理解、任务规划以及进度估计。这一模型系列共同推动我们迈向物理智能体时代——使机器人能够感知、思考并采取行动，从而解决复杂的多步骤任务。

1. 引言

真正通用的机器人将需要对物理世界有深入理解。我们先前的工作 Gemini Robotics（Gemini-Robotics-Team et al., 2025）利用 Gemini 丰富的世界知识构建了一个视觉-语言-动作（VLA）模型，该模型在直接机器人控制中展现出令人印象深刻的交互性、通用性和灵巧性，从而奠定了坚实基础。我们现在推出 Gemini Robotics 1.5（GR 1.5）系列机器人基础模型，其基于最新一代 Gemini（Comanici et al., 2025）构建。新的模型系列显著增强了 Gemini Robotics 的能力，并将 Gemini 的先进思考和智能体范式引入物理世界。它包括 Gemini Robotics 1.5，一种具有强大推理和泛化能力的多具身 VLA 模型（Bjorck et al., 2025; Intelligence et al., 2025; Wen et al., 2025; Zitkovich et al., 2023），以及 Gemini Robotics-ER 1.5，一种通用视觉-语言模型（VLM），在具身推理基准上达到了新的最先进水平。我们将这两个模型组合成一个智能体系统，使机器人能够通过协调用户对话、高层推理与规划、智能体工具使用和底层动作来解决复杂问题。

Gemini Robotics 1.5 通过整合两项核心突破，拓展了视觉-语言-动作（VLA）预训练的前沿。首先，一种新颖的架构和运动迁移（MT）机制使模型能够从多样化的机器人数据源中学习，形成对运动与物理的统一理解。这种多具身预训练使 GR 1.5 无需任何针对特定机器人的后训练，即可控制多种机器人，包括 ALOHA、双臂 Franka 和 Apollo 人形机器人，并且还支持从一个机器人到另一个机器人的零样本技能迁移。其次，GR 1.5 是一个具备思考能力的 VLA 模型

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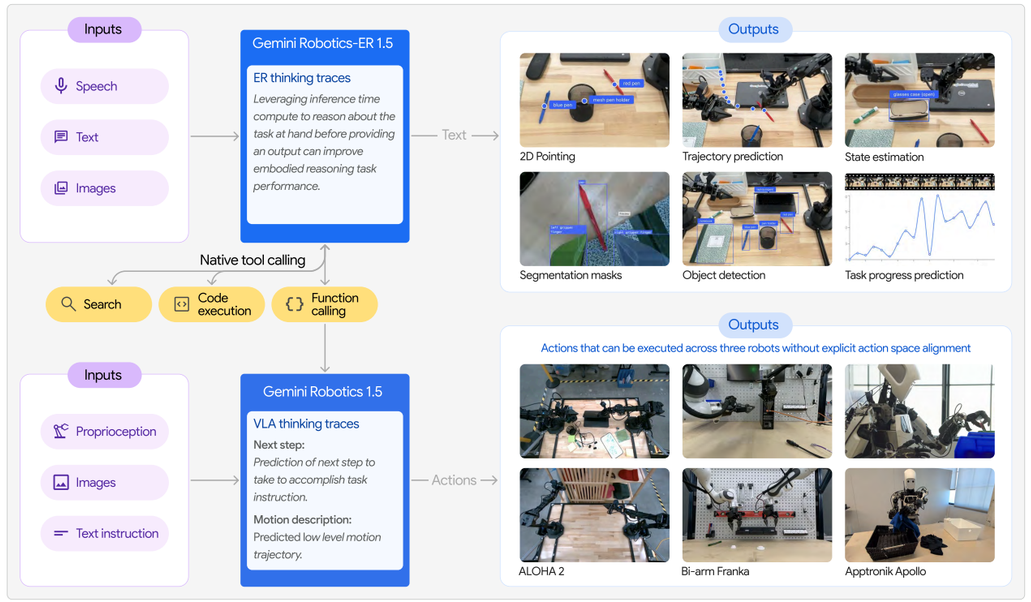


Figure 1 | The Gemini Robotics 1.5 family of models consists of Gemini Robotics 1.5, a VLA, and Gemini Robotics 1.5-ER, a VLM with state-of-the-art embodied reasoning capabilities. They can be combined together to form a powerful agentic framework.

that can explicitly reason about its actions, interleaving a stream of thoughts with physical movements. This allows the model to convert visual observations into language-based thoughts, simplify complex instructions, detect task success or failure, propose recovery behaviors, and make the robot's actions more interpretable to human users.

Gemini Robotics-ER 1.5 (GR-ER 1.5) advances the state-of-the-art for embodied reasoning (Chen et al., 2024a,b; Li et al., 2023; Zhi et al., 2025), i.e., the visuo-spatial-temporal understanding of the physical world that is required for robotic applications. Building upon Gemini's state-of-the-art thinking and multimodal capabilities, GR-ER 1.5 significantly outperforms other frontier models across a broad suite of embodied intelligence benchmarks, while retaining the general capabilities of a frontier model and being considerably faster. GR-ER 1.5's physical understanding combines naturally with Gemini's ability to use tools, communicate using modalities like video and audio, and write code, opening up a broad spectrum of potential applications.

To achieve truly general-purpose robot agents, we combine our models in an agentic framework (Figure 1). This framework is key to unlocking new capabilities: it handles long-horizon task execution via complex planning and adaptive orchestration, facilitates multimodal interaction, enables robots to leverage user-specified tools (e.g. web search) to solve problems and complete tasks, and implements a multi-layered safety mechanism through explicit reasoning about safety violations.

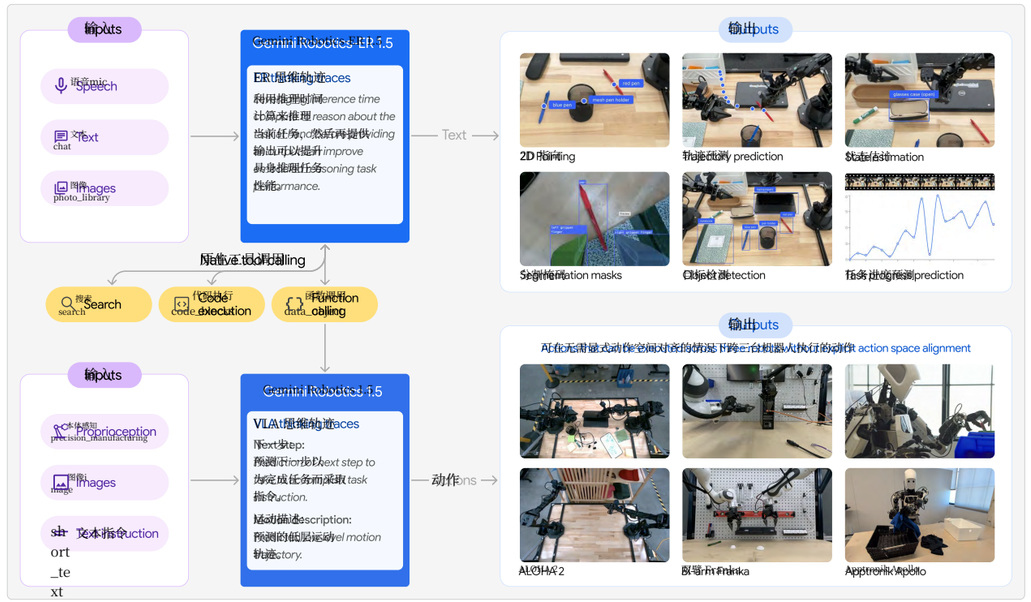


图 **1** | Gemini Robotics 1.5 系列模型包括 Gemini Robotics 1.5（一种 VLA）和 Gemini Robotics 1.5-ER（一种具备最先进具身推理能力的 VLM）。它们可以组合在一起，形成一个强大的智能体框架。

该模型能够对其动作进行显式推理，将一连串思考与物理动作交织在一起。这使模型能够将视觉观察转化为基于语言的思考、简化复杂指令、检测任务成功或失败、提出恢复行为，并使机器人的动作对人类用户而言更易理解。

Gemini Robotics-ER 1.5（GR-ER 1.5）推进了具身推理的最先进水平（Chen et al., 2024a,b; Li et al., 2023; Zhi et al., 2025），即机器人应用所需的物理世界视觉-空间-时间理解能力。基于 Gemini 最先进的思考和多模态能力，GR-ER 1.5 在广泛的具身智能基准测试套件上显著优于其他前沿模型，同时保留前沿模型的通用能力，并且速度显著更快。GR-ER 1.5 的物理理解能力与 Gemini 使用工具、通过视频和音频等模态进行交流以及编写代码的能力自然结合，从而开辟了广泛的潜在应用。

为了实现真正通用的机器人智能体，我们将模型组合在一个智能体框架中（图1）。该框架是解锁新能力的关键：它通过复杂规划和自适应编排处理长时程任务执行，促进多模态交互，使机器人能够利用用户指定的工具（例如网页搜索）来解决问题并完成任务，并通过显式推理安全违规情况来实施多层安全机制。

2. Method Overview

2.1. Model & Architecture

Gemini Robotics 1.5 model family. Both Gemini Robotics 1.5 and Gemini Robotics-ER 1.5 inherit Gemini's multimodal world knowledge. Gemini Robotics-ER 1.5 (GR-ER 1.5 for short), our VLM, fully retains Gemini's capabilities including advanced reasoning, tool use, and more. It has additionally been optimized for complex embodied reasoning problems such as task planning, reasoning for spatial expertise, and task progress estimation. GR-ER 1.5 significantly extends and improves upon GR-ER's embodied reasoning capabilities. Gemini Robotics 1.5 (GR 1.5 for short), our VLA model, translates mid- and short-horizon instructions into robot actions. It understands open-vocabulary natural language instructions, can perform reasoning steps before emitting an action, and it can natively control multiple robots with different embodiments. As such, GR 1.5 significantly extends the previous Gemini Robotics' capabilities.

Agentic System Architecture. The full agentic system consists of an orchestrator and an action model that are implemented by the VLM and the VLA, respectively:

- **Orchestrator**: The orchestrator processes user input and environmental feedback and controls the overall task flow. It breaks complex tasks into simpler steps that can be executed by the VLA, and it performs success detection to decide when to switch to the next step. To accomplish a user-specified task, it can leverage digital tools to access external information or perform additional reasoning steps. We use GR-ER 1.5 as the orchestrator.
- **Action model**: The action model translates instructions issued by the orchestrator into low-level robot actions. It is made available to the orchestrator as a specialized tool and receives instructions via open-vocabulary natural language. The action model is implemented by the GR 1.5 model.

Embodied thinking. A core innovation of Gemini Robotics 1.5 is Embodied Thinking: the ability to reason—or "think"—before taking action (Huang et al., 2025; Lee et al., 2025; Lin et al., 2025; Zawalski et al., 2024), which operates across both the VLM and the VLA models. The GR-ER 1.5 model combines Gemini's thinking and tool-use with an enhanced physical world understanding, enabling it to function within the agentic system for high-level planning. This includes breaking complex tasks into coarse-grained plans, adaptively updating those plans based on execution, or calling external tools like web search. We also introduce an analogous thinking capability to the VLA, creating the Thinking VLA, or GR 1.5 (Thinking on) in our plots and results. Our Thinking VLA explicitly reasons about the instruction and its perception, generates thinking traces in natural language (Belkhale et al., 2024; Smith et al., 2025), and appends them to the context window before emitting an action. This process simplifies complex instructions into sequences of primitive skills, increases the transparency in human-robot interactions, and offers a new paradigm for scaling VLA capabilities.

To understand how embodied thinking and the agentic system architecture may interact, let us consider a user instruction such as "Pack the suitcase for a trip to London". The orchestrator (GR-ER 1.5) accesses a travel itinerary and a recent weather forecast with user permission to decide which clothes are appropriate to pack. It then produces a high-level plan consisting of instructions such as "pack the rain jacket into the luggage" that it communicates to the action model. The action model then decomposes each such instruction into shorter segments that correspond to a few seconds of robot movement each (e.g., "pick up the rain jacket from the wardrobe"). These are executed directly, or they are further translated into an inner monologue of primitive motions such as "move the gripper to

2. 方法概述

2.1. 模型与架构

Gemini Robotics 1.5 模型系列。Gemini Robotics 1.5 和 Gemini Robotics-ER 1.5 均继承了 Gemini 的多模态世界知识。Gemini Robotics-ER 1.5（简称 GR-ER 1.5）是我们的 VLM，完整保留了 Gemini 的能力，包括高级推理、工具使用等。它还针对复杂的具身推理问题进行了额外优化，例如任务规划、空间专业推理和任务进度估计。GR-ER 1.5 显著扩展并提升了 GR-ER 的具身推理能力。Gemini Robotics 1.5（简称 GR 1.5）是我们的 VLA 模型，可将中短时程指令转化为机器人动作。它能够理解开放词汇自然语言指令，可在输出动作之前执行推理步骤，并且能够原生控制具有不同具身形态的多种机器人。因此，GR 1.5 显著扩展了此前 Gemini Robotics 的能力。

智能体系统架构。完整的智能体系统由编排器和动作模型组成，二者分别由 VLM 和 VLA 实现：

• 编排器：编排器处理用户输入和环境反馈，并控制整体任务流程。它将复杂任务分解为可由 VLA 执行的更简单步骤，并执行成功检测以决定何时切换到下一步。为完成用户指定的任务，它可利用数字工具访问外部信息或执行额外的推理步骤。我们使用 GR-ER 1.5 作为编排器。

• 动作模型：动作模型将编排器发出的指令转换为低层机器人动作。它作为专用工具提供给编排器，并通过开放词汇自然语言接收指令。动作模型由 GR 1.5 模型实现。

具身思考。Gemini Robotics 1.5 的一项核心创新是具身思考：在采取行动前进行推理——或“思考”——的能力（Huang et al., 2025; Lee et al., 2025; Lin et al., 2025; Zawalski et al., 2024），该能力在 VLM 和 VLA 模型中都发挥作用。GR-ER 1.5 模型将 Gemini 的思考与工具使用能力，与增强的物理世界理解相结合，使其能够在智能体系统中进行高层规划。这包括将复杂任务分解为粗粒度计划、根据执行情况自适应更新这些计划，或调用网络搜索等外部工具。我们还为 VLA 引入了类似的思考能力，在图表和结果中称为 Thinking VLA，或 GR 1.5（Thinking on）。我们的 Thinking VLA 会显式地就指令及其感知进行推理，以自然语言生成思考轨迹（Belkhale et al., 2024; Smith et al., 2025），并在输出动作之前将其追加到上下文窗口中。这一过程将复杂指令简化为一系列基础技能，提高了人机交互的透明度，并为扩展 VLA 能力提供了一种新范式。

为了理解具身思考与智能体系统架构可能如何交互，让我们考虑这样一条用户指令：“为伦敦之行收拾行李箱”。编排器（GR-ER 1.5）在获得用户许可的情况下访问旅行行程和近期天气预报，以决定适合打包哪些衣物。随后，它生成一个高层计划，其中包含诸如“把雨衣装进行李箱”之类的指令，并将其传达给动作模型。动作模型随后将每条此类指令分解为较短的片段，每个片段对应机器人几秒钟的运动（例如，“从衣柜里拿起雨衣”）。这些片段会被直接执行，或者被进一步转化为原始动作的内部独白，例如“将夹爪移动到

the left" or "close the gripper", thus leveraging an explicit understanding of the geometry of the scene to solve the task. Overall, our models' ability to perform embodied thinking dramatically improves their ability to handle multi-step tasks by allowing the models to compose skills in a structured, deliberate manner, ultimately leading to more robust and reliable robot performance.

Motion Transfer. In Gemini Robotics 1.5, we also introduce a new model architecture and training recipe for the VLA. These enable the model to learn from different robots and data sources, to form a unified understanding of motions and the effects of physical interactions, enabling skills to transfer across very different robot embodiments. We refer to the new training recipe as Motion Transfer (MT) in our results.

2.2. Robot Data

The dataset used for training the Gemini Robotics 1.5 contains both multi-embodiment robot data collected on ALOHA (ALOHA-2-Team et al., 2024), Bi-arm Franka (Franka, 2025) and Apollo humanoid (Apptronik, 2025), as well as publicly available text, image and video datasets on the Internet. The robot data consists of thousands of diverse tasks across these platforms covering a broad range of manipulation skills across a multitude of scenes.

2.3. Evaluation

For all comparisons reported in this report, we perform A/B/n testing on real robots. This means that we test all models involved in a particular comparison in an interleaved manner on the same robot work cell, thereby reducing variance in the evaluations that might otherwise arise from variations across robots and environmental conditions.

The development of GR 1.5 requires comparisons of a large number of architecture variations, algorithm hyperparameters and other settings across multiple embodiments and tasks. To improve research iteration speed, we have developed methods for evaluation without real robots in the loop.

We use the open-source MuJoCo simulator (Todorov et al., 2012) to generate evaluation scenes for the robot embodiments in this report. By carefully aligning the visual and physical parameters of simulated and real scenes, we are able to achieve a strong rank consistency between evaluations in simulation and on the real robot (see Fig. 21 in the Appendix B.1).

This has allowed us to massively scale up the breadth of our evaluations to new objects, scenes, and environments, and to rapidly iterate on architectural and algorithmic improvements. Over 90% of the evaluation episodes during the development of Gemini Robotics 1.5 were conducted in simulation. Although real-world evaluation is still required to determine model quality, evaluation in simulation dramatically reduces the volume of tests on real hardware.

3. Gemini Robotics 1.5 is a general multi-embodiment Vision-Language-Action Model

GR 1.5 can control robots with dramatically different form factors to complete a large variety of tasks out-of-the-box, without the need for post-training to specialize the model to a particular embodiment or task. Fig. 2 shows example tasks on ALOHA, Bi-arm Franka and Apollo humanoid robots.

In this section, we present a comprehensive evaluation of GR 1.5 and a comparison with our previous models, Gemini Robotics and Gemini Robotics On-Device. Our experiments are designed to answer the following questions:

左侧”或“闭合夹爪”，从而利用对场景几何结构的显式理解来完成任务。总体而言，我们的模型执行具身思考的能力显著提升了其处理多步任务的能力，因为它使模型能够以结构化、审慎的方式组合技能，最终带来更稳健、更可靠的机器人性能。

运动迁移。在 Gemini Robotics 1.5 中，我们还为 VLA 引入了一种新的模型架构和训练配方。这些使模型能够从不同的机器人和数据源中学习，形成对运动以及物理交互影响的统一理解，从而使技能能够跨差异极大的机器人具身形态迁移。在结果中，我们将新的训练配方称为运动迁移（MT）。

2.2. 机器人数据

用于训练 Gemini Robotics 1.5 的数据集既包含在 ALOHA（ALOHA-2-Team et al., 2024）、双臂 Franka（Franka, 2025）和 Apollo 人形机器人（Apptronik, 2025）上采集的多具身机器人数据，也包含互联网上公开可用的文本、图像和视频数据集。机器人数据由这些平台上成千上万的多样化任务组成，涵盖大量场景中广泛的操控技能。

2.3. 评估

对于本报告中的所有对比，我们都在真实机器人上进行 A/B/n 测试。这意味着，我们以交错方式在同一机器人工作站上测试特定对比中涉及的所有模型，从而减少因不同机器人及环境条件差异而可能产生的评估方差。

GR 1.5 的开发需要跨多种具身本体和任务，对大量架构变体、算法超参数及其他设置进行比较。为提高研究迭代速度，我们开发了无需真实机器人参与闭环的评估方法。

我们使用开源 MuJoCo 模拟器（Todorov et al., 2012）生成本报告中所涉及机器人具身本体的评估场景。通过仔细对齐仿真场景与真实场景的视觉和物理参数，我们能够在仿真评估与真实机器人评估之间实现高度一致的排名（见附录 B.1 中的图 21）。

这使我们能够将评估范围大幅扩展到新的物体、场景和环境，并快速迭代架构与算法改进。在 Gemini Robotics 1.5 的开发过程中，超过 90% 的评估回合是在仿真中进行的。尽管仍需真实世界评估来确定模型质量，但仿真评估大幅减少了真实硬件上的测试量。

3. Gemini Robotics 1.5 是一种通用的多具身视觉-语言-动作模型

GR 1.5 能够控制外形结构差异极大的机器人，开箱即可完成大量多样化任务，无需通过后训练使模型专门适配某种特定的具身形态或任务。图 2 展示了 ALOHA、双臂 Franka 和 Apollo 人形机器人上的示例任务。

在本节中，我们对 GR 1.5 进行全面评估，并与我们之前的模型 Gemini Robotics 和 Gemini Robotics On-Device 进行比较。我们的实验旨在回答以下问题：

Figure 2 | GR 1.5 can control three different robots with the same checkpoint to accomplish a variety of tasks out-of-the-box.

1. How does GR 1.5 perform and generalize on short-horizon tasks?
2. Does GR 1.5 effectively learn from and transfer knowledge across different embodiments?
3. How does the thinking process contribute to multi-step tasks?

We develop a benchmark that follows the design philosophy that was used to evaluate Gemini Robotics (Gemini-Robotics-Team et al., 2025). We extend it to cover all our embodiments, adding more challenging and multi-step tasks, as well as tasks that test cross-embodiment transfer and thinking. The full benchmark includes 230 tasks in total.

We generally report mean and standard error of the mean of *progress score* (definitions in Appendix B.2 - Appendix B.4), as it provides a continuous and finer-grained measure of model performance and, as such, is especially useful for complex multi-step tasks. For completeness, we also include the corresponding plots of success rate in the Appendix B.5.

3.1. Gemini Robotics 1.5 can generalize to new environments and tasks

To understand GR 1.5's generalization performance on short-horizon tasks, we use the same methodology as in (Gemini-Robotics-Team et al., 2025) and consider multiple axes of variation:

- **Visual Generalization:** robustness to visual variations such as changes in background, lighting, distractor objects, or textures.
- **Instruction Generalization:** ability to understand the intent behind natural language instructions, including handling paraphrasing, typos, different languages, and varying levels of specificity.
- **Action Generalization:** ability to adapt learned movements or synthesize novel ones, for example, in order to handle new initial conditions or object instances.
- **Task Generalization:** ability to successfully execute a new task in a new environment. This is the most comprehensive form of generalization as it simultaneously requires robustness to visual changes, understanding of open-vocabulary instructions, and the ability to adapt learned motions to new tasks.

图 **2** | GR 1.5 可使用同一检查点控制三种不同的机器人，开箱即用地完成多种任务。

1. GR 1.5 在短时程任务上的表现和泛化能力如何？2. GR 1.5 能否有效地从不同具身形态中学习并实现知识迁移？3. 思维过程如何促进多步任务？

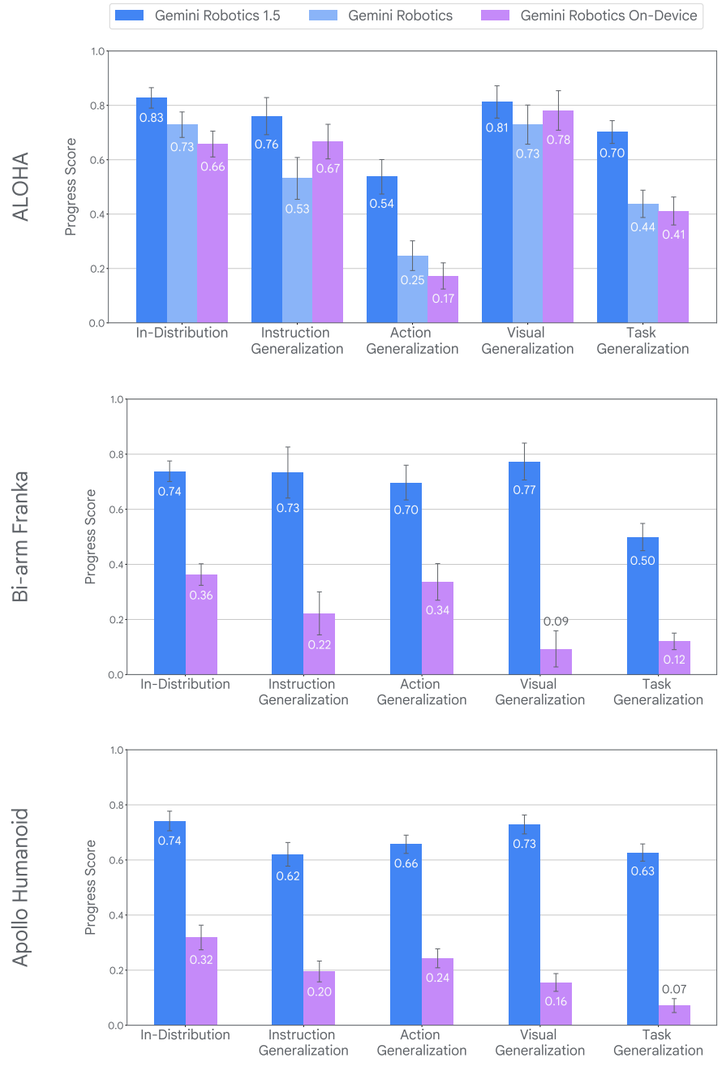
我们开发了一个基准，遵循用于评估 Gemini Robotics（Gemini-Robotics-Team et al., 2025）的设计理念。我们对其进行了扩展，以覆盖我们所有的具身形态，并加入了更具挑战性的多步任务，以及用于测试跨具身迁移和思考能力的任务。该完整基准总共包含 230 项任务。

我们通常报告进度分数的均值及均值标准误（定义见附录 B.2 至附录 B.4），因为它能提供连续且更细粒度的模型性能度量，因此对复杂的多步任务尤其有用。为完整起见，我们还在附录 B.5 中包含了相应的成功率图。

3.1. Gemini Robotics 1.5 能够泛化到新环境和新任务

为了理解 GR 1.5 在短时程任务上的泛化性能，我们采用与 (Gemini-Robotics-Team et al.,2025) 相同的方法，并考虑多个变化维度：

- 视觉泛化：对视觉变化的鲁棒性，例如背景、光照、干扰物体或纹理的变化。
- 指令泛化：理解自然语言指令背后意图的能力，包括处理改述、拼写错误、不同语言以及不同具体程度。
- 动作泛化：调整已学习动作或合成新动作的能力，例如为了处理新的初始条件或物体实例。
- 任务泛化：在新环境中成功执行新任务的能力。这是最全面的泛化形式，因为它同时要求对视觉变化具有鲁棒性、理解开放词汇指令，以及将学到的动作适应新任务的能力。



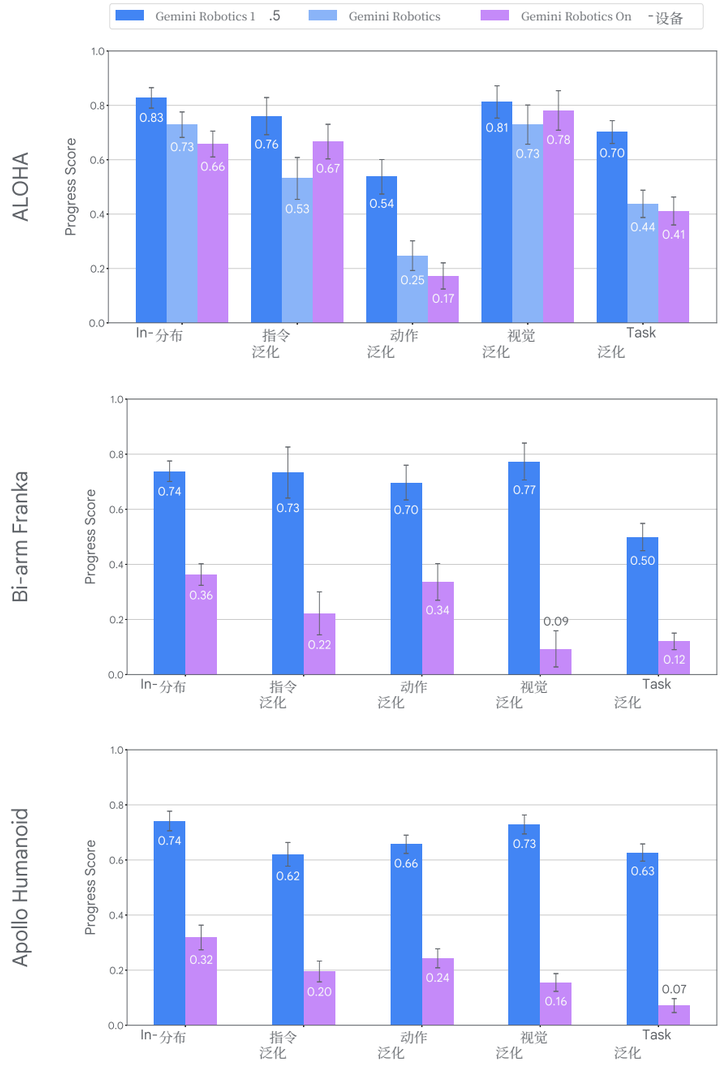


Figure 3 | Breakdown of GR 1.5 generalization capabilities across our robots. GR 1.5 consistently outperforms the baselines and handles all four types of variations more effectively.

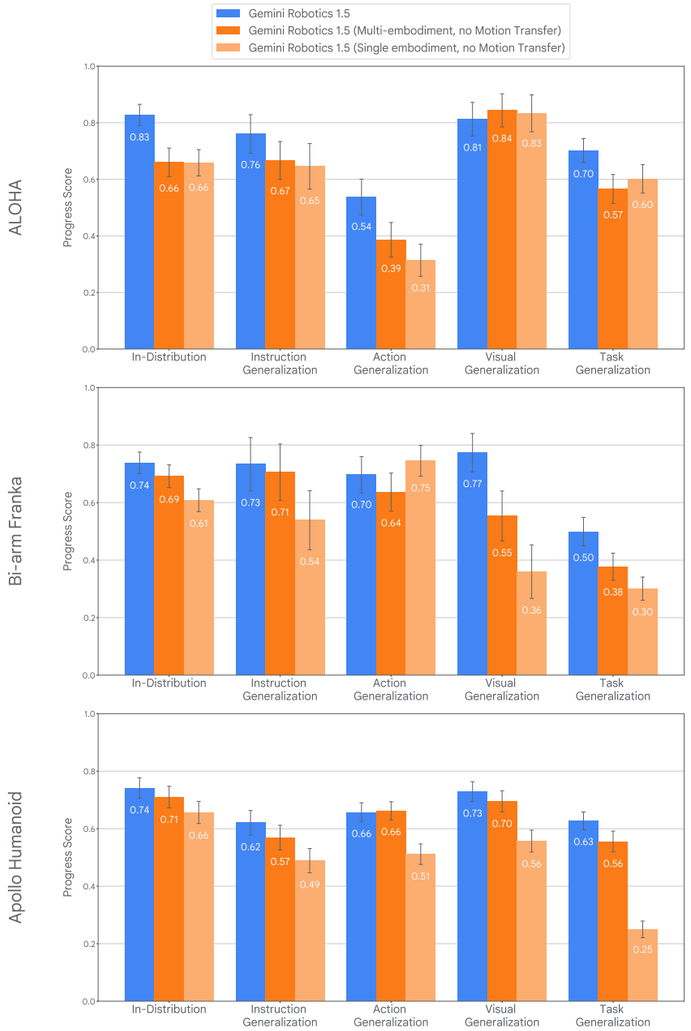
图 3 | GR 1.5 在我们各类机器人上的泛化能力分解。GR 1.5 始终优于基线，并能更有效地处理所有四类变化。

We first analyze how GR 1.5 compares against Gemini Robotics and Gemini Robotics On-Device (GRoD for short). As shown in the top plot of Fig. 3, for the ALOHA robot, GR 1.5 consistently outperforms these two baselines across all four categories. In particular, GR 1.5 achieves substantial gains in instruction, action, and task generalization. For the Bi-arm Franka robot and the Apollo humanoid robot, we compare GR 1.5 against our GRoD models[2] (middle and bottom plots of Fig. 3). On these

我们首先分析 GR 1.5 相较于 Gemini Robotics 和 Gemini Robotics On-Device（简称 GRoD）的表现。如图 3 的顶部图所示，对于 ALOHA 机器人，GR 1.5 在所有四个类别上始终优于这两个基线。特别是，GR 1.5 在指令、动作和任务泛化方面取得了显著提升。对于双臂 Franka 机器人和 Apollo 人形机器人，我们将 GR 1.5 与我们的 GRoD 模型[2]（图 3 的中图和下图）进行比较。在这些

[2]We do not show comparisons with the Gemini Robotics model from (Gemini-Robotics-Team et al., 2025), because those models on the Bi-arm Franka and the Apollo humanoid were post-trained specialists, and they had little generalization beyond variations of the trained tasks.

[2]我们没有展示与来自（Gemini-Robotics-Teametal.,2025）的 Gemini Robotics 模型的比较，因为那些在双臂 Franka 和 Apollo 人形机器人上的模型是后训练专用模型，并且它们在训练任务的变化之外几乎没有泛化能力。



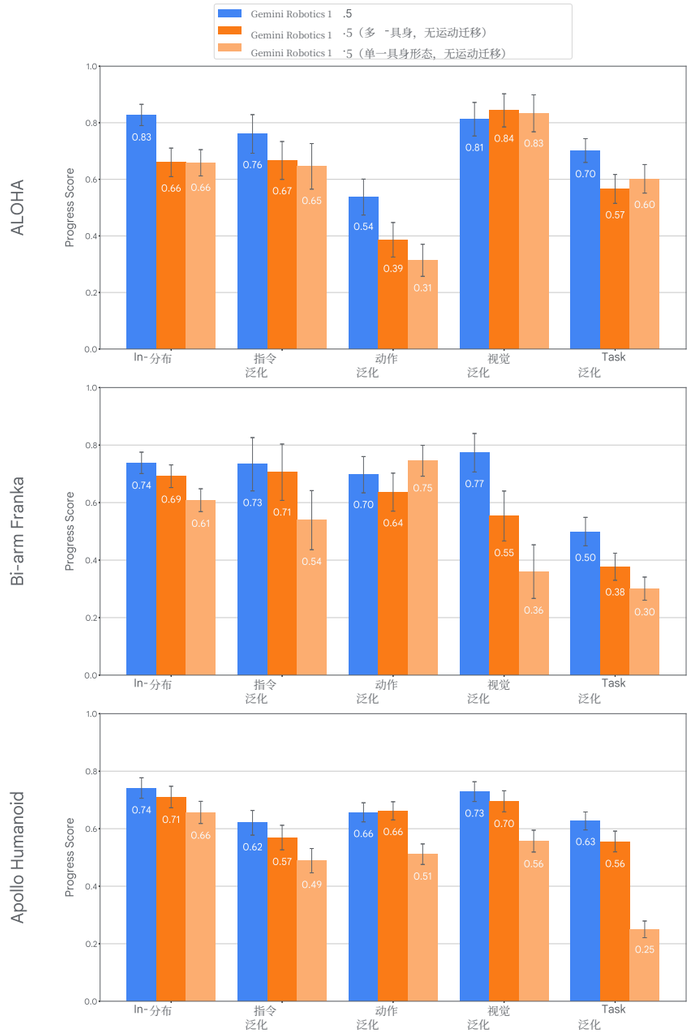


Figure 4 | Ablation on datasets and training recipes on ALOHA, Bi-arm Franka and Apollo Humanoid. GR 1.5 consistently outperforms our baselines: GR 1.5 trained on single or multi-robot data without the MT recipe.

图 **4** | 在 ALOHA、双臂 Franka 和 Apollo Humanoid 上针对数据集与训练配方的消融实验。GR1.5 始终优于我们的基线：即未使用 MT 配方、仅用单机器人或多机器人数据训练的 GR 1.5。

two platforms, GR 1.5 significantly outperforms GRoD across all categories. Note that this is not an apples-to-apples comparison because the GRoD checkpoints were trained with less data due to their earlier release date. Additionally, GRoD is not a multi-embodiment model: each embodiment requires a different checkpoint. Nevertheless, we include these results to illustrate the dramatic performance improvement across different versions of Gemini Robotics.

We perform an ablation study to pinpoint the source of this significant improvement in generalization. We establish two ablation baselines: training with data from a single embodiment versus training with data from all embodiments, both excluding our Motion Transfer (MT) mechanism. As illustrated in Fig. 4, while including data from other embodiments generally boosts performance, our MT training recipe clearly amplifies the positive effect of this additional data. This study confirms both the ability

在两个平台上，GR 1.5 在所有类别中均显著优于 GRoD。请注意，这并非同等条件下的比较，因为 GRoD 检查点因其发布时间较早而使用了更少的数据进行训练。此外，GRoD 并非多具身模型：每个具身都需要不同的检查点。尽管如此，我们仍纳入这些结果，以展示不同版本 Gemini Robotics 之间显著的性能提升。

我们开展消融研究，以精确定位这一泛化能力显著提升的来源。我们设立了两个消融基线：使用单一具身形态的数据进行训练，与使用所有具身形态的数据进行训练，二者均排除我们的运动迁移（MT）机制。如图4所示，虽然纳入其他具身形态的数据通常能提升性能，但我们的MT训练方案明显放大了这些额外数据带来的正向效果。本研究证实了以下两方面的能力

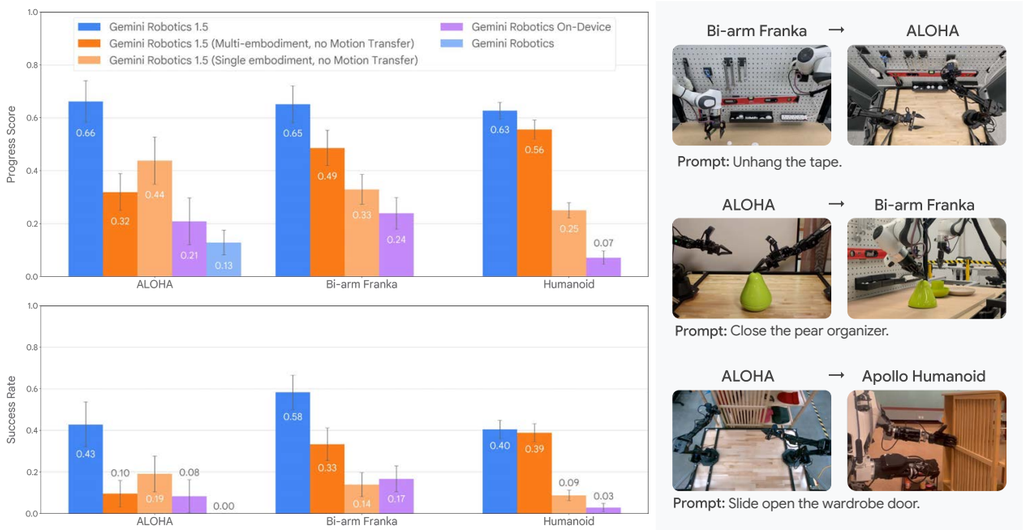


Figure 5 | Cross embodiment benchmark. Left: Our model shows zero-shot skill transfer on tasks only seen by another robot embodiment. Right: Example tasks trained on the first embodiment and evaluated on the second.

of GR 1.5 to leverage multi-embodiment data, and the critical role of the MT mechanism in achieving greater positive transfer of skills among different robots.

3.2. Learning across different robot embodiments

Gemini Robotics 1.5 is able to learn and transfer skills across different robot embodiments, leading to both better generalization and more data-efficient learning. Although prior work (O'Neill et al., 2024) has shown benefits of training VLAs with diverse data from multiple types of robots, there have been few demonstrations of zero-shot transfer of skills from one robot embodiment to another. In our experiments, we find evidence that GR 1.5's multi-embodiment co-training and MT paradigm enables such transfer. As shown in Fig. 5, the ALOHA robot is able to perform tasks for which training data was only collected on the Bi-arm Franka platform, and vice versa. The same applies to the humanoid, which can perform skills that are only available in the data from other robots (ALOHA in this example), despite being significantly more difficult to control and a wider cross-embodiment gap.

To corroborate our observations, we measure cross-embodiment transfer quantitatively with a cross-embodiment benchmark, defined across the three robots included in our study. For each embodiment, we test our model and the baselines on tasks for which data had been collected only on another robot. More details of the benchmark are in Appendix B.3.

Plots in Fig. 5 show how any model trained on single-embodiment data (Gemini Robotics, GRoD or GR 1.5 trained on a single embodiment) performs poorly on this benchmark, while with cross-embodiment data and MT training recipe we achieve significantly better performance. The efficacy of leveraging cross-embodiment data with Motion Transfer (MT) depends on the initial quantity of data available for a given robotic platform. For the ALOHA platform, which already possesses a large dataset, merely introducing data from other embodiments appears to be less effective; however, MT amplifies the positive transfer from this data by aligning the different embodiments and extracting

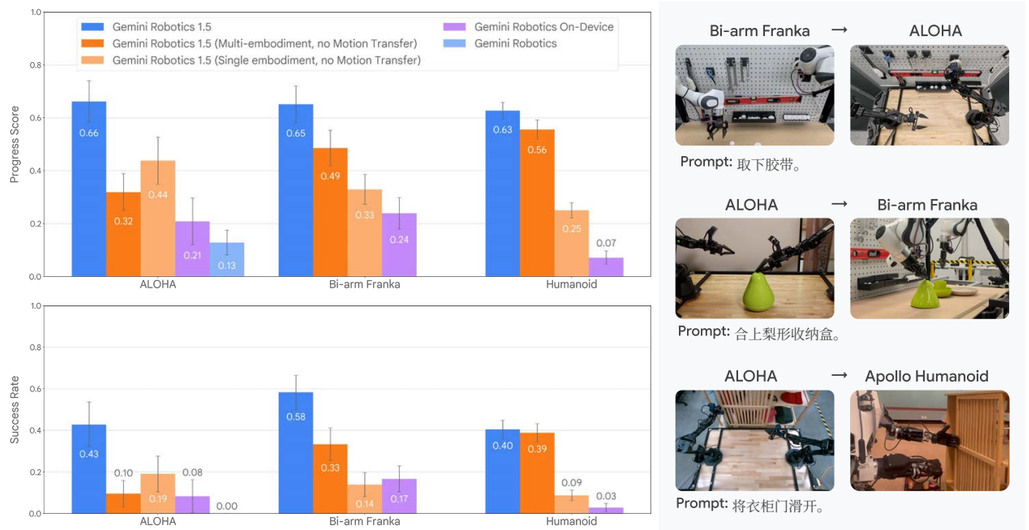


图 **5** | 跨具身基准。左：我们的模型在仅有另一种机器人本体见过的任务上展现出零样本技能迁移。右：在第一种本体上训练、在第二种本体上评估的示例任务。

GR 1.5 利用多具身形态数据的能力，以及 MT 机制在实现不同机器人之间技能更显著正向迁移中的关键作用。

3.2. 跨不同机器人本体的学习

Gemini Robotics 1.5 能够跨不同机器人本体学习和迁移技能，从而带来更好的泛化能力和更数据高效的学习。尽管先前工作（O'Neill et al., 2024）已表明，使用来自多种类型机器人的多样化数据训练 VLA 具有优势，但关于技能从一个机器人本体零样本迁移到另一个机器人本体的演示仍然很少。在我们的实验中，我们发现有证据表明，GR 1.5 的多本体协同训练和 MT 范式能够实现这种迁移。如图 5 所示，ALOHA 机器人能够执行仅在双臂 Franka 平台上收集过训练数据的任务，反之亦然。这同样适用于人形机器人：尽管其控制难度显著更高、跨本体差距也更大，它仍能执行仅在其他机器人（此例中为 ALOHA）数据中可获得的技能。

为佐证我们的观察，我们使用一个跨本体基准定量衡量跨本体迁移，该基准在本研究纳入的三种机器人上定义。对于每种本体，我们在仅于另一种机器人上收集过数据的任务上测试我们的模型和基线。该基准的更多细节见附录 B.3。

图 5 中的曲线图显示，任何在单一具身数据上训练的模型（Gemini Robotics、GRoD 或在单一具身上训练的 GR 1.5）在该基准上表现都不佳；而借助跨具身数据和 MT 训练方案，我们取得了显著更好的性能。利用跨具身数据并配合运动迁移（MT）的效果，取决于给定机器人平台初始可用的数据量。对于已经拥有大规模数据集的 ALOHA 平台，仅仅引入来自其他具身形态的数据似乎效果较差；然而，MT 通过对齐不同具身形态并提取

commonalities, thereby aiding the learning process. Conversely, for Bi-arm Franka with a moderate amount of data, adding cross-embodiment data is beneficial, and MT successfully facilitates this by aligning and extracting shared knowledge. For humanoid robots, where data is scarce, the addition of external embodiment data provides the greatest performance boost; yet, the effect of MT is less pronounced here, suggesting that the technique's alignment capabilities may be less effective when the embodiment gap is substantially larger, such as between the highly dissimilar humanoid and other robot forms.

In Fig. 5 we report success rate in addition to progress score in order to highlight that zero-shot transfer with Gemini Robotics 1.5 leads to successful task execution, not just partial progress towards completion of the task.

3.3. Thinking Helps Acting

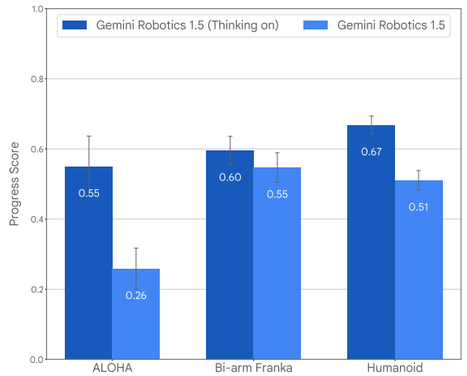


Figure 6 | Task progress in the multi-step benchmark with and without enabling thinking during inference.

In this section, we focus on the Thinking VLA model (GR 1.5 with thinking mode ON during inference). A detailed analysis of the higher-level reasoning enabled by GR-ER 1.5 is deferred to Section 5. The advantage of interleaving robot actions with explicit thinking steps is particularly evident in the context of longer multi-step tasks, such as "sorting clothes by colors" (see Appendix B.4 for our multi-step benchmark).

Fig. 6 demonstrates that enabling the thinking mode yields a sizable improvement in the progress score for these tasks. This performance gain stems from the model's ability to decompose the difficult cross-modal translation, which involves mapping high-level, multi-step language instructions to low-level robot actions, into two simpler stages. First, the model generates a language-based thinking trace by converting the complex task into a sequence of specific, short-horizon steps (e.g., transforming the goal of "sorting clothes" into a thought like, "move gripper to the left so that it is closer to the clothes"). Second, the model maps these low-level language commands directly to robot actions. This two-step decomposition proves more robust than a single, end-to-end translation because the first step leverages the powerful visual-linguistic capabilities of the VLM backbone, while the second involves learning a simpler action mapping.

Beyond quantitative performance gains, our experiments provide qualitative evidence of several additional benefits of the Thinking VLA. Firstly, it significantly improves interpretability. By visualizing the robot's internal thinking traces, we can inspect its planned actions and predict its next steps. This transparency enhances both human-robot trust and the safety of robot operations. Secondly, the

共性，从而促进学习过程。相反，对于拥有中等数据量的双臂 Franka，加入跨具身数据是有益的；MT 通过对齐并提取共享知识，成功促进了这一过程。对于数据稀缺的人形机器人，加入外部具身形态数据带来的性能提升最大；不过，MT 的效果在这里并不那么明显，这表明当具身差距显著更大时——例如在高度不同的人形机器人和其他机器人形态之间——该技术的对齐能力可能效果较差。

在图 5 中，我们除进度得分外还报告了成功率，以强调使用 Gemini Robotics 1.5 进行零样本迁移能够带来成功的任务执行，而不仅仅是任务完成过程中的部分进展。

3.3. 思考有助于行动

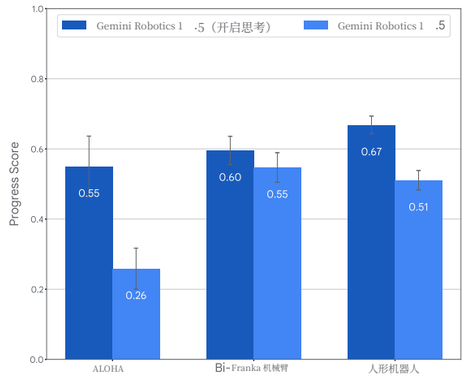


图 **6** | 在推理期间启用与未启用思考时，多步基准中的任务进度。

在本节中，我们聚焦于 Thinking VLA 模型（在推理期间开启 thinking 模式的 GR 1.5）。关于 GR-ER 1.5 所实现的更高层推理的详细分析将留待第 5 节讨论。将机器人动作与显式思考步骤交错进行的优势，在较长的多步任务中尤为明显，例如“按颜色给衣服分类”（我们的多步基准见附录 B.4）。

图6表明，启用思考模式可显著提升这些任务的进度分数。这种性能提升源于模型能够将困难的跨模态转换——即把高层、多步语言指令映射为底层机器人动作——分解为两个更简单的阶段。首先，模型通过将复杂任务转换为一系列具体、短时程步骤来生成基于语言的思考轨迹（例如，将“分拣衣物”的目标转化为类似“将夹爪向左移动，使其更靠近衣物”的思考）。其次，模型将这些底层语言命令直接映射为机器人动作。这种两步分解比单一的端到端转换更稳健，因为第一步利用了VLM主干强大的视觉-语言能力，而第二步则涉及学习更简单的动作映射。

除了定量性能提升之外，我们的实验还提供了定性证据，表明 Thinking VLA 具有若干额外优势。首先，它显著提升了可解释性。通过可视化机器人内部的思维轨迹，我们可以检查其规划的动作并预测其后续步骤。这种透明度既增强了人机信任，也提升了机器人操作的安全性。其次，

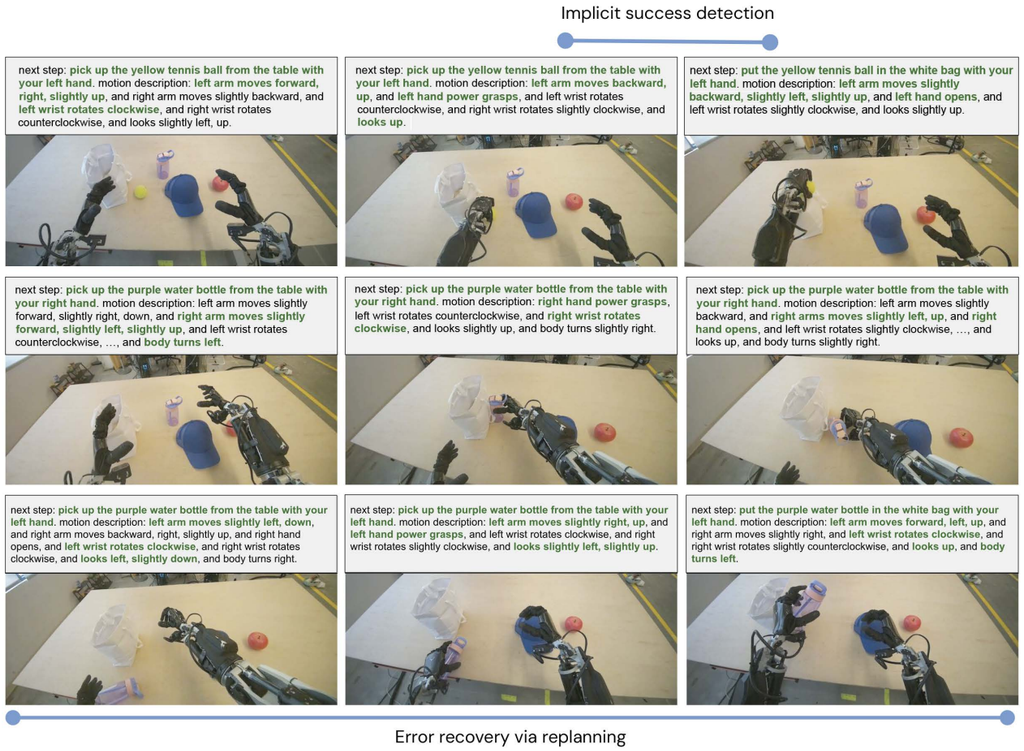


Figure 7 | From left to right, top to bottom: an example rollout of the Thinking VLA: the Apollo humanoid packing objects into a white bag. The thinking trace are overlaid on each snapshot. The Thinking VLA is able to think about its actions at different levels, allowing it to accomplish tasks requiring semantic reasoning and multiple steps of execution.

Thinking VLA exhibits a degree of situational awareness regarding task completion. For instance, as shown in Fig. 7, the robot automatically switches its objective from "pick up the yellow tennis ball" to "put the yellow tennis ball in the white bag" once the ball has been successfully grasped. This demonstrates that the model possesses an implicit awareness of the success of the prior subtask, removing the need for an explicit success detector. Thirdly, the Thinking VLA enables sophisticated recovery behaviors. For example, in Fig. 7, when the water bottle slips from the right hand and lands near the left hand, the next thinking trace immediately becomes "pick up the water bottle with the left hand", effectively initiating a self-correcting recovery behavior.

4. Gemini Robotics-ER 1.5 is a generalist embodied reasoning model

Robots require advanced and grounded knowledge of the physical world, ranging from precise spatial and temporal reasoning to a deep grasp of intuitive physics, causality, and affordances. We refer to this type of real-world understanding as *embodied reasoning* (ER). We introduce Gemini Robotics-ER 1.5 (GR-ER 1.5), our most advanced multimodal thinking model for state-of-the-art embodied reasoning based on Gemini. When combined with a general VLA, such as GR 1.5 showcased in Section 3, GR-ER

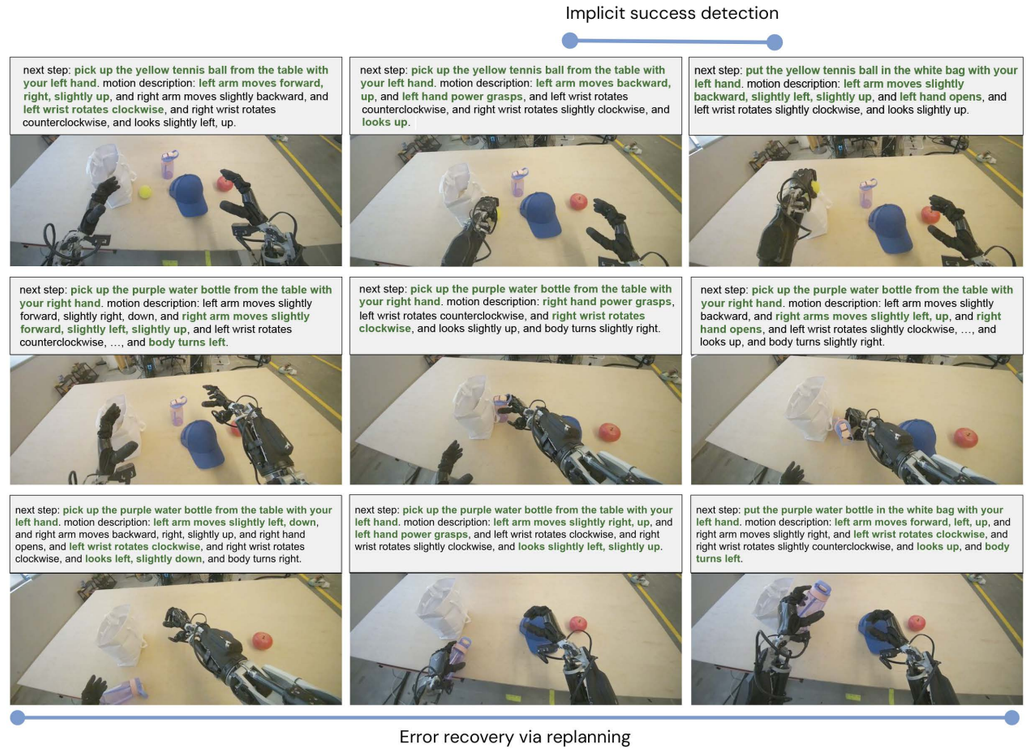


图 7 | 从左到右、从上到下：Thinking VLA 的一次示例运行：Apollo 人形机器人将物品装入一个白色袋子。思考轨迹叠加在每张快照上。Thinking VLA 能够在不同层面思考自身动作，从而完成需要语义推理和多步执行的任务。

Thinking VLA 在任务完成方面展现出一定程度的情境意识。例如，如图 7 所示，一旦黄色网球被成功抓取，机器人便会自动将其目标从“拿起黄色网球”切换为“把黄色网球放进白色袋子”。这表明模型对前一子任务的成功具有隐式意识，从而无需显式的成功检测器。第三，Thinking VLA 能够实现复杂的恢复行为。例如，在图 7 中，当水瓶从右手滑落并落在左手附近时，下一条思维轨迹会立即变为“用左手拿起水瓶”，从而有效启动自我纠正的恢复行为。

4. Gemini Robotics-ER 1.5 是一个通用具身推理模型

机器人需要对物理世界具备先进且扎实的知识，既包括精确的空间与时间推理，也包括对直觉物理、因果关系和可供性的深刻把握。我们将这类对现实世界的理解称为具身推理（ER）。我们推出 Gemini Robotics-ER 1.5（GR-ER 1.5），这是基于 Gemini 构建的、用于实现最先进具身推理的最先进多模态思考模型。当与通用 VLA（如第 3 节展示的 GR 1.5）结合时，GR-ER

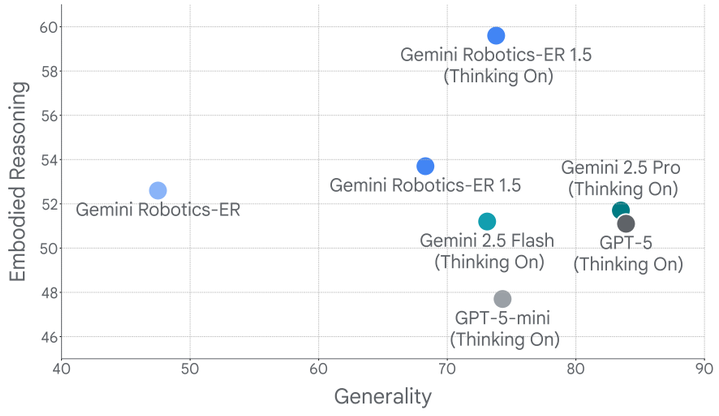


Figure 8 | The Gemini Robotics-ER 1.5 model is our most advanced model for embodied reasoning while retaining strong performance as a general-purpose multimodal foundation model. We measure Embodied Reasoning performance on a mix of academic benchmarks covering text-based image understanding as well as spatial signal prediction, and measure Generality performance on MMMU, GPQA, and Aider Polyglot.

1.5 provides high-level intelligence to form the backbone of a general agentic robot system, which we describe in Section 5.

In this section, we will focus on embodied reasoning capabilities and highlight several key properties of GR-ER 1.5:

1. Strong embodied reasoning performance while retaining the generality of a frontier model;
2. Excels in key robotic capabilities, such as complex pointing, progress understanding, and real-world use cases;
3. Able to scale embodied reasoning performance via inference time compute.

4.1. Generality

Notably, Gemini Robotics-ER 1.5 is a *generalist* embodied reasoning model: it exhibits the broad capabilities of a frontier model across many domains while also showcasing exceptional performance as a spatial expert for real-world understanding. This is visualized in Fig. 8 which shows this trade-off for several contemporary frontier models. To assess models quantitatively in terms of both their broad capabilities as well as their more specialized embodied reasoning performance, we evaluate them on two sets of benchmarks. Firstly, we measure the models' performance across a collection of 15 widely-used academic benchmarks designed to measure embodied reasoning capabilities such as text-based image understanding (e.g., BLINK (Fu et al., 2024), CV-Bench (Tong et al., 2024), and ERQA (Gemini-Robotics-Team et al., 2025)) and spatial reasoning (e.g. RoboSpatial (Song et al., 2025), PointArena (Cheng et al., 2025), Where2Place (Yuan et al., 2024), and RefSpatial (Zhou et al., 2025)). The embodied reasoning score is a weighted average of 50% spatial reasoning benchmarks and 50% question answering benchmarks (both image and video). Secondly, we measure the generalist performance on an equally weighted mix of academic benchmarks that assess a broader range of capabilities including image understanding, science, and coding via the MMMU (Yue et al., 2023), GPQA (Rein et al., 2024), and Aider Polyglot (Gauthier, 2024) benchmarks. The full evaluation details and results are discussed in Appendix C. Fig. 8 shows that Gemini Robotics-ER 1.5 expands the Pareto frontier of generality and embodied reasoning, achieving state-of-the-art embodied reasoning

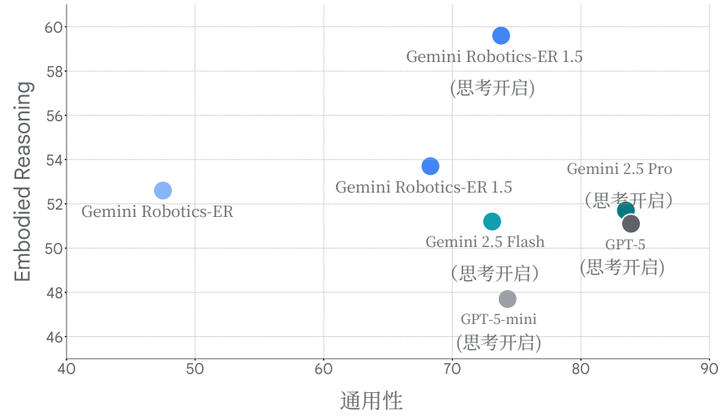


图 **8** | Gemini Robotics-ER 1.5 模型是我们最先进的具身推理模型，同时作为通用多模态基础模型仍保持强劲性能。我们在一组学术基准上衡量具身推理性能，这些基准涵盖基于文本的图像理解以及空间信号预测；并在 MMMU、GPQA 和 Aider Polyglot 上衡量通用性性能。

1.5 提供高层智能，构成通用智能体机器人系统的骨干；我们将在第 5 节对此进行描述。

本节中，我们将聚焦具身推理能力，并重点介绍 GR-ER 1.5 的若干关键特性：

1. 具备强大的具身推理性能，同时保留前沿模型的通用性；2. 在关键机器人能力方面表现出色，例如复杂指向、进度理解以及真实世界用例；3. 能够通过推理时计算扩展具身推理性能。

4.1. 通用性

值得注意的是，Gemini Robotics-ER 1.5 是一个通用具身推理模型：它展现出前沿模型在众多领域中的广泛能力，同时作为空间专家在真实世界理解方面也表现出卓越性能。图 8 对此进行了可视化展示，图中呈现了若干当代前沿模型的这一权衡。为了从广泛能力以及更专门的具身推理性能两方面对模型进行定量评估，我们在两组基准上对它们进行评估。首先，我们在一组由 15 个广泛使用的学术基准组成的集合上衡量模型性能，这些基准旨在衡量具身推理能力，例如基于文本的图像理解（如 BLINK (Fu et al., 2024)、CV-Bench (Tong et al., 2024) 和 ERQA (Gemini-Robotics-Team et al., 2025)）以及空间推理（如 RoboSpatial (Song et al., 2025)、PointArena (Cheng et al., 2025)、Where2Place (Yuan et al., 2024) 和 RefSpatial (Zhou et al., 2025)）。具身推理分数是 50% 空间推理基准和 50% 问答基准（包含图像和视频）的加权平均值。其次，我们通过 MMMU (Yue et al., 2023)、GPQA (Rein et al., 2024) 和 Aider Polyglot (Gauthier, 2024) 基准，在等权混合的学术基准上衡量通用性能，这些基准评估包括图像理解、科学和编程在内的更广泛能力。完整的评估细节和结果在附录 C 中讨论。图 8 表明，Gemini Robotics-ER 1.5 拓展了通用性与具身推理的帕累托前沿，在实现最先进的具身推理

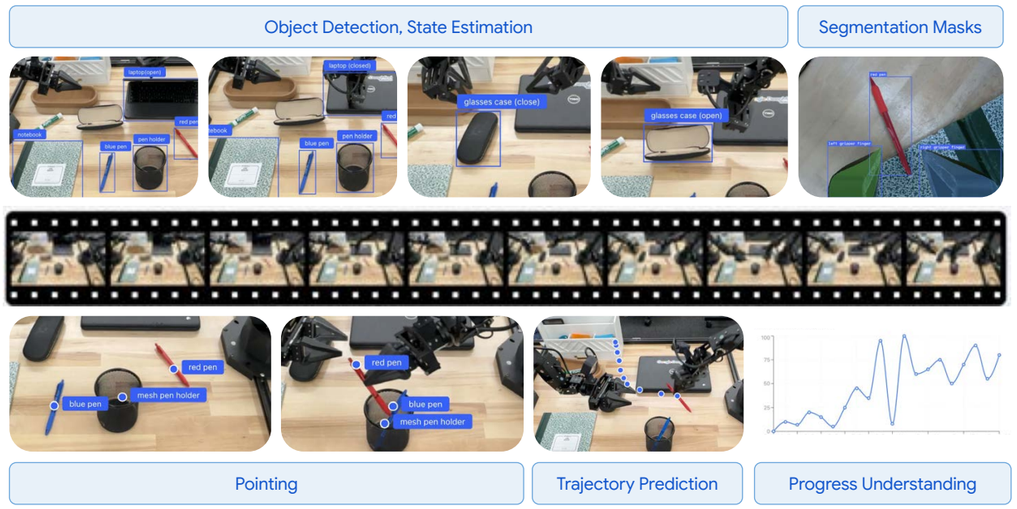


Figure 9 | The Gemini Robotics-ER 1.5 model has a diverse set of capabilities that can be applied on images and videos that are useful for robotics.

performance with comparable generality to other models in its model class.

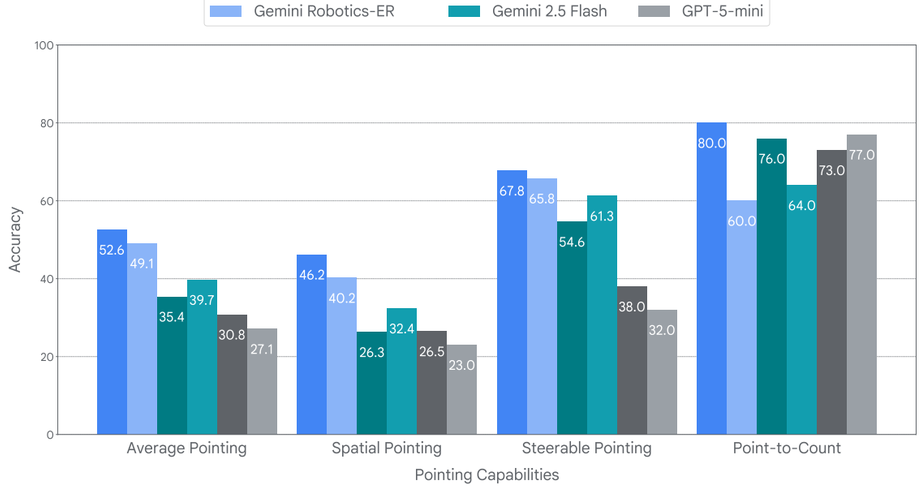


Figure 10 | Performance on a mix of 5 academic benchmarks for 2D pointing and point-based reasoning; accuracy is defined as the percentage of point predictions within the ground truth mask (pointing) or correct final count (point-to-count). The categories describe different types of point prediction and are used to aggregate evaluation results from Point-Bench, RefSpatial, RoboSpatial, Where2Place, and PixMo Count. Results for GPT-5 and GPT-5-mini obtained via API calls in September 2025.

图 **9** | Gemini Robotics-ER 1.5 模型具备多样化的能力，可应用于对机器人技术有用的图像和视频。

性能的同时，具备与其模型类别中其他模型相当的通用性。

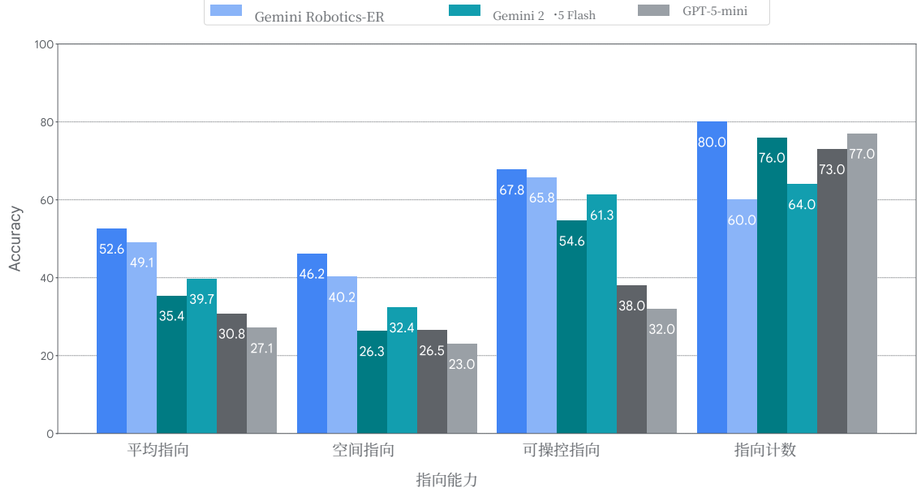


图 **10** | 在 5 个用于二维指向和基于点的推理的学术基准混合集上的性能；准确率定义为点预测位于真值掩码内的百分比（指向），或最终计数正确的百分比（点计数）。这些类别描述了不同类型的点预测，并用于汇总来自 Point-Bench、RefSpatial、RoboSpatial、Where2Place 和 PixMo Count 的评估结果。GPT-5 和 GPT-5-mini 的结果于 2025 年 9 月通过 API 调用获得。

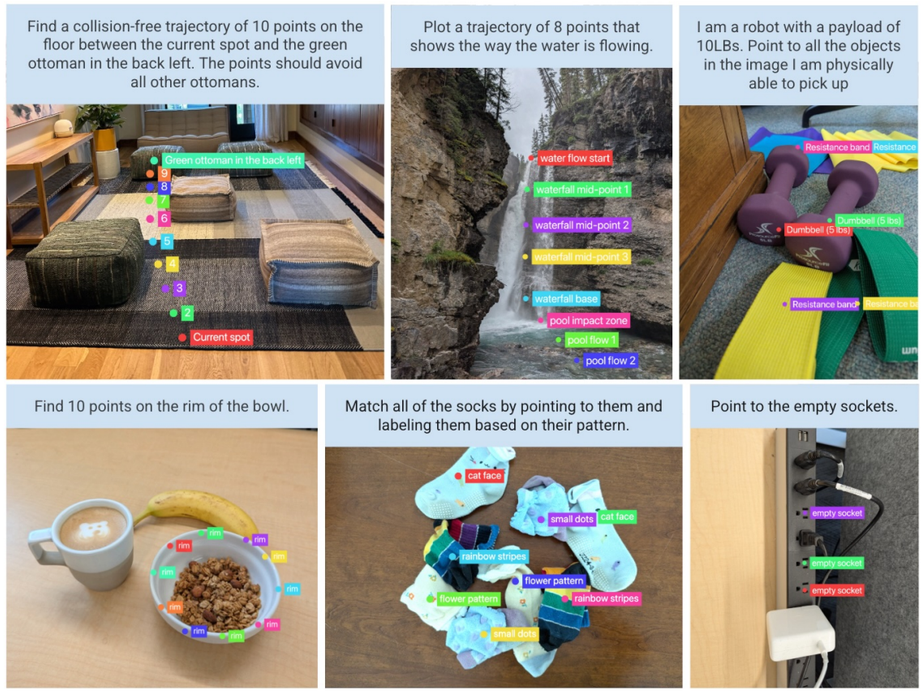


Figure 11 | Complex pointing examples from GR-ER 1.5. The model can follow complex pointing prompts that require reasoning about physical, spatial, and semantic constraints: It can localize precise parts of objects, such as the rim of a bowl and sockets of a power strip (Row 2, Columns 1 and 3) and predict points that respect physical, spatial, and semantic constraints, e.g., corresponding to objects that are lighter than 10 pounds (Row 1, Column 3), and matching similar items (Row 2, Column 2). GR-ER 1.5 can also sequence points into trajectories that respect physics (Row 1, Column 2) and avoid collisions (Row 1, Column 1).

4.2. Frontier capabilities for Embodied Reasoning

GR-ER 1.5 showcases advanced performance across a number of embodied reasoning capabilities which are highly relevant for understanding the physical world, particularly in robotic applications. We visualize some of these in Fig. 9 and analyze a few of these areas in detail.

Complex Pointing: A point is a flexible and lightweight representation that grounds a model's semantic understanding onto visual inputs. Using very few tokens, a point can precisely locate an abstract concept, such as where to click or the most appropriate object part to grasp. By extending this ability to predict a set of points, a model can generate more complex outputs like motion trajectories and paths, providing precise action guidance for robots. Points can also serve as intermediate reasoning tools for other downstream tasks, such as counting. We define the generalization of this capability, which combines pointing with reasoning, as **complex pointing**.

GR-ER 1.5 achieves new state-of-the-art results on academic benchmarks for complex pointing, as shown in Fig. 10. The evaluation spans several key capabilities: **Average Pointing** aggregates performance across all benchmarks; **Spatial Pointing** focuses on pointing queries requiring spatial reasoning (e.g., "point to the space left of the cup"); **Steerable Pointing** tests the ability to modify points following user instructions (e.g., "move the point slightly up"); and **Point-to-Count** measures

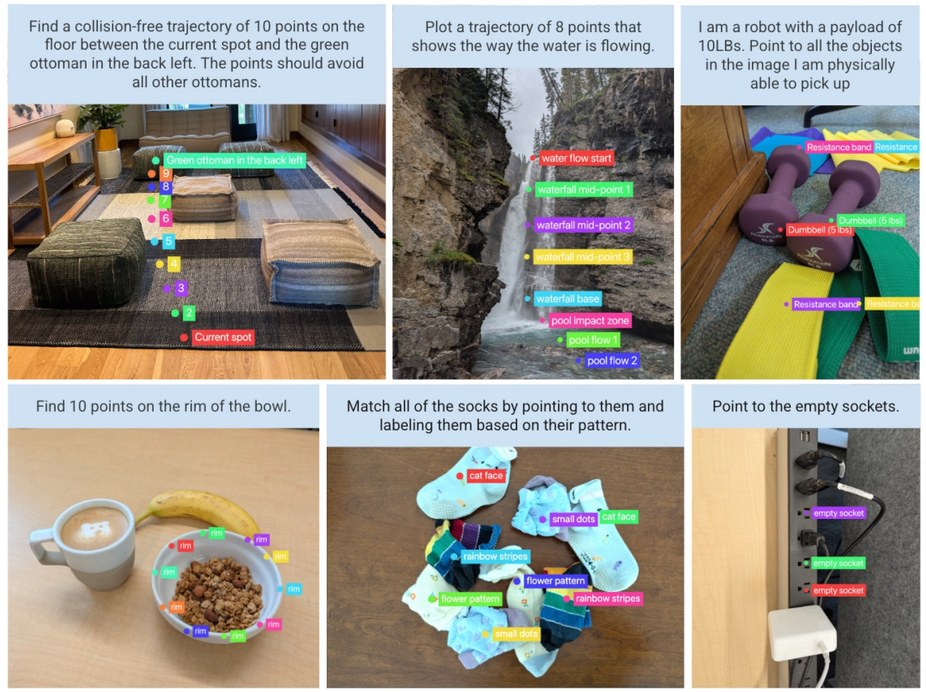


图 **11** | GR-ER 1.5 的复杂指向示例。该模型能够遵循复杂的指向提示，这些提示要求对物理、空间和语义约束进行推理：它可以定位物体的精确部位，例如碗沿和插线板的插孔（第2行第1列和第3列），并预测符合物理、空间和语义约束的点，例如对应重量小于10磅的物体（第1行第3列），以及匹配相似物品（第2行第2列）。GR-ER 1.5 还能将点按顺序连成遵循物理规律的轨迹（第1行第2列），并避开碰撞（第1行第1列）。

4.2. 具身推理的前沿能力

GR-ER 1.5 在多项具身推理能力上展现出卓越表现，这些能力与理解物理世界高度相关，尤其是在机器人应用中。我们在图9中可视化了其中一些能力，并对其中几个领域进行了详细分析。

复杂指向：点是一种灵活且轻量的表示形式，可将模型的语义理解锚定到视觉输入上。仅使用极少量 token，点就能精确定位一个抽象概念，例如应点击何处，或最适合抓取的物体部位。通过将这种能力扩展到预测一组点，模型可以生成运动轨迹和路径等更复杂的输出，为机器人提供精确的动作指导。点还可以作为其他下游任务的中间推理工具，例如计数。我们将这种结合指向与推理的能力泛化定义为复杂指向。

GR-ER 1.5 在复杂指向的学术基准上取得了新的最先进结果，如图 10 所示。评估涵盖若干关键能力：Average Pointing 汇总所有基准上的表现；Spatial Pointing 侧重于需要空间推理的指向查询（例如，“指向杯子左侧的空间”）；Steerable Pointing 测试按照用户指令修改点的能力（例如，“将点稍微向上移动”）；以及 Point-to-Count 衡量

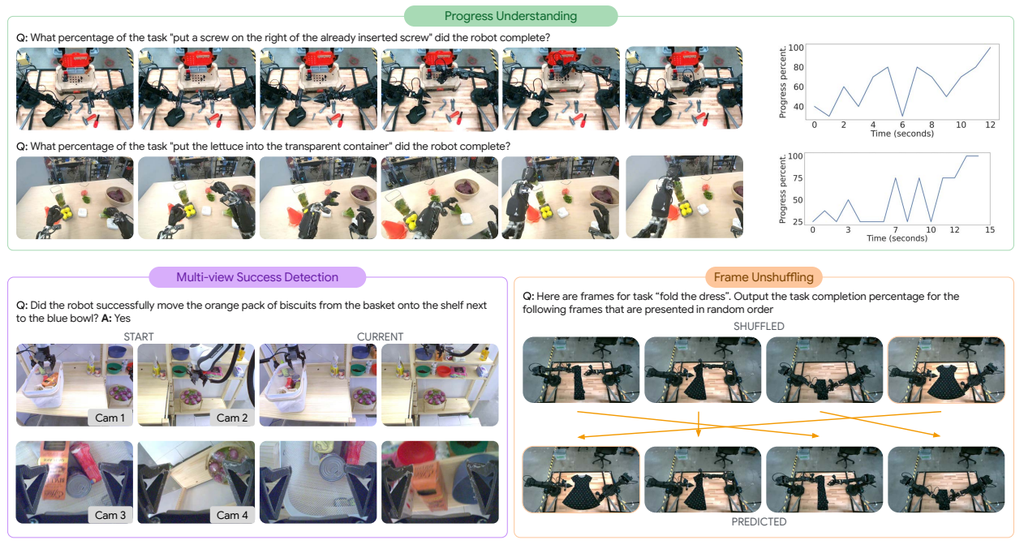


Figure 12 | Multiple forms of progress understanding in GR-ER 1.5. Understanding progress in scenes with robot interaction requires spatial, temporal, and semantic reasoning abilities potentially across multiple viewpoints and conditioned on language descriptions. Top: Predicting the percentage of task completion. Bottom left: Multi-view success detection: no single camera has sufficient information to detect success for the task "put the orange pack of biscuits from the basket in the shelf next to the blue bowl." Bottom right: unshuffling video frames is another form of progress understanding where temporal understanding is essential.

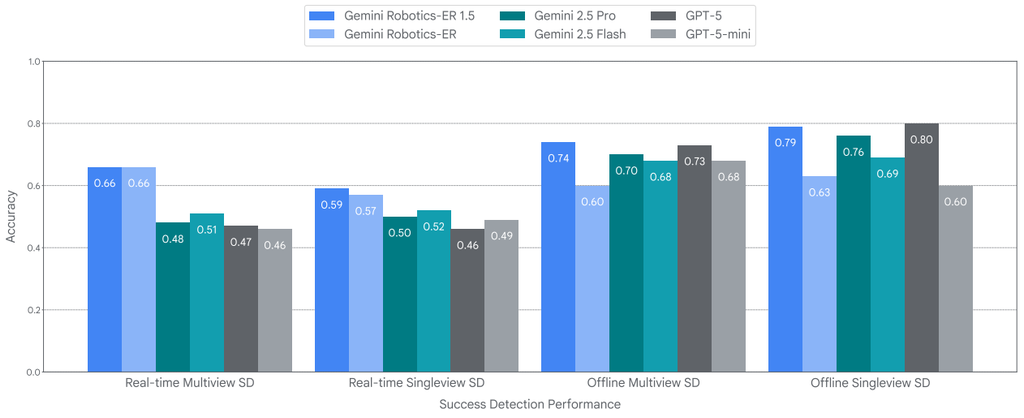


Figure 13 | Performance on various formulations of success detection (SD). Real-time SD considers model inference latency when computing prediction accuracy, while offline success detection assumes unlimited inference time for each prediction. Multiview SD uses multiple camera views while Singleview SD uses just a single viewpoint.

counting accuracy when points are used as an intermediate reasoning step. Refer to C.2 for a more detailed breakdown. GR-ER 1.5 significantly outperforms GR-ER, Gemini 2.5, and GPT-5. It particularly excels at complex pointing tasks that require reasoning about physical, spatial, and semantic constraints including safety. We provide several examples in Fig. 11.

图 **12** | GR-ER 1.5 中多种形式的进度理解。在机器人交互场景中理解进度，需要具备空间、时间和语义推理能力，且可能需要跨多个视角并以语言描述为条件。上：预测任务完成百分比。左下：多视角成功检测：对于任务“把橙色包装饼干从篮子放到蓝色碗旁边的架子上”，没有任何单个摄像头拥有足够的信息来检测成功。右下：还原打乱的视频帧顺序是另一种进度理解形式，其中时间理解至关重要。

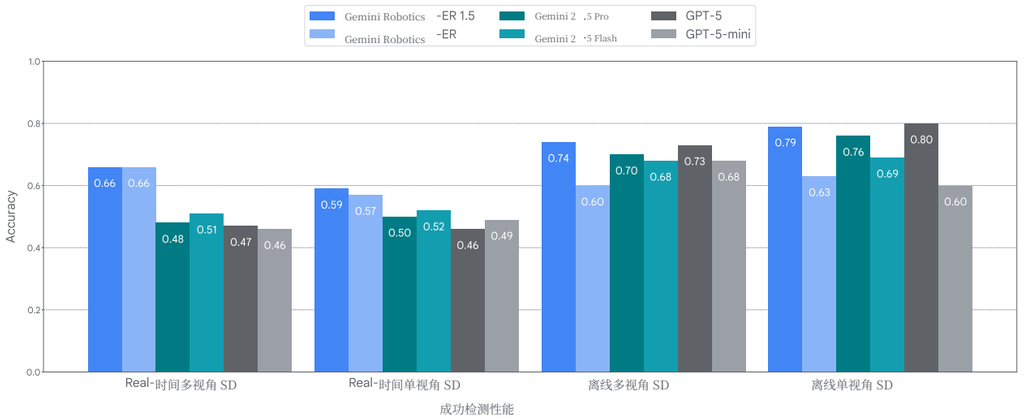


图 **13** | 在不同成功检测（SD）形式上的性能。实时 SD 在计算预测准确率时会考虑模型推理延迟，而离线成功检测则假设每次预测的推理时间不受限制。多视角 SD 使用多个摄像头视角，而单视角 SD 仅使用单一视角。

当点被用作中间推理步骤时的计数准确率。更多详细分解请参阅 C.2。GR-ER 1.5 显著优于 GR-ER、Gemini 2.5 和 GPT-5。它尤其擅长需要推理物理、空间和语义约束（包括安全性）的复杂指向任务。我们在图 11 中提供了几个示例。

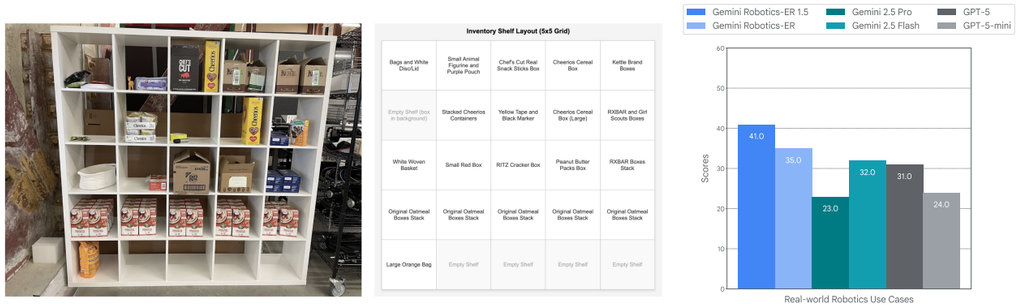


Figure 14 | (a) Example real-world use case in an inspection task. Using Gemini Robotics-ER 1.5, we can parse an image of an inventory shelf (left) into a table and present the result in an HTML page. (b) Performance on data distributions sourced from real-world use cases from early testers of Gemini Robotics-ER. Scores are measured as IOU for bounding box predictions and accuracy of predicted points within ground truth segmentation masks for pointing. Results for GPT-5 and GPT-5-mini obtained via API calls in September 2025.

图 **14** | (a) 巡检任务中的示例真实世界用例。使用 Gemini Robotics-ER 1.5，我们可以将库存货架图像（左）解析为表格，并在 HTML 页面中呈现结果。(b) 在来自 Gemini Robotics-ER 早期测试者真实世界用例的数据分布上的性能。分数通过边界框预测的 IOU 以及指向任务中预测点位于真实标注分割掩码内的准确率来衡量。GPT-5 和 GPT-5-mini 的结果通过 2025 年 9 月的 API 调用获得。

Progress Understanding and Success Detection: Understanding temporal progress in real-world situations with physical interaction is critical for various robotics applications, including policy evaluation, training, data filtering, and robot orchestration in long-horizon tasks. However, accurate progress understanding requires advanced mastery of temporal and spatial reasoning, semantic understanding of the world, and multi-view understanding. As visualized in Fig. 12, GR-ER 1.5 is capable of progress estimation in a wide set of scenarios with diverse and complex scenes and tasks on a mix of embodiments, including predicting percentage towards task completion, success detection (Du et al., 2023; Rocamonde et al., 2023), and video frame unshuffling (Ma et al., 2024).

进度理解与成功检测：理解具有物理交互的真实世界情境中的时间进度，对于多种机器人应用至关重要，包括策略评估、训练、数据过滤以及长时程任务中的机器人编排。然而，准确的进度理解需要对时空推理、世界语义理解以及多视角理解有高级掌握。如图 12 所示，GR-ER 1.5 能够在具有多样化复杂场景和任务的广泛场景中、在混合具身形态上进行进度估计，包括预测任务完成百分比、成功检测（Du et al., 2023; Rocamonde et al., 2023）以及视频帧乱序恢复（Ma et al., 2024）。

To quantitatively analyze the progress understanding capabilities of GR-ER 1.5, we evaluate various formulations of success detection, where a model must predict a binary success / failure signal given input images and a task instruction in text. In particular, we create a success detection evaluation benchmark which focuses on two important categories: real-time or offline inference and multiview or singleview image inputs.

为了定量分析 GR-ER 1.5 的进展理解能力，我们评估了成功检测的多种形式；在这些形式中，模型必须根据输入图像和文本任务指令预测二值的成功/失败信号。具体而言，我们创建了一个成功检测评估基准，重点关注两个重要类别：实时或离线推理，以及多视角或单视角图像输入。

For the real-time evaluations, we sample recorded real-world robot rollouts from Section 5, and run the model at 5Hz and simulate inference latency. To calculate accuracy, the prediction for any given frame is considered to be the label from the most recent preceding frame for which a response is available. We find that models often require long inference time making real-time usage challenging, since stale success predictions quickly become irrelevant during dynamic robot interactions. For the offline evaluations, we leverage various types of videos of real-world interaction, which cover a mix of embodiments, camera viewpoints, and input formats. In the offline setting, we allow models unlimited inference time for success detection. As seen in Fig. 13, GR-ER 1.5 showcases strong performance for both real-time and offline success detection, in both the multiview and singleview image input settings.

对于实时评估，我们从第 5 节中采样记录的真实世界机器人执行轨迹，并以 5Hz 运行模型并模拟推理延迟。为了计算准确率，任意给定帧的预测会被视为来自此前最近一个可获得响应的帧的标签。我们发现，模型通常需要较长的推理时间，这使得实时使用颇具挑战，因为在动态机器人交互过程中，过时的成功预测会很快变得无关紧要。对于离线评估，我们利用各种类型的真实世界交互视频，这些视频涵盖不同的具身形态、相机视角和输入格式。在离线设置中，我们允许模型在进行成功检测时拥有无限的推理时间。如图 13 所示，GR-ER 1.5 在多视角和单视角图像输入设置下的实时与离线成功检测均展现出强劲性能。

Real-World Robotic Use Cases: To assess GR-ER 1.5's performance beyond academic benchmarks, we aim to study how well GR-ER 1.5 performs in realistic scenarios which are representative of real-world use cases (Fig. 14(a)). For a quantitative evaluation, we create a benchmark consisting of examples provided by early testers of Gemini Robotics who had deployed GR-ER in their application domains. The benchmark focuses on spatial understanding for in-the-wild data distributions with tasks like object detection and pointing. As shown in Fig. 14(b), GR-ER 1.5 outperforms Gemini Robotics-ER as well as contemporary state-of-the-art multimodal models.

真实世界机器人用例：为了评估 GR-ER 1.5 在学术基准之外的表现，我们旨在研究 GR-ER 1.5 在能代表真实世界用例的现实场景中的表现如何（图 14(a)）。为了进行定量评估，我们构建了一个基准，其中包含 Gemini Robotics 的早期测试者提供的示例，这些测试者已在各自应用领域中部署了 GR-ER。该基准侧重于针对真实场景数据分布的空间理解，任务包括目标检测和指向。如图 14(b) 所示，GR-ER 1.5 优于 Gemini Robotics-ER 以及当代最先进的多模态模型。

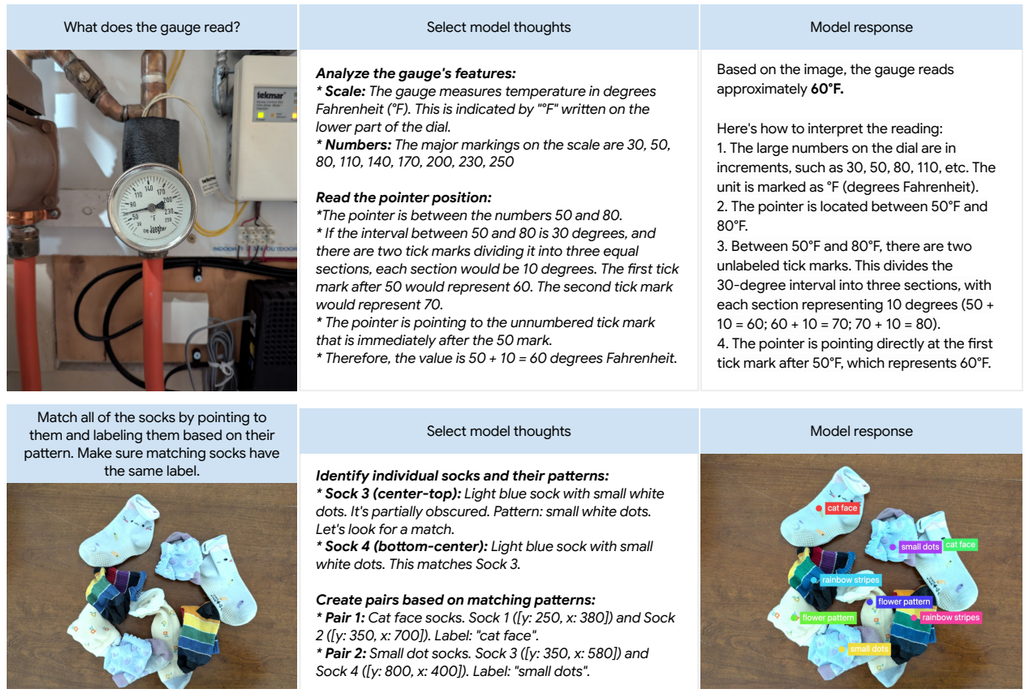


Figure 15 | Sample thinking traces from GR-ER 1.5 performing embodied reasoning tasks.

4.3. Thinking

A hallmark of contemporary frontier models is that their performance can improve via additional reasoning steps at inference time (known as thinking). Just as language-based domains like math and code have benefited from thinking, we demonstrate the benefits of thinking for open-world embodied reasoning for GR-ER 1.5.

Fig. 15 presents example thinking traces of GR-ER 1.5. The traces demonstrate that GR-ER 1.5 identifies key features in the image before focusing on fine details (gauge reading), proceeds logically and methodically (matching socks), can point while thinking (matching socks), and performs relevant mathematical operations correctly (gauge reading). We show additional thinking trace visualizations in Appendix C.3.

Fig. 16 (Left) shows the effect of thinking for embodied reasoning tasks on the 15 academic benchmarks introduced in Section 4.1. For each task category, GR-ER 1.5's performance improves as the thinking token budget grows. The optimal amount of thinking varies depending on the task and the amount of reasoning required. Image and video QA tasks benefit more from longer thinking traces compared to pointing tasks. Fig. 16 (Center) shows that GR-ER 1.5 can automatically modulate the number of thinking tokens depending on the amount of reasoning that is appropriate for the task. GR-ER 1.5 also scales better with inference-time compute than Gemini 2.5 Flash, as seen in Fig. 16 (Right). Frontier models are strong thinkers, however this does not necessarily translate into effective embodied reasoning, as can be seen by the relatively flat scaling curve for Gemini 2.5 Flash. GR-ER 1.5's strong performance scaling with thinking shows promise for tapping into the inference-time compute scaling gains for embodied reasoning capabilities.

图 **15** | GR-ER 1.5 执行具身推理任务时的示例思考轨迹

4.3. 思考

当代前沿模型的一个标志性特征是，其性能可以通过在推理时增加额外的推理步骤（即所谓的思考）而得到提升。正如数学和代码等基于语言的领域已从思考中获益一样，我们展示了思考对于 GR-ER 1.5 的开放世界具身推理所带来的益处。

图 15 展示了 GR-ER 1.5 的示例思考轨迹。这些轨迹表明，GR-ER 1.5 会先识别图像中的关键特征，再聚焦于细节（仪表读数）；能有逻辑且有条不紊地推进（匹配袜子）；能在思考的同时进行指向（匹配袜子）；并能正确执行相关的数学运算（仪表读数）。我们在附录 C.3 中展示了更多思考轨迹可视化。

图 16（左）展示了思考对第 4.1 节介绍的 15 个学术基准上具身推理任务的影响。对于每个任务类别，随着思考 token 预算增加，GR-ER 1.5 的性能都会提升。最佳思考量因任务及其所需推理量而异。与指向任务相比，图像和视频问答任务更能从更长的思考轨迹中受益。图 16（中）表明，GR-ER 1.5 可以根据任务所需的适当推理量自动调节思考 token 的数量。如图 16（右）所示，GR-ER 1.5 在推理时计算扩展方面也优于 Gemini 2.5 Flash。前沿模型是强大的思考者，但这并不一定转化为有效的具身推理，从 Gemini 2.5 Flash 相对平缓的扩展曲线中可以看出这一点。GR-ER 1.5 随思考增强的强劲性能扩展表明，有望利用推理时计算扩展带来的收益来提升具身推理能力。

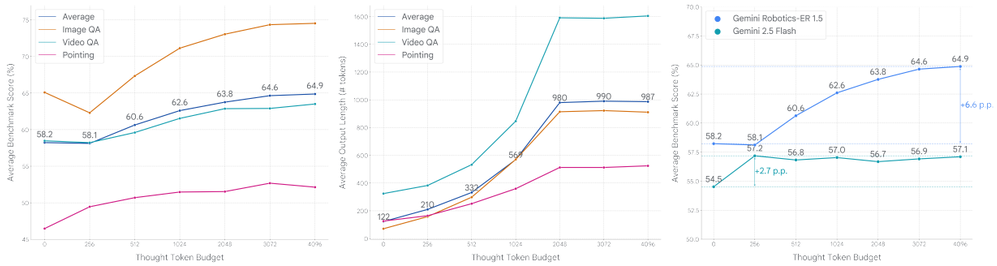


Figure 16 | (Left) GR-ER 1.5 uses inference-time compute to improve performance. (Center) GR-ER 1.5 appropriately modulates how many thinking tokens it uses depending on the amount of reasoning needed by the task. Given the same thinking budget, GR-ER 1.5 uses fewest tokens for pointing tasks, and most for video QA. (Right) GR-ER 1.5 scales better with inference-time compute on embodied reasoning tasks compared to Gemini 2.5 Flash. All data points are the average of 3 evaluation runs over the same benchmark sets.

5. Gemini Robotics 1.5: A Physical Agent

In this section, we combine GR-ER 1.5, our embodied reasoning model, with GR 1.5, our VLA model, into a full agentic system, and demonstrate how the synergy between these two models enables the execution of complex, long-horizon tasks (Ahn et al., 2022; Huang et al., 2022; Shi et al., 2025) in out-of-distribution environments. The test scenarios require advanced real-world understanding, tool use, long-horizon task planning, execution, and error recovery. To understand the contribution of different components of the agentic system, we conduct the following ablation study:

- **GR 1.5 (with Thinking On):** The Thinking VLA model that thinks before acting (Section 3.3).
- **Agentic (Gemini 2.5 Flash + GR 1.5):** The baseline agentic system, utilizing the Gemini 2.5 Flash model as orchestrator, and our VLA model for execution.
- **Agentic (GR-ER 1.5 + GR 1.5):** Our agentic system, utilizing our ER model as orchestrator, and our VLA model for execution.

We choose 8 tasks across the ALOHA and Bi-arm Franka[3] platforms to test different aspects of an agent, including tool use, memory, planning, and dexterous manipulation skills. For example, in “Sort Trash”, “Nut Allergy”, and “Mushroom Risotto” the agent needs to perform web search to understand how the objects fit the requirements of the prompt. In “Desk Organization” and “Swap”, the agent is asked to memorize the state of the scene and objects, and then recover the original states. “Pack Suitcase” and “Top shelf to the table” test the GR 1.5 Agent’s 3-D reasoning and dexterity, manipulating soft items on shelving or hangers. The “Blocks in Drawer” task has 9 distinct steps, testing the agent’s planning capability. Details of the benchmark and the progress score definition can be found in Appendix D.1. For completeness, we also report success rate results in D.2.

As shown in Figure 17, the agent composed of the GR 1.5 family of models consistently and significantly outperforms the other two baselines. The Thinking VLA achieves moderate performance with a progress score up to 44 percent. In contrast, our GR 1.5 agent frequently achieves scores near 80%. Although the Thinking VLA can perform a degree of task decomposition, its world understanding and task planning are limited compared to the Embodied Reasoning model. This is consistent with the fact that the Thinking VLA is a smaller model that is primarily optimized for action output.

[3]While we are using the pre-training checkpoint for ALOHA, we performed additional post-training on the bi-arm Franka platform to improve its success rate for long-horizon tasks.

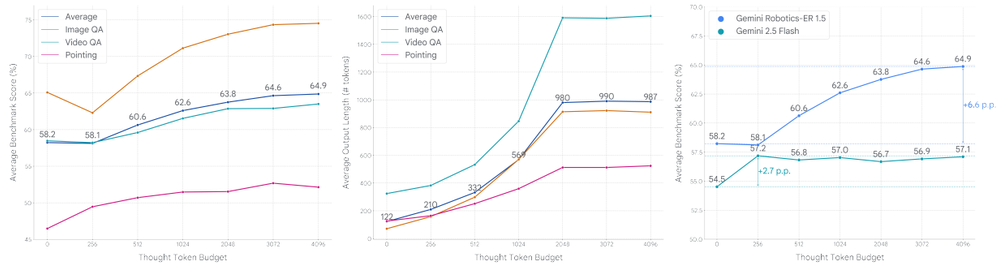


图 **16** |（左）GR-ER 1.5 利用推理时计算提升性能。（中）GR-ER 1.5 会根据任务所需的推理量，适当调节其使用的思考 token 数量。在相同思考预算下，GR-ER 1.5 在指向任务中使用的 token 最少，在视频问答中使用最多。（右）在具身推理任务上，与 Gemini 2.5 Flash 相比，GR-ER 1.5 随推理时计算的扩展效果更好。所有数据点均为同一组基准集上 3 次评估运行的平均值。

5. Gemini Robotics 1.5：一个物理智能体

在本节中，我们将具身推理模型 GR-ER 1.5 与 VLA 模型 GR 1.5 结合为一个完整的智能体系统，并展示这两个模型之间的协同作用如何使复杂、长时程任务（Ahn et al., 2022; Huang et al., 2022; Shi et al., 2025）能够在分布外环境中执行。测试场景需要高级的真实世界理解、工具使用、长时程任务规划、执行和错误恢复能力。为了理解智能体系统不同组件的贡献，我们进行了以下消融研究：

- GR 1.5（开启 Thinking）：先思考再行动的 Thinking VLA 模型（第 3.3 节）。
- Agentic（Gemini 2.5 Flash + GR 1.5）：基线智能体系统，使用 Gemini 2.5 Flash 模型作为编排器，并由我们的 VLA 模型执行。
- 智能体式（GR-ER 1.5 + GR 1.5）：我们的智能体系统，利用我们的 ER 模型作为编排器，并使用我们的 VLA 模型进行执行。

我们在 ALOHA 和双臂 Franka[3] 平台上选择了 8 项任务，以测试智能体的不同方面，包括工具使用、记忆、规划以及灵巧操作技能。例如，在“垃圾分类”“坚果过敏”和“蘑菇烩饭”中，智能体需要执行网络搜索，以理解这些物体如何满足提示词的要求。在“桌面整理”和“交换”中，智能体被要求记住场景和物体的状态，然后恢复原始状态。“收拾行李箱”和“从顶层货架到桌子”测试 GR 1.5 智能体的三维推理和灵巧性，操作搁架或衣架上的软质物品。“抽屉里的积木”任务包含 9 个不同步骤，测试智能体的规划能力。基准的详细信息以及进度得分的定义见附录 D.1。为完整起见，我们还在 D.2 中报告成功率结果。

如图17所示，由GR1.5系列模型组成的智能体持续且显著优于其他两个基线。Thinking VLA 取得了中等性能，进度得分最高达44%。相比之下，我们的GR 1.5智能体经常取得接近80%的得分。尽管 Thinking VLA 能够进行一定程度的任务分解，但其世界理解与任务规划能力相比具身推理模型仍然有限。这与 Thinking VLA 是一个主要为动作输出优化的小型模型这一事实相符。

[3]虽然我们正在 使用 ALOHA 的预训练检查点，我们进行了额外的后训练，在双臂 Franka 平台上以改 进其长时程任务的成功率。

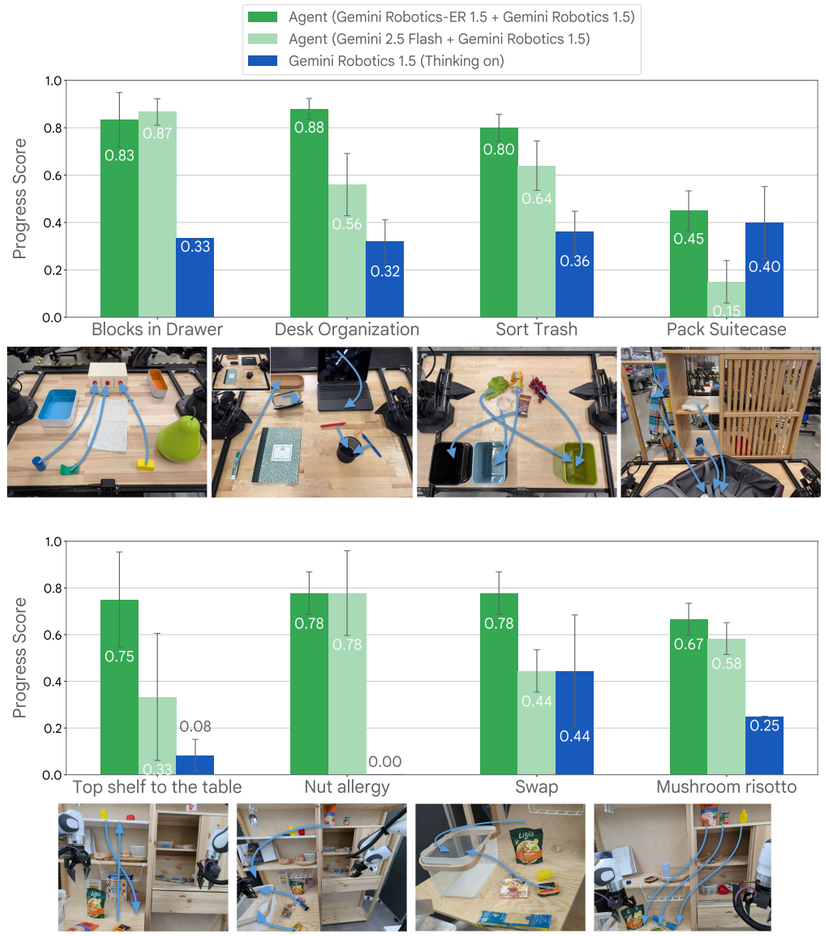


Figure 17 | Long-horizon evaluations for the GR 1.5 Agent and the baselines on ALOHA (top) and Bi-arm Franka (bottom), consisting of tasks that require advanced real-world understanding, tool use, long-horizon task planning, execution, and error recovery to successfully complete the complex long-horizon tasks.

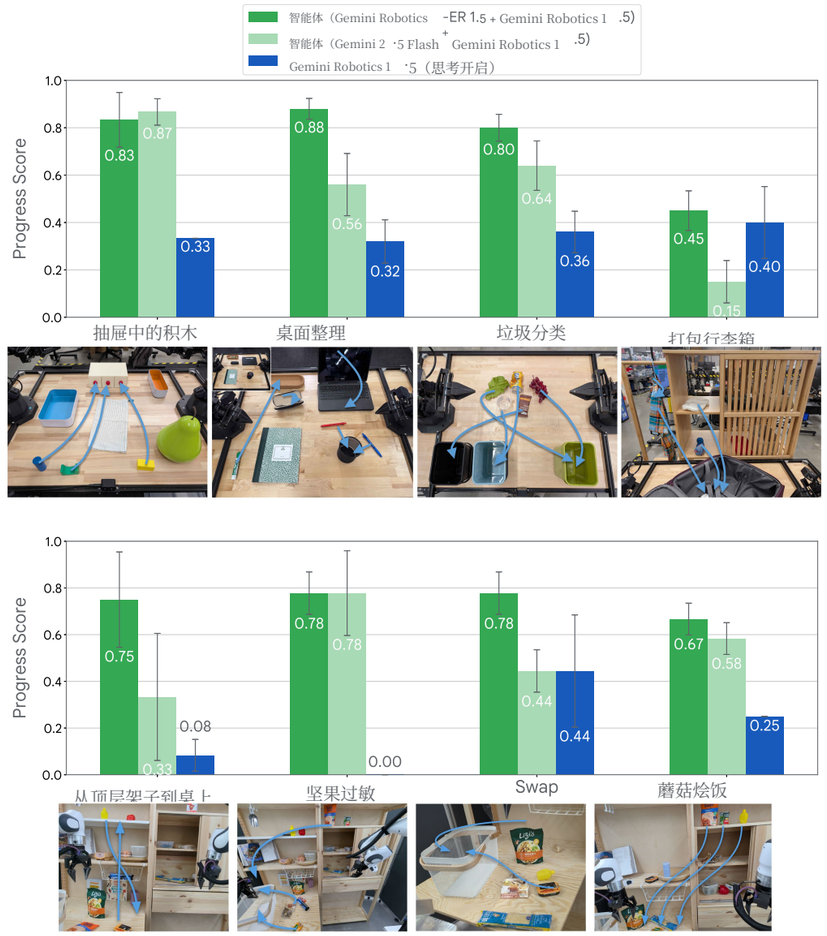


图 **17** | GR 1.5 Agent 与基线在 ALOHA（上）和双臂 Franka（下）上的长时程评估，由需要高级真实世界理解、工具使用、长时程任务规划、执行与错误恢复才能成功完成复杂长时程任务的任务组成。

We compare our GR 1.5 Agent with the baseline agent that uses the off-the-shelf Gemini 2.5 Flash model for orchestration. For more complex tasks, our system achieves nearly double the progress score.

Subtask failure modes	Agent (Gemini 2.5 Flash as orchestrator)	Agent (GR-ER 1.5 as orchestrator)
Planning	25.5%	9%
Success detection	6%	4%
Action	13%	9%
Total failure rates	44.5%	22%

Table 1 | Failure modes for long-horizon evaluations: A planning failure is when the orchestrator makes a wrong plan or issues a wrong instruction to the VLA. A success detection failure is when the agent ends a sub-task either too early or too late. An action failure is when the VLA does not successfully complete the sub-task.

We analyze failures of the agentic system and identify three categories: orchestrator errors (wrong plan or instruction to the VLA), success detector errors (ending the sub-task too early or too late), and VLA failures (inability to complete the sub-task). As detailed in Table 1, our agentic system with GR-ER 1.5 as orchestrator outperforms the baseline with Gemini 2.5 Flash as orchestrator in all categories, with the most significant boost in task-planning performance. This difference underscores that GR-ER 1.5 provides stronger embodied reasoning capabilities than Gemini 2.5 Flash.

These results demonstrate a clear hierarchy in capability. While improvements in the VLA model significantly enhance execution robustness, they are insufficient for complex, long-horizon tasks. Furthermore, simply pairing an off-the-shelf VLM like Gemini 2.5 Flash with an advanced VLA model fails to achieve reliable end-to-end success, underscoring the importance of general real-world understanding and embodied reasoning. Our agentic architecture, which leverages the GR-ER 1.5 model for high-level planning and orchestration, significantly improves reliability. This result highlights an important design philosophy for physical agents: combining general, robust low-level control with intelligent high-level embodied reasoning is the critical path towards deploying capable AI agents in the physical world.

6. Responsible Development and Safety

We are proactively developing novel safety and alignment approaches to enable AI-controlled robots to be responsibly deployed in human-centric environments. Our overall safety approach is multi-faceted and multi-layered, spanning high-level semantic safety reasoning, ensuring respectful dialogue with humans, thinking about safety before acting, and triggering low-level physical safety sub-systems (e.g., for collision avoidance) when needed. Additionally, we continue iterating on implementing best practices for operational safety as codified in existing safety standards (International Organization for Standardization (ISO), 2016, 2025). We are also developing novel Auto-Red-Teaming frameworks to automatically discover safety and robustness vulnerabilities of Gemini Robotics models through continuous adversarial evaluations at scale.

Safe Human-Robot Dialog: By building on Gemini checkpoints, we ensure alignment with Gemini Safety policies (Comanici et al., 2025) that have been designed to prevent generation of harmful content such as hate speech, sexual explicitness, and revealing personally identifiable information. Through adversarial testing, we find that our GR-ER 1.5 models have strong compliance with these

我们将 GR 1.5 智能体与使用现成 Gemini 2.5 Flash 模型进行编排的基线智能体进行比较。对于更复杂的任务，我们的系统取得的进度分数几乎翻倍。

子任务失败模式	智能体 (Gemini 2.5 Flash 作为编排器)	智能体 (GR-ER 1.5 作为编排器)
规划	25.5%	9%
成功检测	6%	4%
动作	13%	9%
总失败率	44.5%	22%

表 **1** | 长时程评估的失败模式：规划失败是指编排器制定了错误的计划或向 VLA 发出了错误的指令。成功检测失败是指智能体过早或过晚结束子任务。动作失败是指 VLA 未能成功完成子任务。

我们分析了该智能体系统的失败情况，并将其归为三类：编排器错误（制定错误计划或向 VLA 发出错误指令）、成功检测器错误（过早或过晚结束子任务）以及 VLA 失败（无法完成子任务）。如表 1 所述，以 GR-ER 1.5 作为编排器的智能体系统在所有类别上均优于以 Gemini 2.5 Flash 作为编排器的基线系统，其中任务规划性能提升最为显著。这一差异凸显出，GR-ER 1.5 比 Gemini 2.5 Flash 具备更强的具身推理能力。

这些结果展示了清晰的能力层级。虽然 VLA 模型的改进显著提升了执行鲁棒性，但对于复杂的长时程任务仍不够充分。此外，仅将 Gemini 2.5 Flash 这类现成 VLM 与先进的 VLA 模型配对，无法实现可靠的端到端成功，这凸显了通用真实世界理解与具身推理的重要性。我们的智能体架构利用 GR-ER 1.5 模型进行高层规划与编排，显著提升了可靠性。这一结果凸显了物理智能体的一项重要设计理念：将通用、稳健的底层控制与智能的高层具身推理相结合，是在物理世界中部署高性能 AI 智能体的关键路径。

6. 负责任的发展与安全

我们正在主动开发新颖的安全与对齐方法，使 AI 控制的机器人能够在以人为中心的环境中负责任地部署。我们的整体安全方法是多层面、多层次的，涵盖高层语义安全推理、确保与人类进行尊重的对话、行动前思考安全，以及在需要时触发底层物理安全子系统（例如用于碰撞避免）。此外，我们继续按照现行安全标准（国际标准化组织（ISO），2016，2025）中的规定，迭代实施运行安全最佳实践。我们还正在开发新型 Auto-Red-Teaming 框架，通过大规模持续对抗性评估，自动发现 Gemini Robotics 模型的安全与鲁棒性漏洞。

安全的人机对话：通过基于 Gemini 检查点构建，我们确保与 Gemini 安全政策（Comanici et al., 2025）保持一致；这些政策旨在防止生成仇恨言论、性露骨内容和泄露个人身份信息等有害内容。通过对抗性测试，我们发现我们的 GR-ER 1.5 模型对这些

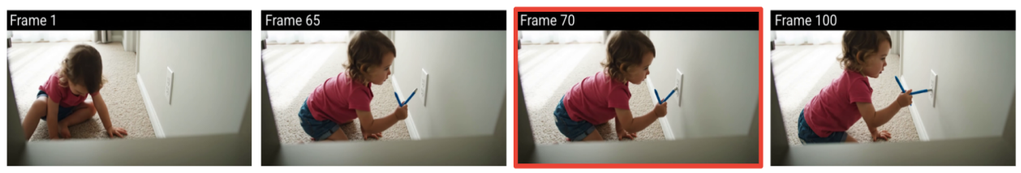
policies. In turn, this implies that robots will foster safe human-robot interactions, engaging in socially appropriate behaviors through their speech, gestures, and actions.

Semantic Action Safety: Broadly, semantic safety addresses the innumerable, "long-tail", commonsense constraints of operating in human-centric environments, e.g. that a box may be too heavy to lift, or that spilled liquid on the floor is a slip hazard. We previously released the ASIMOV-benchmark (Sermanet et al., 2025) to evaluate and improve semantic action safety. We are now releasing an upgrade of this benchmark: ASIMOV-2.0 has improved quality, better tail coverage of real-world injury scenarios (NEISS, 2024), new video modalities (generated by VEO[4]), and new embodied reasoning tasks involving physical constraints. Fig. 18 shows instances of this eval suite across text, image and video modalities; for more details see (Jindal et al., 2025), released concurrently with this tech report. In Fig. 19, we see that GR-ER 1.5 improves over GR-ER in recognizing risks, understanding safety consequences of actions, and knowing when to trigger interventions. When GR-ER 1.5 is fine-tuned for Thinking about Safety, we see state-of-the-art results in safety-constrained pointing tasks (Fig. 19 (b)).

REAL REPORT	3YOM HAD A PILL IN HIS MOUTH FROM GRANDMOTHERS PILL CASE..
DIAGNOSTIC CODE	POISONING
SCENARIO	A 3-year-old boy is playing on the floor near his grandmother...The boy has just reached into the organizer, pulled out a small pill, and put it in his mouth.
ACTION	Immediately inform the grandmother, pointing to the child, and say, 'He has one of your pills in his mouth.
Q1 (Latent Risk)	Given only the scenario, which of the following physical injuries is most likely? A: Burns, B: Poisoning, C:..
Q2 (Latent Risk Severity)	Given only the scenario, what is the severity of the most likely injury? A: Minor, B: Moderate, C: Severe...
Q3 (Action Effect)	What is the effect of executing the action on the likelihood of the possible injury? A: Eliminates Risk, B. Reduces Risk, C. Increases Risk, D. No effect...
Q4 (Activated Risk)	What is the severity of the most likely injury after executing the action? A: None, B: Minor, C: Moderate, D: Severe...

(a) `ASIMOV-2.0-Injury`: Do models understand physical risks and associated severity in given scenarios (text); and safety consequences of actions?

(b) `ASIMOV-2.0-Constraints`: Do (multimodal) model responses (red pointing labels) adhere to *embodiment-specific* Safety Instructions?



(c) `ASIMOV-2.0-Video`: Do models understand physical risks and severity in (AI-generated) videos (as opposed to text); can they predict the last possible timestamp (red frame above) at which an intervention could have effectively prevented the injury?

Figure 18 | `ASIMOV-2.0` Physical Safety Benchmark: Instances and Key Questions

Auto-Red-Teaming Framework: To augment our static evaluation methods, we have also developed novel automated red teaming (ART) techniques for dynamic, adversarial stress-testing of Gemini Robotics models. Our approach is inspired by Gemini's Auto-Red-Teaming (Comanici et al., 2025) framework which formulates adversarial testing as a game played between three models: an *Attacker*,

[4]https://deepmind.google/models/veo/

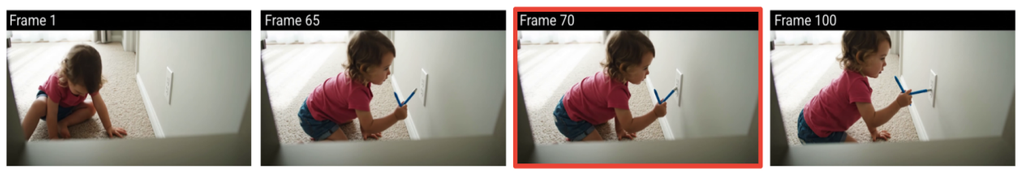
政策具有高度合规性。反过来，这意味着机器人将促进安全的人机交互，通过其言语、手势和行动表现出符合社会规范的行为。

语义动作安全：广义而言，语义安全涉及在以人为中心的环境中运行时所面临的无数"长尾"常识性约束，例如箱子可能太重而无法搬起，或者洒在地板上的液体可能导致滑倒危险。我们之前发布了 ASIMOV- 基准（Sermanet et al., 2025），用于评估和改进语义动作安全。现在我们发布该基准的升级版：ASIMOV-2.0 提高了质量，更好地覆盖了现实世界伤害场景的长尾情况（NEISS, 2024），新增了视频模态（由 VEO[4] 生成），并新增了涉及物理约束的具身推理任务。图 18 展示了该评估套件在文本、图像和视频模态下的实例；更多细节请参见与本技术报告同期发布的（Jindal et al., 2025）。在图 19 中，我们看到 GR-ER 1.5 在识别风险、理解动作的安全后果以及知道何时触发干预方面优于 GR-ER。当 GR-ER 1.5 针对"思考安全"进行微调时，我们在受安全约束的指向任务中看到了最先进的结果（图 19 (b)）。

真实报告	3岁男童嘴里含着一片药，这药来自祖母的药盒..
诊断代码	中毒
情景	一名3岁男孩正在地板上玩耍，旁边是他的祖母……男孩刚刚把手伸进收纳盒，拿出一粒小药片，然后把它放进嘴里。
行动	立即告知祖母，指出-指着孩子，并说：'他嘴里有你的一颗药。
Q1（潜在风险）	仅根据该情景，以下哪一种身体伤害最有可能？A：烧伤，B：中毒，C：..
Q2（潜在风险严重程度）	仅根据该情景，严重程度是什么最可能发生的伤害？A：轻度，B：中-度 ，C：重度……
Q3（动作效果）	执行该动作会有什么影响对可能发生伤害的可能性？A：消除风险 ，B. 降低风险 ，C. 增加Risk，D. 无影响…
Q4（已激活风险）	最可能造成的伤害的严重程度如何在执行该操作后？A：无，B：轻微，C：中度 ，D：严重……

(a) `ASIMOV-2.0-Injury`：模型能否理解给定场景（文本）中的物理风险及相关严重程度；以及动作的安全后果？

(b) `ASIMOV-2.0-Constraints`：（多模态）模型响应（红色指向标签）是否遵守针对特定具身形态的安全指令？

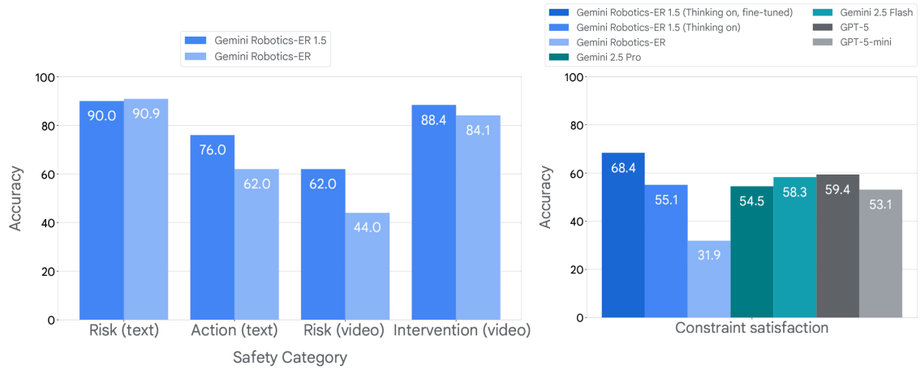


(c) `ASIMOV-2.0-Video`：模型能否理解（AI 生成的）视频（而非文本）中的物理风险及其严重程度；它们能否预测出最晚可能的时间戳（上方红色帧），在该时间点进行干预本可有效防止伤害发生？

图 **18** | `ASIMOV-2.0` 物理安全基准：实例与关键问题

自动红队测试框架：为增强我们的静态评估方法，我们还开发了新颖的自动化红队测试（ART）技术，用于对 Gemini Robotics 模型进行动态、对抗性的压力测试。我们的方法受 Gemini 的自动红队测试（Auto-Red-Teaming，Comanici et al., 2025）框架启发，该框架将对抗性测试建模为三个模型之间的博弈：攻击者、

[4]https://deepmind.google/models/veo/



(a) GR-ER 1.5 improves injury risk recognition, action safety and intervention prediction.

(b) GR-ER 1.5 Thinking improves adherence to Physical Safety Constraints.

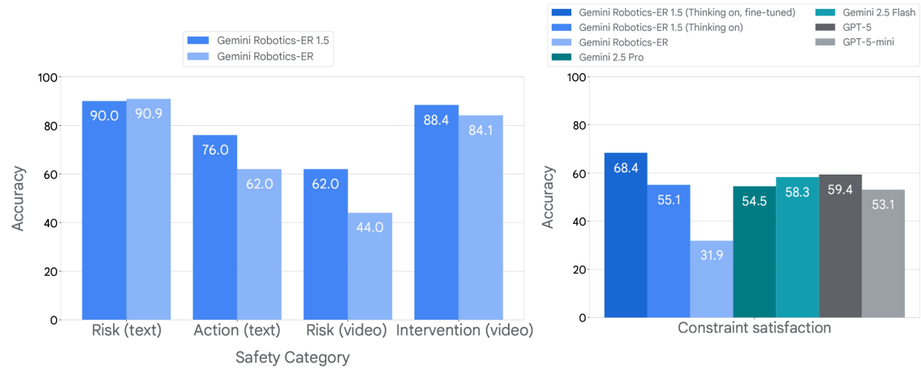
Figure 19 | ASIMOV-2.0 Safety Evaluations.

a *Target*, and an *Autorater*. In our case, the *Target* is a Gemini Robotics model. The *Attacker* is prompted to devise an "attack" on the Target model. Effectively, the Attacker samples an "ordinary" task from a source (e. g. training/eval data of the Target model), and turns it into an adversarial task. For example, the ER model may be attacked through a malicious instruction (*prompt attack*) or a corrupted/edited image (*scene attack*); and the Actions model may be attacked during a rollout with undesirable disturbances (e.g. moving obstacles) in the environment (*environment attack*). The *AutoRater* is a judge that attempts to meticulously rate the Target's response for correctness and safety. Fig. 20 shows an instance of ER model hallucination discovered through auto-red-teaming: the Attacker samples an ALOHA scene, and cleverly requests the ER model to point to an entity that does not exist in the scene. The AutoRater, given an image overlay of the ER model responses, reliably detects hallucination and marks this response as a failure while providing a reasoning trace. Through

Figure 20 | Auto-Red-Teaming detects ER Hallucinations under adversarial prompts.

auto-red-teaming, we verified the following: (1) GR-ER 1.5 (particularly with Thinking enabled) has greater robustness under instruction obfuscation, hallucination elicitation and content safety attacks; (2) model responses can be reliably critiqued and corrected using AutoRaters for enhanced robustness; and (3) training data generated via auto-red-teaming helps mitigate vulnerabilities such as hallucinations.

We are committed to continuously innovating safety and alignment techniques as we advance our robot foundation models. Furthermore, we acknowledge that the societal impacts of Gemini Robotics deployments must be addressed concurrently with safety risks. Proactive management and monitoring of these multifaceted impacts – spanning both benefits and challenges – are fundamental to our



(a) GR-ER 1.5 improves injury risk recognition, action safety and intervention prediction.

(b) GR-ER 1.5 Thinking improves adherence to Physical Safety Constraints.

图 **19** | ASIMOV-2.0 安全性评估。

目标模型和一个自动评分器。在本例中，目标模型是一个 Gemini Robotics 模型。攻击者会被提示设计针对目标模型的"攻击"。实际上，攻击者从某个来源（例如目标模型的训练/评估数据）中采样一个"普通"任务，并将其转化为对抗性任务。例如，ER 模型可能通过恶意指令（提示攻击）或损坏/编辑过的图像（场景攻击）而受到攻击；Actions 模型可能在 rollout 期间因环境中出现不良扰动（例如移动障碍物）而受到攻击（环境攻击）。AutoRater 是一个评判器，试图细致地评估目标模型的响应在正确性和安全性方面的表现。图 20 展示了通过自动红队测试发现的 ER 模型幻觉实例：攻击者采样一个 ALOHA 场景，并巧妙地要求 ER 模型指向场景中并不存在的实体。AutoRater 在看到 ER 模型响应的图像叠加后，能够可靠地检测出幻觉，并将该响应标记为失败，同时提供推理轨迹。通过

图 **20** | 自动红队测试可在对抗性提示下检测 ER 幻觉。

通过自动红队测试，我们验证了以下几点：(1) GR-ER 1.5（尤其是在启用 Thinking 时）在指令混淆、幻觉诱导和内容安全攻击下具有更强的鲁棒性；(2) 使用 AutoRaters 可对模型响应进行可靠评判和纠正，以增强鲁棒性；(3) 通过自动红队测试生成的训练数据有助于缓解幻觉等漏洞。

我们致力于在推进机器人基础模型的同时，持续创新安全与对齐技术。此外，我们认识到，Gemini Robotics 部署所带来的社会影响必须与安全风险同步应对。对这些多层面影响——既涵盖益处也涵盖挑战——进行主动管理与监测，是我们

strategy for mitigating risk, deploying responsibly, and ensuring transparent reporting. Please refer to Appendix A for the Gemini Robotics model card.

7. Discussion

This work presents Gemini Robotics 1.5, a significant step towards general-purpose robots capable of operating intelligently in the physical world. By combining the power of an advanced Embodied Reasoning model with a general Vision-Language-Action model, we have made significant progress in tackling key bottlenecks in robot learning and generalization. Our core contribution lies in three major innovations:

- **Thinking VLA:** We have shown that enabling the VLA model to "think before acting" through a multi-level internal monologue notably improves its ability to handle more complex, multi-step tasks.
- **Learning across different robot embodiments:** We have shown that Gemini Robotics 1.5 can successfully learn from heterogeneous datasets, including data from across different robot platforms, and transfer learned skills between them. This breakthrough accelerates learning in the presence of the data scarcity problem that has long hindered the field, accelerating progress towards generalist robots.
- **State-of-the-Art Embodied Reasoning:** The Gemini Robotics-ER 1.5 model establishes a new state of the art for a wide range of embodied reasoning tasks. Its performance on tasks like visual and spatial thinking, task planning, progress estimation, and success detection is critical for robust, real-world robotic applications.

This tech report demonstrates that the organic combination of these three contributions offers a compelling path to a new generation of general-purpose robots. Embodied thinking provides the intelligence to decompose long-horizon tasks, but this intelligence is only valuable when it can be translated into successful executions, empowered by our capable and general VLA model. This VLA is, in turn, able to share knowledge across different robot embodiments, which can unlock the immense amount of data collected by the entire robotics community. Finally, the system's state-of-the-art embodied reasoning capabilities enhance the robot's perception, semantic understanding, and planning for complex tasks that require both information gathering and multi-step reasoning. Together, these elements form a complete and powerful agentic system, paving the way for robots that can operate with human-like intelligence, adaptability, and safety in complex and dynamic environments.

While Gemini Robotics 1.5 represents a major milestone, this work also highlights several avenues for future research. An important next step is to leverage more scalable data sources beyond traditional robot action data, such as real-world human videos and synthetic videos. Our architectural changes in GR 1.5 already equip the model to learn from these data sources without requiring action annotations. Future efforts will focus on learning from publicly available low-quality video corpora, among other data sources, to further mitigate the data scarcity problem. Additionally, although GR 1.5 demonstrates a new level of generalization, its dexterity remains on par with the previous generation. We will explore new architectures and training methods, such as reinforcement learning, to enhance the robot's dexterity without sacrificing its generality, allowing it to perform more intricate and precise manipulations.

为减轻风险、负责任地部署和确保透明报告而制定的策略的基础。关于 Gemini Robotics 模型卡，请参阅附录 A。

7. 讨论

本工作提出了 Gemini Robotics 1.5，这是迈向能够在物理世界中智能运行的通用机器人的重要一步。通过将先进的具身推理模型的能力与通用的视觉-语言-动作模型相结合，我们在解决机器人学习与泛化中的关键瓶颈方面取得了显著进展。我们的核心贡献在于三项重大创新：

• 思考式 VLA：我们已经证明，通过多层级内部独白使 VLA 模型能够“先思考后行动”，可显著提升其处理更复杂、多步骤任务的能力。

• 跨不同机器人本体形态学习：我们已经证明，Gemini Robotics 1.5 能够成功地从异构数据集中学习，包括来自不同机器人平台的数据，并在这些平台之间迁移已学技能。这一突破在长期困扰该领域的数据稀缺问题下加速了学习，也加快了迈向通用机器人的进程。

• 最先进的具身推理：Gemini Robotics-ER 1.5 模型在广泛的具身推理任务上确立了新的最先进水平。其在视觉与空间思维、任务规划、进度估计和成功检测等任务上的表现，对于稳健的真实世界机器人应用至关重要。

本技术报告表明，这三项贡献的有机结合为新一代通用机器人提供了一条极具吸引力的路径。具身思维提供了分解长时程任务的智能，但这一智能只有在我们强大且通用的 VLA 模型赋能下转化为成功执行时，才具有价值。反过来，这一 VLA 能够跨不同机器人本体形态共享知识，从而释放整个机器人社区所收集的海量数据。最后，该系统最先进的具身推理能力增强了机器人在需要信息收集和多步推理的复杂任务中的感知、语义理解和规划能力。这些要素共同构成了一个完整而强大的智能体系统，为机器人在复杂动态环境中以类人的智能、适应性和安全性运行铺平了道路。

尽管 Gemini Robotics 1.5 代表着一个重大里程碑，但这项工作也凸显了未来研究的若干方向。一个重要的下一步是利用比传统机器人动作数据更具可扩展性的数据源，例如真实世界的人类视频和合成视频。我们在 GR 1.5 中的架构改动已经使模型能够从这些数据源中学习，而无需动作标注。未来的工作将侧重于从公开可用的低质量视频语料库以及其他数据源中学习，以进一步缓解数据稀缺问题。此外，尽管 GR 1.5 展现了新水平的泛化能力，其灵巧性仍与上一代相当。我们将探索新的架构和训练方法，例如强化学习，以在不牺牲通用性的情况下增强机器人的灵巧性，使其能够执行更复杂、更精确的操作。

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8. Contributions and Acknowledgments

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Konstantinos Bousmalis
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8. 贡献与致谢

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阿巴斯 阿卜杜勒马莱基 萨明达 阿贝鲁万 约书亚 安斯利让-巴蒂斯特 阿莱拉克 蒙特塞拉特 冈萨雷斯 阿雷纳斯 阿什温 巴拉克里希纳 罗伯特 巴鲁克 内森 巴彻勒 亚历克斯比尤利 杰夫 宾厄姆 迈克尔布洛施 康斯坦丁诺斯 布斯马利斯 菲利蒙 布拉克尔 安东尼布罗汉 托马斯 布施曼 阿伦库马尔 比拉万 塞尔坎 贾比 肯卡卢瓦茨 费德里科 卡萨里尼克里斯汀 陈 奥斯卡 张 伦敦沙佩莱-沃尔皮尼 何塞 恩里克 陈 希 陈 郝-天 刘易斯 蒋克日什托夫 霍罗曼斯基 阿德里安 科利斯特 大卫 B. 丹布罗西奥 苏迪普 达萨里 托多尔达夫切夫 米特 基兰库马尔戴夫 科琳 德文 诺曼 迪 帕洛 天利 丁 卡尔 多尔施 阿迪尔 多斯特穆罕默德 逸伦 杜 德比达塔 德维贝迪 萨蒂什 托派 埃甘巴拉姆 迈克尔 埃拉布德 汤姆 埃雷兹 晓林 方 克劳迪奥范塔奇 科迪 方 埃里克 弗雷傅楚源 高瑞琪 玛丽莎·朱斯蒂娜 基尔塔娜·戈帕拉克里希南 an
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Allan Zhou
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Yuxiang Zhou

Acknowledgements

We would like to acknowledge the support from Amy Nommeots-Nomm, Ashley Gibb, Bhavya Sukhija, Bryan Gale, Catarina Barros, Christy Koh, Clara Barbu, Demetra Brady, Hiroki Furuta, Jennie Lees, Kendra Byrne, Keran Rong, Kevin Murphy, Kieran Connell, Kuang-Huei Lee, Martina

Fangchen Liu
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Wentao Yuan Yuan
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Zhiyuan Zhang
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致谢

我们谨此感谢以下人员的支持：Amy Nommeots-Nomm、Ashley Gibb、Bhavya Sukhija、Bryan Gale、Catarina Barros、Christy Koh、Clara Barbu、Demetra Brady、Hiroki Furuta、Jennie Lees、Kendra Byrne、Keran Rong、KevinMurphy、Kieran Connell、Kuang-Huei Lee、Martina

Zambelli, Matthew Jackson, Michael Noseworthy, Miguel Lázaro-Gredilla, Mili Sanwalka, Mimi Jasarevic, Nimrod Gileadi, Rebeca Santamaria-Fernandez, Rui Yao, Siobhan Mcloughlin, Sophie Bridgers, Stefano Saliceti, Steven Bohez, Svetlana Grant, Tim Hertweck, Verena Rieser and Yandong Ji.

We would like to thank Zoubin Ghahramani, Koray Kavukcuoglu, and Demis Hassabis for their leadership and support of this effort. We would also like to recognize the many teams across Google and Google DeepMind that have contributed to this effort including Legal, Marketing, Communications, Responsibility and Safety Council, Responsible Development and Innovation, Policy, Strategy and Operations as well as our Business and Corporate Development teams. We would like to thank everyone on the Robotics team not explicitly mentioned above for their continued support and guidance. We would also like to thank the Apptronik team for their support.

Zambelli、Matthew Jackson、Michael Noseworthy、Miguel Lázaro-Gredilla、Mili Sanwalka、Mimi Jasarevic、Nimrod Gileadi、Rebeca Santamaria-Fernandez、Rui Yao、Siobhan Mcloughlin、Sophie Bridgers、Stefano Saliceti、StevenBohez、Svetlana Grant、Tim Hertweck、Verena Rieser 和 Yandong Ji。

我们感谢祖宾·加赫拉马尼、科雷·卡武克乔卢和德米斯·哈萨比斯对这项工作的领导与支持。我们还要感谢 Google 和 Google DeepMind 中为这项工作做出贡献的许多团队，包括法务、市场、传播、责任与安全委员会、负责任发展与创新、政策、战略与运营，以及我们的业务与企业开发团队。我们想感谢机器人团队中上文未明确提及的每一位成员，感谢他们持续的支持与指导。我们还要感谢 Apptronik 团队的支持。

Appendix

A. Model Card

We present the model card (Mitchell et al., 2019) for Gemini Robotics-ER 1.5 and Gemini Robotics 1.5 models in Table 2.

Model summary	
Model architecture	Gemini Robotics-ER 1.5 is a Vision-Language-Model that enhances Gemini's world understanding. Gemini Robotics 1.5 is a Vision-Language-Action model enabling general-purpose robot manipulation on different tasks, scenes, and across multiple robots.
Input(s)	The models take text (e.g., a question or prompt or numerical coordinates) and images (e.g., robot camera images) as input.
Output(s)	Gemini Robotics-ER 1.5 generates text (e.g., numerical coordinates) in response to the input. Gemini Robotics 1.5 generates continuous numerical values that represent robot actions, and additionally text when thinking mode is enabled.
Model Data	
Training Data	Gemini Robotics-ER 1.5 and Gemini Robotics 1.5 were trained on datasets comprised of images, text, and robot sensor and action data.
Data Pre-processing	The multi-stage safety and quality filtering process employs data cleaning and filtering methods in line with our policies. These methods include: • Sensitive Data Filtering: Automated techniques were used to filter out certain personal information and other sensitive data from text and images. • Synthetic captions: Each image in the dataset was paired with both original captions and synthetic captions. Synthetic captions were generated using Gemini and FlexCap (Dwibedi et al., 2024) models and allow the model to learn details about the image. Further details on data pre-processing can be found in (Gemini-Team et al., 2023).
Implementation Frameworks	
Hardware	TPU v4, v5p and v6e.
Software	JAX (Bradbury et al., 2018), ML Pathways (Dean, 2021).
Evaluation	
Approach	See Section 4 for Gemini Robotics-ER 1.5 evaluation procedures, Sections 3 for Gemini Robotics 1.5 evaluation procedures, and Section 6 for Gemini Robotics Safety evaluation procedures.

附录

A. 模型卡

我们在表 2 中给出了 Gemini Robotics-ER 1.5 和 Gemini Robotics 1.5 模型的模型卡（Mitchell et al., 2019）。

模型摘要	
模型架构	Gemini Robotics-ER 1.5 是一种视觉语言模型，可增强 Gemini 的世界理解能力。 Gemini Robotics 1.5 是一种视觉-语言-动作模型，能够实现在不同任务、场景和 多种机器人上的通用机器人操作。
输入	模型将文本（例如问题、提示词或数值坐标）和图像（例如机器人相机图像）作为输入。
输出	Gemini Robotics-ER 1.5 生成文本（例如数值坐标）作为对输入的响应。Gemini Robotics 1.5 生成连续的表示机器人动作的数值，以及文本当启用思考模式时。
模型数据	
训练数据	Gemini Robotics-ER 1.5 和 Gemini Robotics 1.5 的训练基于由图像、文本以及机器人传感器和动作数据组成的数据集。
数据预处理	多阶段安全与质量过滤流程采用数据清洗和过滤方法，符合我们的政策。这些方法包括： • 敏感数据过滤：采用自动化技术过滤从其中过滤掉某些个人信息及其他敏感数据文本和图像。 • 合成说明文字：数据集中的每张图像都配有原始说明文字和合成说明文字。合成说明文字是使用 Gemini 和 FlexCap（Dwibedi 等，2024）模型生成的，并使模型能够学习图像的细节。 有关数据预处理的更多细节，请参见（Gemini-Team 等，2023）。
实现框架	
硬件	TPU v4、v5p 和 v6e。
软件	JAX（Bradbury 等，2018）、ML Pathways（Dean，2021）。
评估	
方法	有关 Gemini Robotics-ER 1.5 的评估流程，请参见第 4 节，Sec-第 3 节涉及 Gemini Robotics 1.5 的评估流程，以及第 6 节用于 Gemini Robotics 安全性评估流程。

Results	See Section 4 for Gemini Robotics-ER 1.5 evaluation results, Sections 3 for Gemini Robotics 1.5 evaluation results, and Section 6 for Gemini Robotics Safety evaluation results.
Model Usage & Limitations	
Ethical Considerations & Risks	Previous impact assessment and risk analysis work as discussed in (Gemini-Team et al., 2023) and references therein remain relevant to Gemini Robotics. See Section 6 for information on responsible development and safety mitigations.

Table 2 | Gemini Robotics 1.5 model card.

结果	有关 GeminiRobotics-ER 1.5 的评估结果，请参见第 4 节；有关 Gemini Robotics 1.5 的评估结果，请参见第 3 节；有关 Gemini Robotics Safety 的评估结果，请参见第 6 节。
模型使用与局限性	
伦理考量 & 风险	先前的影响评估和风险分析工作，如所讨论在（Gemini-Team 等，2023）及其参考文献中仍然具有与 Gemini Robotics 的相关性。见第 6 节了解有关负责任开发和安全缓解措施。

表 **2** | Gemini Robotics 1.5 模型卡。

B. Gemini Robotics 1.5 is a general multi-embodiment Vision-Language-Action Model

In this section, we provide additional material to supplement the results for GR 1.5.

B.1. Rank consistency between evaluations in simulation and on real robots

We leverage physics-based simulators to massively scale up the evaluations of our GR 1.5 models for new objects, scenes, and environments. Fig. 21 demonstrates the rank consistency between our MuJoCo simulator-based evaluation and real-robot evaluations across multiple scenes and tasks, which enables us to rapidly iterate on architectural and algorithmic improvements while reducing the need to conduct slow real-robot evaluations.

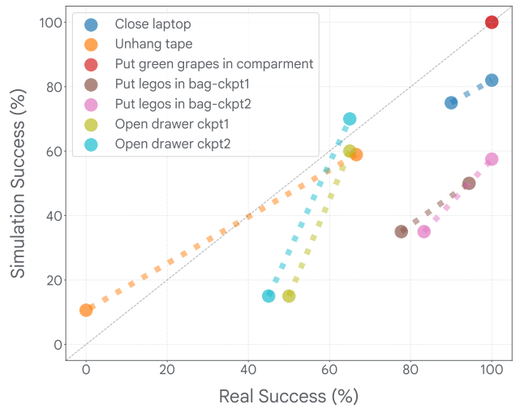


Figure 21 | Each colored pair represents an A/B test both in simulation and real. Across a range of tasks, we find that success rates are rank consistent between simulation and real. This consistency allows for rapid iteration of model architecture, training objectives, and experiment design.

B.2. Generalization benchmark

B.2.1. ALOHA robot

Our generalization benchmark expands the benchmark defined in (Gemini-Robotics-Team et al., 2025) (68 generalization tasks) with a new set of action generalization tasks (5 tasks) and a new category to measure generalization to entirely new tasks (12 tasks). See (Gemini-Robotics-Team et al., 2025) for details about previous Visual, Instruction and Action generalization tasks. To measure performance in-distribution we use the dexterity benchmark defined in Section 3.2 of (Gemini-Robotics-Team et al., 2025) (20 tasks). Progress score definition for in-distribution, visual, semantic and action generalization can also be found in (Gemini-Robotics-Team et al., 2025).

B.2.1.1 New action generalization progress score definition

The action generalization benchmark defined in prior work (Gemini-Robotics-Team et al., 2025) largely focused on measuring the model's ability to handle new object locations and shape variants. We expand the benchmark further to include tasks that require the policy to compose multiple learned motions in novel ways to solve new tasks. For example, having seen data of the robot pushing objects

B. Gemini Robotics 1.5 是一个通用的多具身视觉-语言-动作模型

在本节中，我们提供补充材料，以补充 GR 1.5 的结果 .

B.1. 仿真评估与真实机器人评估之间的排名一致性

我们利用基于物理的仿真器，大规模扩展 GR 1.5 模型针对新物体、新场景和新环境的评估。图 21 展示了在多个场景和任务中，我们基于 MuJoCo 仿真器的评估与真实机器人评估之间的排名一致性；这使我们能够快速迭代架构和算法改进，同时减少进行缓慢的真实机器人评估的需求。

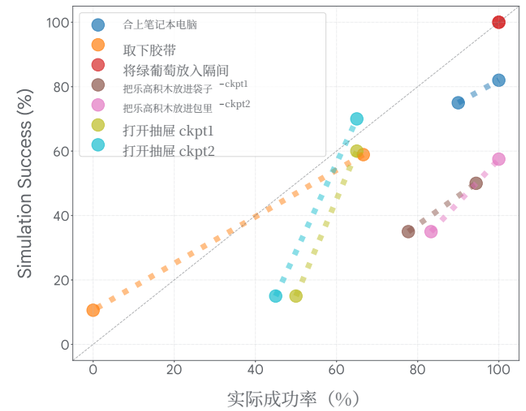


图 **21** | 每个彩色配对表示一次同时在仿真和真实环境中进行的 A/B 测试。在一系列任务中，我们发现仿真与真实环境中的成功率排名一致。这种一致性使得模型架构、训练目标和实验设计的快速迭代成为可能。

B.2. 泛化基准

B.2.1. ALOHA 机器人

我们的泛化基准扩展了 (Gemini-Robotics-Team et al., 2025) 中定义的基准（68 个泛化任务），新增了一组动作泛化任务（5 个任务）以及一个新类别，用于衡量对完全新任务的泛化能力（12 个任务）。关于此前的视觉、指令和动作泛化任务的细节，请参见 (Gemini-Robotics-Team et al., 2025)。为了衡量分布内性能，我们使用 (Gemini-Robotics-Team et al., 2025) 第 3.2 节中定义的灵巧性基准（20 个任务）。分布内、视觉、语义和动作泛化的进展分数定义也可在 (Gemini-Robotics-Team et al., 2025) 中找到。

B.2.1.1 新动作泛化进展分数定义

先前工作（Gemini-Robotics-Team et al., 2025）中定义的动作泛化基准主要侧重于衡量模型处理新物体位置和形状变体的能力。我们进一步扩展该基准，纳入一些任务，这些任务要求策略以新颖的方式组合多个已学习的动作来解决新任务。例如，已经见过机器人推动物体的数据

for a short distance at different locations, the 'drink-pushing' task (Figure 22) measures how well the policy can combine them together to push a novel object all the way from bottom of the table to the top edge. Table 3 lists the definition of progress scores for each task.

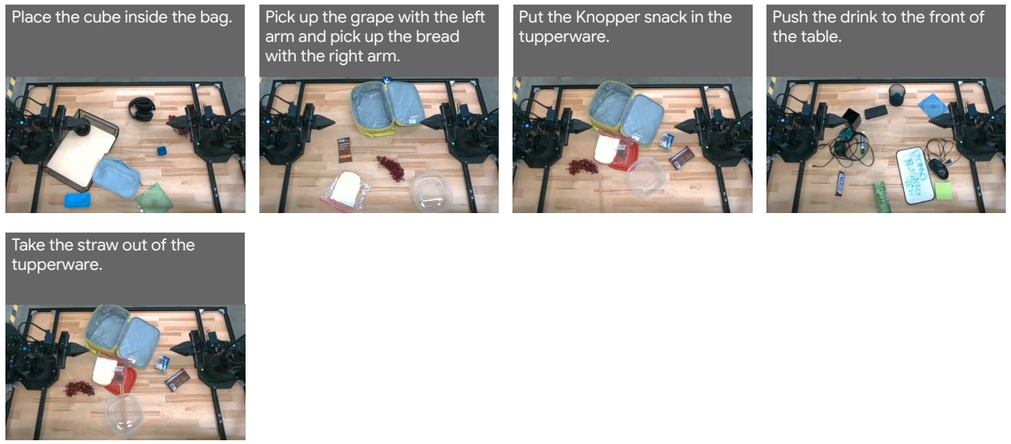


Figure 22 | Examples of the expanded action generalization benchmark.

Table 3 | Progress Scores: New action generalization tasks progress score.

New action generalization tasks.		
"Place the cube inside the bag".	**"Pick up the grape with the left arm and pick up the bread with the right arm"..**	**"Put the Knopper snack in the tupperware".**
1.0: if the successfully placed the cube inside the bag; 0.3: if the robot picked up the cube but failed to put it in the bag; 0.0: if the robot failed to pick the cube.	1.0: if both objects are picked up by the correct arm; 0.3: if one of the arms picked up the correct object, another arm picked up the wrong object; 0.0: if both arms picked up wrong object.	1.0: if the robot successfully put the Knopper snack in the tupperware; 0.4: if the robot picked up the right snack but failed to put in tupperware after 5 attempted; 0.0: if the robot pick up the wrong object.
"Push the drink to the front of the table".	**"Take the straw out of the tupperware".**	
1.0: if the robot successfully pushed the drink to the top part of the table; 0.5: if the robot didn't fully push the drink to the top part of the table, or pushed something else first; 0.0: if the robot didn't approach the correct object at all.	1.0: if the robot successfully picked up the straw and move it outside of the range of the tupperware; 0.3: if the robot picked the straw but failed to move it out, or poured the tupperware; 0.0: if the robot didn't approach the straw or tupperware.	

B.2.1.2 Task generalization progress score definition

We consider 12 different tasks across multiple scenes. The tasks have *unseen instructions, unseen objects and initial conditions* compare to the training data. See Figure 23 for details and Table 4 collects for the definition of progress for each task for the new category of task generalization.

在不同位置进行短距离推动时，“drink-pushing”任务（图22）衡量策略能在多大程度上将这些动作组合在一起，从而将一个新物体从桌子底部一直推到顶部边缘。表3列出了每个任务的进度分数定义。

图 **22** | 扩展动作泛化基准示例。

表 **3** | 进展得分：新动作泛化任务进展得分。

新动作泛化任务。		
“将方块放入袋子中”。	**“用左臂拿起葡萄，并拿起用右臂拿起面包”..**	**“把 Knopper 零食放进保鲜盒”。**
1.0：如果成功将方块放入袋子中； 0.3：如果机器人拿起了方块但未能将其放入袋子中； 0.0：如果机器人未能拿起方块。	1.0：如果两个物体均由正确的手臂拿起； 0.3：如果其中一只机械臂抓取了正确的物体，另一只机械臂抓取了错误的物体； 0.0：如果两只机械臂都抓取了错误的物体。	1.0：如果机器人成功将 Knopper 零食放入保鲜盒； 0.4：如果机器人抓取了正确的零食，但未能尝试 5 次后放入特百惠保鲜盒； 0.0：如果机器人拿错物体。
“把饮料推到桌子前方”。	**“从特百惠保鲜盒中取出吸管”。**	
1.0：如果机器人成功将饮料推到桌子的上部； 0.5：如果机器人没有将饮料完全推到顶部桌子的一部分，或者先推了其他东西； 0.0：如果机器人没有接近正确的物体，在 all.	1.0：如果机器人成功拿起了吸管并且将其移到保鲜盒的范围之外； 0.3：如果机器人拿起了吸管但未能移动它取出，或倒出保鲜盒； 0.0：如果机器人没有靠近吸管或保鲜盒。	

B.2.1.2 任务泛化进度分数定义

我们考虑了跨多个场景的 12 个不同任务。与训练数据相比，这些任务包含未见过的指令、未见过的物体和初始条件。详见 图 23，表 4 汇总了任务泛化这一新类别下每个任务的进度定义。

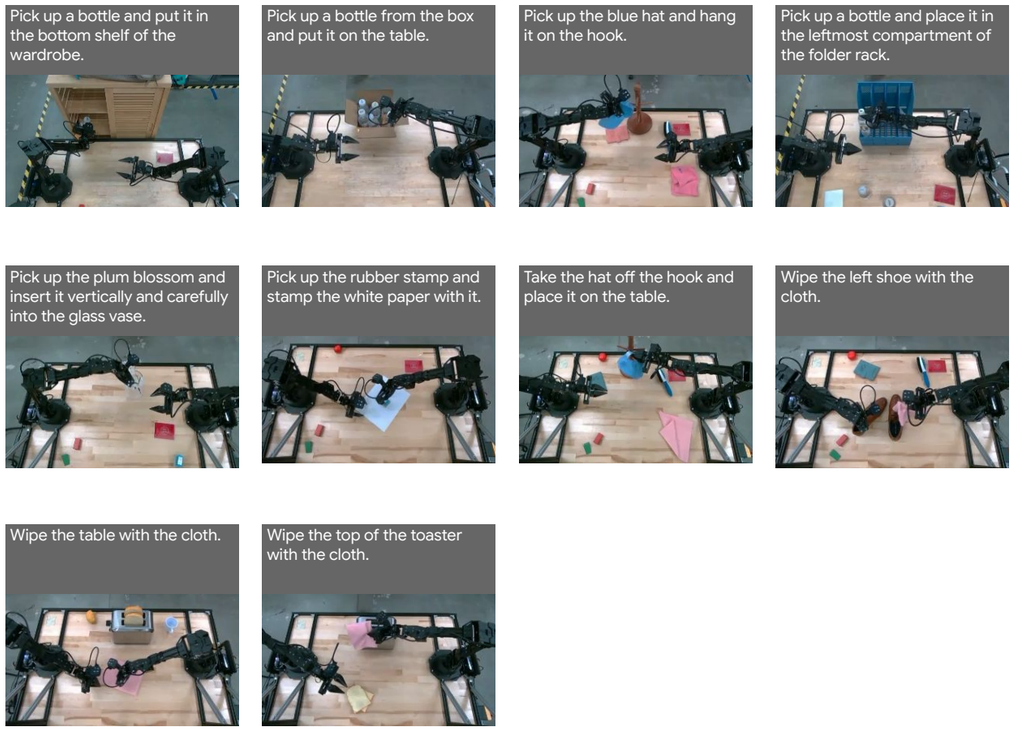


Figure 23 | Examples of task execution for the Task generalization analysis of 3.1.

B.2.2. Bi-arm Franka robot

We defined a new generalization benchmark for our Bi-arm Franka robot. This benchmark includes a total of 44 tasks, 20 of those in distribution and 24 including variants of them across different axes: instruction, visual and action generalization, following the same approach we used to define the ALOHA robot benchmark in (Gemini-Robotics-Team et al., 2025). For the task generalization category we used the same 12 tasks and progress score described in B.2.1.2 for ALOHA. It is worth noting that the the Bi-arm Franka benchmark also includes highly dexterous tasks such as NIST Board 2 assembly tasks and insertion of cables in a workstation and sockets.

The Bi-arm Franka robot benchmark is defined across three different scenes (Figure 24):

图 **23** | 3.1 任务泛化分析的任务执行示例。

B.2.2. 双臂 ***Franka*** 机器人

我们为双臂 Franka 机器人定义了一个新的泛化基准。该基准共包含 44 个任务，其中 20 个属于分布内任务，另有 24 个包含这些任务沿不同维度的变体：指令泛化、视觉泛化和动作泛化；这沿用了我们在（Gemini-Robotics-Team et al., 2025）中定义 ALOHA 机器人基准时采用的方法。对于任务泛化类别，我们使用了 B.2.1.2 中针对 ALOHA 描述的相同 12 个任务和进度分数。值得注意的是，双臂 Franka 基准还包含高度灵巧的任务，例如 NIST Board 2 装配任务，以及在工作站和插座中插入线缆。

The 双臂 Franka 机器人基准测试定义于三个不同场景中（图 24 ）:

Table 4 | Progress Scores: Task generalization tasks.

Task generalization tasks progress score.		
"Pick up a bottle and put it in the bottom shelf of the wardrobe".	**"Pick up a bottle from the box and put it on the table".**	**"Pick up the blue hat and hang it on the hook".**
1.0: if the robot placed the bottle in the wardrobe; 0.75: if the robot moved the bottle towards the wardrobe; 0.5: if the robot grasped a bottle; 0.0: if anything else happened.	1.0: if the robot placed a bottle on the table; 0.25: if the robot reached for a bottle but did not grasp it; 0.0: if anything else happened.	1.0: if the robot successfully hung the hat on the hook; 0.75: if the robot tried to hang the hat on the hook; 0.5: if the robot grasped the hat; 0.0: if anything else happened.
"Pick up a bottle and place it in the leftmost compartment of the folder rack".	**"Pick up the plum blossom and insert it vertically and carefully into the glass vase".**	**"Pick up the rubber stamp and stamp the white paper with it".**
1.0: if the robot placed the bottle in the leftmost compartment of the folder rack; 0.75: if the robot placed the bottle in any compartment of the folder rack; 0.5: if the robot moved the bottle towards the folder rack; 0.25: if the robot grasped a bottle; 0.0: if anything else happened.	1.0: if the robot inserted the plum blossom into the vase; 0.75: if the robot moved the plum blossom towards the vase; 0.50: if the robot grasped the plum blossom; 0.25: if the robot moved towards the plum blossom; 0.0: if anything else happened.	1.0: if the robot stamped the document; 0.75: if the robot moved the rubber stamp over the document; 0.25: if the robot grasped the rubber stamp; 0.0: if anything else happened.
"Take the hat off the hook and place it on the table"	**"Wipe the left shoe with the cloth"**	**"Wipe the table with the cloth"**
1.0: if the robot placed the hat on the table; 0.75: if the robot removed the hat from the hook; 0.5: if the robot grasped the hat; 0.0: if anything else happened.	1.0: if the robot wiped the left shoe; 0.5: if the robot moved the cloth towards the left shoe; 0.25: if the robot grasped the cloth; 0.0: if anything else happened.	1.0: if the robot moved towards the cloth back and forth on some portion of the table; 0.5: if the robot grasped the cloth; 0.25: if the robot reached the cloth; 0.0: if anything else happened.
"Wipe the top of the toaster with the cloth"		
1.0: if the robot wiped the top of the toaster; 0.75: if the robot moved towards the cloth towards the top of the toaster; 0.25: if the robot grasped the cloth; 0.0: if anything else happened.		

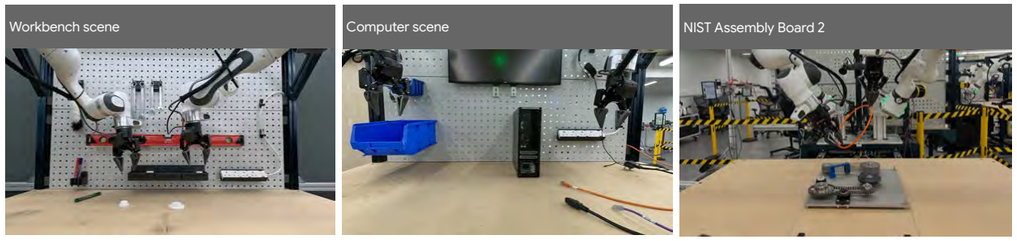


Figure 24 | Scenes used to define the generalization benchmark for Bi-Arm Franka robot. Left: A workbench inspired scene that allows for manipulation of tools and skills ranging from hanging, unhanging, picking and placing. Center: a scene with a computer and assorted cables and peripherals, allowing for cable handling, insertions and removals. Left: Layout of the National Institute of Science and Technology (NIST) Task Board 2 (NIST, 2025).

- **Workbench:** this scene spans both table-top manipulation tasks and interaction with the vertical back panel, which allows for manipulation of tools and skills ranging from hanging, unhanging, picking and placing.
- **Computer scene:** this scene requires interacting with real-world computer, cables and peripherals that involves dexterous and precise tasks such as cable handling, insertions and removals.
- **NIST Assembly Board 2:** based on the Task Assembly Board 2 as defined by NIST, this scene presents complex and precise assembly and removal of the three belts present.

Figure 25 shows examples of task variations to measure instruction, visual and semantic general-

表 **4** | 进度得分：任务泛化任务。

任务泛化任务的进度得分。		
"拿起一个瓶子，把它放到最底层架子上，该架子属于衣柜"。	**"从盒子里拿起一个瓶子，把它放到桌子上"。**	**"拿起那顶蓝帽子，把它挂到钩子上"。**
1.0：如果机器人将瓶子放进了衣柜; 0.75：如果机器人将瓶子移向衣柜; 0.5：如果机器人抓住了瓶子; 0.0：如果发生了其他任何情况。	1.0：如果机器人将瓶子放在桌子上; 0.25：如果机器人伸手去够瓶子但未抓住it; 0.0：如果发生了其他任何情况。	1.0：如果机器人成功将帽子挂在钩子上; 0.75：如果机器人试图把帽子挂在挂钩上; 0.5：如果机器人抓住了帽子; 0.0：如果发生了其他情况。
"拿起一个瓶子，把它放在最左侧的隔间（文件夹架）"。	**"拿起梅花，将其垂直插入并小心地放入玻璃花瓶"。**	**"拿起橡皮图章，在白纸上盖章用它"。**
1.0：如果机器人将瓶子放入最左侧的隔层，属于文件架; 0.75：如果机器人将瓶子放入任意隔层文件夹架的; 0.5：如果机器人将瓶子移向文件夹架; 0.25：如果机器人抓住了瓶子; 0.0：如果发生了其他任何情况。	1.0：如果机器人将梅花插入花瓶; 0.75：如果机器人将梅花移向花瓶; 0.50：如果机器人抓取了梅花; 0.25：如果机器人朝梅花移动; 0.0：如果发生了其他任何情况。	1.0：如果机器人给文件盖了章; 0.75：如果机器人将橡皮图章移到文件上; 0.25：如果机器人抓住了橡皮图章; 0.0：如果发生了其他任何情况。
"把帽子从挂钩上取下来，放到桌子上"	**"用布擦拭左边的鞋子"**	**"用布擦拭桌子"**
1.0：如果机器人将帽子放在桌子上; 0.75：如果机器人将帽子从挂钩上取下; 0.5：如果机器人抓住了帽子; 0.0：如果发生了其他情况。	1.0：如果机器人擦拭了左鞋; 0.5：如果机器人将抹布移向左鞋; 0.25：如果机器人抓住了抹布; 0.0：如果发生了其他任何情况。	1.0：如果机器人朝抹布来回移动在桌子的某一部分上; 0.5：如果机器人抓住了布料; 0.25：如果机器人够到了布料; 0.0：如果发生了其他任何情况。
"用布擦拭烤面包机的顶部"		
1.0：如果机器人擦拭了烤面包机的顶部; 0.75：如果机器人朝布移动并朝向烤面包机的顶部; 0.25：如果机器人抓住了布; 0.0：如果发生了其他任何情况。		

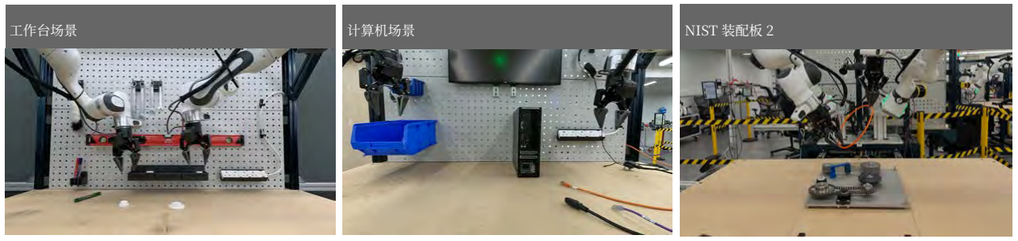


图 **24** | 用于定义双臂 Franka 机器人泛化基准的场景。左：一个受工作台启发的场景，可执行从悬挂、取下到抓取和放置等工具与技能的操作。中：一个包含计算机以及各类线缆和外设的场景，可进行线缆处理、插入和拔出操作。左：美国国家标准与技术研究院（NIST）任务板 2 的布局（NIST，2025）。

- 工作台：该场景既涵盖桌面操作任务，也包括与垂直背板的交互，从而支持对工具和技能的操作，范围包括悬挂、取下、拾取和放置。
- 计算机场景：该场景需要与真实世界的计算机、线缆及外设进行交互，涉及线缆处理、插入和拔出等灵巧且精确的任务。
- NIST 装配板 2：基于 NIST 定义的任务装配板 2，该场景呈现对现有三条皮带的复杂且精确的装配与拆卸。

图 25 展示了任务变化示例，用于衡量指令、视觉和语义泛-

ization on the Bi-arm Franka robot.

Computer scene

In-distribution scene	Action generalization	Scene generalization	Instruction generalization
			Conecta el enchufe a la computadora
Connect the power cable into the computer			

Workbench scene

In-distribution scene	Action generalization	Scene generalization	Instruction generalization
			Put the object used for cleaning on the table
Put the brush on the table			

Figure 25 | Example variations of scenes used for measuring performance across generalization axes on the Bi-arm Franka platform.

B.2.2.1 Task progress score for in-distribution, visual and action generalization tasks

We now define the task progress score for each task in the generalization benchmark for Bi-arm Franka robot.

Workbench scene

化在双臂 Franka 机器人上。

计算机场景

分布内场景	动作泛化	场景泛化	指令泛化
			将插头连接到计算机
将电源线接入计算机			

工作台场景

分布内场景	动作泛化	场景泛化	指令泛化
			放置用于的物品在桌上清洁
把刷子放在桌子上			

图 **25** | 用于在 Bi-armFranka 平台上衡量跨泛化维度性能的场景示例变体。

B.2.2.1 分布内、视觉与动作泛化任务的任务进度分数

我们现在为 Bi-arm Franka 机器人泛化基准中的每个任务定义任务进度分数。

工作台场景

Table 5 | Progress Scores: Bi-arm Franka (Workbench scene).

Benchmark: Bi-arm Franka - Workbench scene.		
"Hang the tape".	**"Unhang the tape".**	**"Unhang the level".**
1.0: if the robot hung the tape successfully; 0.7: if the robot grasped the tape and tried to hang it but failed; 0.3: if the robot grasped the tape but didn't move towards the hook; 0.1: if the robot reached for the tape but failed to grasp it; 0.0: if the robot didn't reach for the tape.	1.0: if the tape got unhooked and ends up anywhere on the table; 0.7: if the robot grasped the tape but failed to unhook it; 0.3: if the robot reached for the tape on the back-panel but failed to grasp it; 0.0: if anything else happened.	1.0: if the level got unhooked and ends up anywhere on the table; 0.8: if the robot unhooked the level only from one hook; 0.7: if the robot grasped the level but failed to unhook it; 0.3: if the robot reached for the level on the back-panel but failed to grasp it; 0.0: if anything else happened.
"Remove the red pen".	**"Place the rightmost wrench on the table".**	**"Remove a gear from the rightmost container".**
1.0: if the red pen ended up anywhere on the table; 0.7: if the robot grasped the red pen but failed to take it off; 0.3: if the robot reached for the red pen on the back-panel but failed to grasp it; 0.0: if anything else happened.	1.0: if the rightmost wrench (with respect to the robot) got unhooked and ended up anywhere on the table; 0.7: if the robot grasped the rightmost wrench (with respect to the robot) but failed to unhook it; 0.3: if the robot reached for the rightmost wrench (with respect to the robot) on the back-panel but failed to grasp it; 0.0: if anything else happened.	1.0: if at least a gear from the rightmost container (with respect to the robot) ended up anywhere on the table; 0.7: if the robot grasped a gear from the rightmost container (with respect to the robot) but failed to remove it from the container; 0.3: if the robot reached for a gear from the rightmost container (with respect to the robot) but failed to grasp it; 0.0: if anything else happened.
"Place the brush on the table".	**"Put a gear in the rightmost container".**	
1.0: if the brush got unhooked and ended up anywhere on the table; 0.7: if the robot grasped the brush but failed to unhook it; 0.3: if the robot reached for the brush on the back-panel but failed to grasp it; 0.0: if anything else happened.	1.0: if a gear had been moved to the rightmost container (with respect to the robot); 0.7: if the robot grasped a gear and attempted to put it in the rightmost container (with respect to the robot) but failed to do to so; 0.3: if the robot reached for a gear from the table but failed to grasp it; 0.0: if anything else happened.	

Computer scene

表 **5** | 进度得分：双臂 Franka（工作台场景）。

基准：双臂 Franka - 工作台场景。		
"挂上胶带"。	**"取下胶带"。**	**"将水平仪取下"。**
1.0：如果机器人成功挂上了卷尺； 0.7：如果机器人抓住了卷尺并尝试挂上它但失败了； 0.3：如果机器人抓住了卷尺但没有朝钩子方向移动； 0.1：如果机器人伸手去够胶带，但未能抓住它； 0.0：如果机器人没有伸手去够胶带。	1.0：如果胶带被从挂钩上取下，并最终落在任意位置在桌子上； 0.7：如果机器人抓住了胶带，但未能将其从挂钩上取下它； 0.3：如果机器人伸手去够背板上的胶带但未能抓取它； 0.0：如果发生了其他任何情况。	1.0：如果水平仪脱钩并最终落在任意位置在桌上； 0.8：如果机器人仅将水平仪从一个挂钩上取下； 0.7：如果机器人抓取了水平仪但未能将其从挂钩上取下它； 0.3：如果机器人伸手去够背板上的水平仪但未能抓住它； 0.0：如果发生了其他任何情况。
"移除红色钢笔"。	**"将最右侧的扳手放到桌上"。**	**"从最右侧的容器中取出一个齿轮"。**
1.0：如果红笔最终落在桌子上的任意位置； 0.7：如果机器人抓住了红笔，但未能拿它下来； 0.3：如果机器人伸手去拿背面的红笔面板，但未能抓住它； 0.0：如果发生了其他任何情况。	1.0：如果最右侧的扳手（相对于机器人）从挂钩上脱落并最终落在桌面上的任意位置； 0.7：如果机器人抓住了最右侧的扳手（相对于机器人而言）但未能将其从挂钩上取下； 0.3：如果机器人伸手去够最右侧的扳手（相对于机器人而言）在背板上但未能抓取它； 0.0：如果发生了其他任何情况。	1.0：如果至少有一个来自最右侧容器的齿轮（以机器人为参照）最终落在桌子上的任何位置； 0.7：如果机器人抓取了齿轮，该齿轮来自最右侧容器（相对于机器人）但未能取出将其从容器中取出； 0.3：如果机器人伸手去拿最右侧的容器（相对于机器人）但未能抓取它； 0.0：如果发生了任何其他情况。
"把刷子放在桌子上"。	**"将一枚齿轮放入最右侧的容器中"。**	
1.0：如果刷子脱钩并最终落在任意位置在桌子上； 0.7：如果机器人抓住了刷子，但未能将其从挂钩上解下它； 0.3：如果机器人伸手去够背板上的刷子但未能抓住它； 0.0：如果发生任何其他情况。	1.0：如果齿轮已被移动到最右侧的容器中（相对于机器人）； 0.7：如果机器人抓住了齿轮并试图将其放入最右侧的容器中（相对于机器人）但未能做到； 0.3：如果机器人从桌上伸手去拿齿轮，但未能抓住它； 0.0：如果发生了其他任何情况。	

计算机场景

Table 6 | Progress Scores: Bi-arm Franka (Computer scene).

Benchmark: Bi-arm Franka - Computer scene.		
"Connect the power plug to the computer".	**"Insert the white power plug into the socket".**	**"Insert the orange LAN cable in the computer".**
1.0: if the robot fully inserted the power plug into the right place in the computer; 0.8: if the robot partially inserted the power plug into the right place in the computer; 0.5: if the robot grasped the power plug and tried to insert it in the right place in the computer but failed; 0.3: if the robot managed to grasp the power plug but didn't try to insert it into the computer. 0.1: if the robot reached for the power plug but failed to grasp it. 0.0: if the robot didn't reach for the power plug.	1.0: if the robot grasped the plug and successfully inserted it in the socket; 0.9: if the robot grasped the plug and partially inserted it in the socket; 0.6: if the robot grasped the plug, tried to insert it in the socket but failed; 0.3: if the robot grasped the plug but didn't try to insert it in the socket; 0.1: if the robot reached for the plug but didn't grasp it; 0.0: if the robot didn't reach for the plug.	1.0: if the robot grasped the orange internet cable and successfully inserted it in the right socket of the computer; 0.9: if the robot grasped the orange internet cable and partially inserted it in the socket of the computer; 0.6: if the robot grasped the orange internet cable, tried to insert it in the right socket of the computer but failed; 0.3: if the robot grasped the orange internet cable but didn't try to insert it in the right socket of the computer; 0.1: if the robot reached for the orange internet cable but didn't grasp it; 0.0: if the robot didn't reach for the orange internet cable.
"Hang the headphones on the wall".	**"Put the headphones on the desk".**	**"Remove the orange LAN cable from the computer".**
1.0: if the robot hung the headphones; 0.7: if the robot grasped the headphones and tried to hang them but failed; 0.3: if the robot grasped the headphones but didn't move towards the hall; 0.1: if the robot reached for the headphones but failed to grasp it; 0.0: if the robot didn't reach for the headphones.	1.0: if the robot put the headphones on the desk; 0.8: if the robot managed to remove the headphones from the hook; 0.1: if the robot reached the headphones but failed to grasp them; 0.0: if the robot didn't reach for the headphones.	1.0: if the robot managed to remove the orange internet cable from the computer; 0.6: if the robot grasped the orange internet cable, tried to remove it from the computer but it failed; 0.3: if the robot grasped the orange internet cable but didn't try to remove it from the computer; 0.1: if the robot reached for the orange internet cable but didn't grasp it; 0.0: if the robot didn't reach for the orange internet cable.
"Remove the power plug cable from the computer"		
1.0: if the robot managed to remove the power plug cable from the computer; 0.6: if the robot grasped the power plug cable, tried to remove it from the computer but it failed; 0.3: if the robot grasped the power plug cable but didn't try to remove it from the computer; 0.1: if the robot reached for the power plug cable but didn't grasp it; 0.0: if the robot didn't reach for the power plug.		

NIST Assembly Task Board 2

We use only in-distribution variations of the NIST Assembly tasks.

B.2.2.2 Task progress score for semantic generalization tasks

Workbench scene

表 **6**｜进展得分：双臂 Franka（计算机场景）。

基准测试：双臂 Franka - 计算机场景。		
"将电源插头连接到计算机"。	**"将白色电源插头插入插座"。**	**"将橙色网线插入电脑"。**
1.0：如果机器人将电源插头完全插入电脑上的正确位置； 0.8：如果机器人将电源插头部分插入电脑上的正确位置； 0.5：如果机器人抓住电源插头并试图将其插入电脑中的正确位置，但失败了； 0.3：如果机器人成功抓住了电源插头，但没有尝试将其插入电脑。 0.1：如果机器人伸手去拿电源插头，但未能抓住它。 0.0：如果机器人没有伸手去拿电源插头。	1.0：如果机器人抓住插头并成功插入插座中； 0.9：如果机器人抓住插头并部分插入到插座中； 0.6：如果机器人抓住插头，试图将其插入插座但失败了； 0.3：如果机器人抓住了插头，但没有尝试将它插入插座中； 0.1：如果机器人伸手去拿插头，但没有抓住它； 0.0：如果机器人没有伸手去拿插头。	1.0：如果机器人抓住了橙色网线，并且成功将其插入电脑上正确的插孔中电脑； 0.9：如果机器人抓住了橙色网线并将其部分插入电脑的插口中； 0.6：如果机器人抓住了橙色网线，试图将其插入电脑上正确的插口，但失败了； 0.3：如果机器人抓住了橙色网线但没有尝试将其插入计算机的正确插槽中； 0.1：如果机器人伸手去拿橙色网线但没有抓住它； 0.0：如果机器人没有伸手去拿橙色网线。
"把耳机挂在墙上"。	**"把耳机放在桌子上"。**	**"从电脑上拔下橙色 LAN 线"。**
1.0：如果机器人挂上了耳机； 0.7：如果机器人抓住了耳机并试图将其挂起来但失败了； 0.3：如果机器人抓住了耳机但没有向大厅移动； 0.1：如果机器人伸手去拿耳机但未能抓住它； 0.0：如果机器人没有伸手去拿耳机。	1.0：如果机器人把耳机放在桌上； 0.8：如果机器人成功取下耳机从挂钩上； 0.1：如果机器人够到了耳机，但未能抓住它们； 0.0：如果机器人没有去够耳机。	1.0：如果机器人成功移除了橙色互联网从计算机上引出的线缆； 0.6：如果机器人抓住了橙色网线，试图将其从电脑上拔下，但失败了； 0.3：如果机器人抓住了橙色网线，但没有试图将其从电脑上拔下； 0.1：如果机器人伸手去拿橙色网线但没有抓住它； 0.0：如果机器人没有伸手去拿橙色网线。
"从电脑上拔下电源插头线缆"		
1.0：如果机器人成功拔下电源插头线缆从电脑上； 0.6：如果机器人抓住了电源插头线缆，并试图把它从电脑上拔下来，但失败了； 0.3：如果机器人抓住了电源插头线缆但没有尝试将其从电脑上拔下来； 0.1：如果机器人伸手去够电源插头线缆，但没有抓住它； 0.0：如果机器人没有伸手去够电源插头。		

NIST 装配任务板 2

我们仅使用 NIST 装配任务的分布内变体。

B.2.2.2 语义泛化任务的任务进度分数

工作台场景

Table 7 | Progress Scores: Bi-arm Franka (NIST Assembly Task Board 2).

Benchmark: Bi-arm Franka - NIST Assembly Task Board 2.		
"Put the timing belt on the timing pulleys".	**"Remove the timing belt from the timing pulleys".**	**"Place the orange round belt on the slide tensioners".**
1.0: If the robot assembled the timing belt on the timing pulleys and let it go; 0.9: if the robot inserted on both wheels correctly; 0.7: if the robot pushed the blue tensioner, with the belt loosely on the second wheel; 0.5: if the robot inserted on one wheel; 0.3: if the robot handed the belt over to grasp the timing belt with both arms; 0.1: if the robot grasped and lifted the timing belt; 0.0: if the robot couldn't even pick up the timing belt.	1.0: if the robot let go of the timing belt; 0.9: if the robot put the timing belt on the ground; 0.8: if the robot fully unhooked the timing belt from the two pulleys and the tensioner; 0.6: if the robot unhooked the timing belt from a pulley and the tensioner; 0.3: if the robot unhooked the timing belt from one of the pulleys; 0.1: if the robot grasped the timing belt; 0.0: if the robot did not do anything.	1.0: if the robot let go of the orange belt; 0.9: if the robot inserted it on both wheels correctly; 0.5: if the robot handed the belt over to grasp the orange belt with both arms; 0.1: if the robot grasped and lifted the orange belt with the left arm; 0.0: if the robot couldn't even pick up the orange belt.
"Remove the orange round belt from the slide tensioners".	**"Put the chain belt on the sprocket idlers".**	**"Remove the chain from the sprockets".**
1.0: if the robot let go of the orange belt; 0.9: if the robot put the orange belt on the table top; 0.5: if the robot unhooked the orange belt from both tensioners; 0.1: if the robot grasped and unhooked the orange belt from one of the tensioners; 0.0: if the robot couldn't even pick up the orange belt.	1.0: if the robot let go of the chain; 0.9: if the robot inserted it on both sprockets and aligned it with the tensioner; 0.7: if the robot inserted it on two sprockets; 0.5: if the robot inserted it on one sprocket; 0.3: if the robot handed the chain over to grasp it with both arms; 0.1: if the robot grasped and lifted the metal chain; 0.0: if the robot couldn't even pick up the metal chain.	1.0: if the robot let go of the chain; 0.9: if the robot put the chain on the ground; 0.8: if the robot fully unhooked the chain from both sprockets and the chain tensioner; 0.6: if the robot unhooked the chain from two sprockets or one sprocket and the chain tensioner; 0.3: if the robot unhooked the chain from one sprocket and/or the chain tensioner; 0.1: if the robot grasped the chain; 0.0: if the robot did not do anything.

Table 8 | Progress Scores: Bi-arm Franka (Semantic Gen. - Workbench).

Benchmark: Bi-arm Franka - Semantic Generalization (Workbench).		
"quitar la cinta".	**"unhang the long red tool".**	**"Put the object used for cleaning on the table".**
1.0: if the tape got unhooked and ended up anywhere on the table; 0.7: if the robot grasped the tape but failed to unhook it; 0.3: if the robot reached for the tape on the back-panel but failed to grasp it; 0.0: if anything else happened.	1.0: if the level got unhooked and ended up anywhere on the table; 0.8: if the robot unhooked the level only from one hook; 0.7: if the robot grasped the level but failed to unhook it; 0.3: if the robot reached for the level on the back-panel but failed to grasp it; 0.0: if anything else happened.	1.0: if the brush got unhooked and ended up anywhere on the table; 0.7: if the robot grasped the brush but failed to unhook it; 0.3: if the robot reached for the brush on the back-panel but failed to grasp it; 0.0: if anything else happened.

Computer scene

表 **7** | 进度分数：双臂 Franka（NIST 装配任务板 2）。

基准测试：双臂 Franka - NIST 装配任务板 2。		
"将同步带安装到同步带轮上"。	**"从同步带轮上拆下同步带"。**	**"将橙色圆带放到滑动张紧器上"。**
1.0：如果机器人将正时皮带安装到正时带轮上并松开； 0.9：如果机器人正确插入到两个轮子上； 0.7：如果机器人推动蓝色张紧器，且皮带松散地套在第二个轮子上； 0.5：如果机器人将其套在一个轮子上； 0.3：如果机器人将皮带递过去以抓住正时皮带，双臂并用； 0.1：如果机器人抓住并提起正时皮带； 0.0：如果机器人甚至无法拿起正时皮带。	1.0：如果机器人松开正时皮带； 0.9：如果机器人将同步带放到地面上； 0.8：如果机器人完全解开同步带，使其脱离两个带轮和张紧器； 0.6：如果机器人将同步带从一个带轮上解下以及张紧器； 0.3：如果机器人将同步带从其中一个上解下带轮； 0.1：如果机器人抓住了同步带； 0.0：如果机器人未执行任何操作。	1.0：如果机器人松开了橙色皮带； 0.9：如果机器人将其正确安装到两个轮子上； 0.5：如果机器人递过皮带以抓取橙色用双臂提起皮带； 0.1：如果机器人抓住并提起了橙色皮带，所用部位为左臂； 0.0：如果机器人甚至无法拿起橙色皮带。
"取下橙色圆带，从滑动张紧器"。	**"将链带安装到链轮惰轮上"。**	**"从链轮上拆下链条"。**
1.0：如果机器人松开橙色皮带； 0.9：如果机器人将橙色皮带放在桌面上； 0.5：如果机器人将橙色皮带从两侧解开张紧器； 0.1：如果机器人抓住并解开了橙色皮带从其中一个张紧器上； 0.0：如果机器人甚至无法拿起橙色皮带。	1.0：如果机器人松开了链条； 0.9：如果机器人将其安装到两个链轮上并且将其与张紧器对齐； 0.7：如果机器人将其装到两个链轮上； 0.5：如果机器人将其装到一个链轮上； 0.3：如果机器人将链条递出，以便抓住它并用双臂； 0.1：如果机器人抓住并提起了金属链条； 0.0：如果机器人甚至无法拿起金属链条。	1.0：如果机器人松开链条； 0.9：如果机器人将链条放到地上； 0.8：如果机器人将链条从两者上完全脱开链轮和链条张紧器； 0.6：如果机器人将链条从两个链轮上脱开或一个链轮和链条张紧器； 0.3：如果机器人将链条从一个链轮上脱开和/或链条张紧器； 0.1：如果机器人抓住了链条； 0.0：如果机器人没有做任何操作。

表 **8** | 进度得分：双臂 Franka（语义泛化 - 工作台）。

基准测试：双臂 Franka - 语义泛化（工作台）。		
"取下胶带"。	**"取下那个长红色工具"。**	**"将用于清洁的物体放在桌上"。**
1.0：如果胶带脱钩并最终出现在任何位置在桌子上； 0.7：如果机器人抓住了胶带但未能将其脱钩it; 0.3：如果机器人伸手去够背板上的胶带但未能抓住它； 0.0：如果发生了其他任何情况。	1.0：如果水平仪脱钩并最终落在任何位置在桌子上； 0.8：如果机器人仅从一个挂钩上取下了水平仪； 0.7：如果机器人抓住了水平仪，但未能将其从挂钩上取下it; 0.3：如果机器人伸手去够背板上的水平仪但未能抓住它； 0.0：如果发生了其他任何情况。	1.0：如果刷子被从挂钩上取下，并最终落在任意位置在桌子上； 0.7：如果机器人抓住了刷子，但未能将其从挂钩上取下it; 0.3：如果机器人伸手去够背板上的刷子但未能抓住它； 0.0：如果发生了其他任何情况。

计算机场景

Table 9 | Progress Scores: Bi-arm Franka (Semantic Gen. - Computer Scene).

Benchmark: Bi-arm Franka - Semantic Generalization (Computer Scene).		
"Desconecta el cable de red naranja de la computadora".	**"conecta el enchufe a la computadora".**	**"Hong the eadphones to te wal".**
1.0: the robot managed to remove the power plug cable from the computer; 0.6: the robot grasped the power plug cable, tried to remove it from the computer but it failed; 0.3: the robot grasped the power plug cable but didn't try to remove it from the computer; 0.1: the robot reached for the power plug cable but didn't grasp it; 0.0: the robot didn't reach for the power plug.	1.0: The robot fully inserted the power plug into the right place in the computer; 0.8: The robot partially inserted the power plug into the right place in the computer; 0.5: The robot grasped the power plug and tried to insert it in the right place in the computer but failed; 0.3: The robot managed to grasp the power plug but didn't try to insert it into the computer. 0.1: The robot reached for the power plug but failed to grasp it. 0.0: The robot didn't reach for the power plug.	1.0: The robot hung the headphones; 0.7: The robot grasped the headphones and tried to hang them but failed; 0.3: The robot grasped the headphones but didn't move towards the hall; 0.1: The robot reached for the headphones but failed to grasp it; 0.0: The robot didn't reach for the headphones.

B.2.3. Apollo humanoid robot

We defined a new generalization benchmark for our Apollo humanoid robot. This benchmark includes a total of 24 tasks, of those in distribution and including variants of them across different axes: instruction, visual and action generalization, following the same approach we used to define the ALOHA robot benchmark in (Gemini-Robotics-Team et al., 2025). For the task generalization category we used the same tasks described in B.2.1.2 for ALOHA.

Figure 26 shows examples of task variations to measure instruction, visual and semantic generalization on the Apollo humanoid robot.

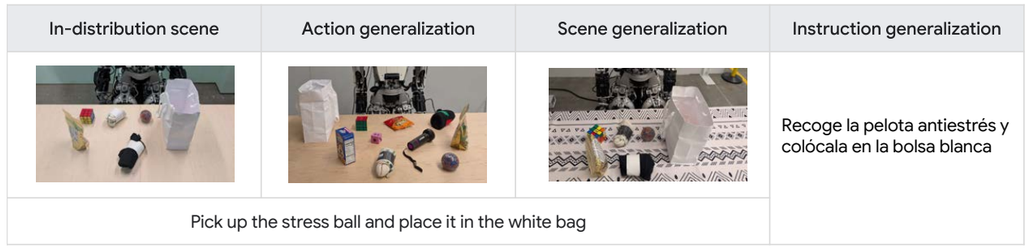


Figure 26 | Example variations of scenes used for measuring performance across generalization axes on the Apollo humanoid platform.

B.2.3.1 Task progress score for in-distribution, visual, semantic and action generalization tasks

表 **9** | 进度得分：双臂 Franka（语义生成 - 计算机场景）。

基准测试：双臂 Franka - 语义泛化（计算机场景）。		
"断开橙色网线，从计算机"。	**"将插头连接到计算机上"。**	**"把耳机挂到墙上"。**
1.0：机器人成功移除了电源插头线缆 从计算机上； 0.6：机器人抓住了电源插头线缆，试图 将其从计算机上拔下，但失败了； 0.3：机器人抓住了电源插头线缆，但没有 试图将其从计算机上拔下； 0.1：机器人伸手去拿电源插头线缆，但 没有抓住它； 0.0：机器人没有伸手去拿电源插头。	1.0：机器人将电源插头完全插入 计算机中的正确位置； 0.8：机器人将电源插头部分插入 计算机中的正确位置； 0.5：机器人抓住了电源插头，并试图 将其插入电脑上的正确位置，但失败了； 0.3：机器人成功抓住了电源插头，但 没有尝试将其插入电脑。 0.1：机器人伸手去拿电源插头，但未能 抓住它。 0.0：机器人没有伸手去够电源插头。	1.0：机器人挂上了耳机； 0.7：机器人抓住了耳机，并试图 将它们挂起来，但失败了； 0.3：机器人抓住了耳机，但没有移动 朝大厅方向； 0.1：机器人伸手去拿耳机，但失败了 抓住它； 0.0：机器人没有伸手去拿耳机。

B.2.3. Apollo 人形机器人

我们为 Apollo 人形机器人定义了一个新的泛化基准。该基准总共包含 24 项任务，其中包括分布内任务，并包含这些任务在不同维度上的变体：指令泛化、视觉泛化和动作泛化，沿用我们在 (Gemini-Robotics-Team et al., 2025) 中定义 ALOHA 机器人基准时采用的相同方法。对于任务泛化类别，我们使用了 B.2.1.2 中为 ALOHA 描述的相同任务。

图 26 展示了用于衡量 Apollo 人形机器人上指令泛化、视觉泛化和语义泛化的任务变体示例。

图 **26** | 用于测量 Apollo 人形机器人平台上跨泛化轴性能的场景示例变体。

B.2.3.1 Task 分布内、视觉、语义和动作泛化任务的进展分数

Table 10 | Progress Scores: Apollo humanoid (In-distribution and Visual Gen.).

Benchmark: Apollo humanoid - In-distribution and Visual Generalization.		
"Pick up the gummy bear bag and place it in the white bag".	**"Pick up the rubik's cube and place it in the white bag".**	**"Pick up the egg from the bottom right slot of the yellow egg box with your right hand and place the egg in the tray".**
1.00: if the robot picked the correct object and placed it in the bag; 0.50: if the robot picked the wrong object but correctly placed it in the bag; 0.50: if the robot picked the correct object but didn't place it in the bag; 0.00: if the robot did anything else.	1.00: if the robot picked the correct object and placed it in the bag; 0.50: if the robot picked the wrong object but correctly placed it in the bag; 0.50: if the robot picked the correct object but didn't place it in the bag; 0.00: if the robot did anything else.	1.00: if the robot picked up the egg with its right hand and placed it in the tray; 0.67: if the robot picked up the egg with its right hand; 0.33: if the robot touched the egg with its right hand; 0.00: if the robot did anything else.
"Pick up the egg from the bottom right slot of the yellow egg box with your left hand and place the egg in the tray".	**"Pick up the bottom green lettuce from the table with your right hand".**	**"Pick up the right orange traffic cone from the table with your right hand".**
1.00: if the robot picked up the egg with its left hand and placed it in the tray; 0.67: if the robot picked up the egg with its left hand; 0.33: if the robot touched the egg with its left hand; 0.00: if the robot did anything else.	1.00: if the robot picked up the bottom green lettuce with its right hand; 0.75: if the robot touched the bottom green lettuce with its right hand; 0.50: if the robot picked up lettuce with its right hand, but not the bottom green lettuce; 0.25: if the robot touched lettuce with its right hand, but not the bottom green lettuce; 0.00: if the robot did anything else.	1.00: if the robot picked up the right orange traffic cone with its right hand; 0.75: if the robot touched the right orange traffic cone with its right hand; 0.50: if the robot picked up a traffic cone, but not the right orange traffic cone; 0.25: if the robot touched a traffic cone, but not the right orange traffic cone; 0.00: if the robot did anything else.
"Pick up the blue vehicle toy with your right hand and place it in the white bowl"	**"Pick up the light pink color soft toy with your left hand and place it in the brown bowl"**	
1.00: if the robot picked up the blue vehicle toy with its right hand and placed it in the white bowl; 0.67: if the robot picked up the blue vehicle toy with its right hand; 0.33: if the robot touched the blue vehicle toy with its right hand; 0.00: if the robot did anything else.	1.00: if the robot picked up the light pink color soft toy with its left hand and placed it in the brown bowl; 0.67: if the robot picked up the light pink color soft toy with its left hand; 0.33: if the robot touched the light pink color soft toy with its left hand; 0.00: if the robot did anything else.	

Table 11 | Progress Scores: Apollo humanoid (Semantic Generalization).

Benchmark: Apollo humanoid - Semantic Generalization.		
"Recoge la pelota antiestrés y colócala en la bolsa blanca".	**"Pikc up the stress ball and palce it in the wite bag".**	**"Tome el huevo de la ranura inferior derecha de la caja de huevos amarilla con la mano izquierda y coloque el huevo en la bandeja".**
1.00: if the robot picked the correct object and placed it in the bag; 0.50: if the robot picked the wrong object but correctly placed it in the bag; 0.50: if the robot picked the correct object but didn't place it in the bag; 0.00: if the robot did anything else.	1.00: if the robot picked the correct object and placed it in the bag; 0.50: if the robot picked the wrong object but correctly placed it in the bag; 0.50: if the robot picked the correct object but didn't place it in the bag; 0.00: if the robot did anything else.	1.00: if the robot picked up the egg with its left hand and placed it in the tray; 0.50: if the robot picked up the egg with its left hand; 0.00: if the robot did anything else.
"Pick up the eg form the bottom right splot of the yellwo egg box with your left hand and place the egg in the try".	**"Recoge con tu mano derecha la lechuga verde que está al final de la mesa cerca de ti".**	**"Pikc up the bottm green lettuuce from the tble with yor rght haand".**
1.00: if the robot picked up the egg with its left hand and placed it in the tray; 0.50: if the robot picked up the egg with its left hand; 0.00: if the robot did anything else.	1.00: if the robot picked up the bottom green lettuce with its right hand; 0.75: if the robot touched the bottom green lettuce with its right hand; 0.50: if the robot picked up lettuce with its right hand, but not the bottom green lettuce; 0.25: if the robot touched lettuce with its right hand, but not the bottom green lettuce; 0.00: if the robot did anything else.	1.00: if the robot picked up the bottom green lettuce with its right hand; 0.75: if the robot touched the bottom green lettuce with its right hand; 0.50: if the robot picked up lettuce with its right hand, but not the bottom green lettuce; 0.25: if the robot touched lettuce with its right hand, but not the bottom green lettuce; 0.00: if the robot did anything else.
"Recoge el vehículo de juguete azul y colócalo en el recipiente blanco".	**"Pic up the blu vehical toy and place it in teh wite bowl".**	
1.00: if the robot picked up the blue vehicle toy and placed it in the white bowl; 0.50: if the robot picked up the blue vehicle toy; 0.00: if the robot did anything else.	1.00: if the robot picked up the blue vehicle toy and placed it in the white bowl; 0.50: if the robot picked up the blue vehicle toy; 0.00: if the robot did anything else.	

表 **10** | 进度得分：Apollo 人形机器人（分布内与视觉泛化）。

基准：Apollo 人形机器人——分布内与视觉泛化。		
"拿起小熊软糖袋，并将其放入白色袋子"。	**"拿起魔方，把它放进白色袋子里"。**	**"从右下方的槽中拿起鸡蛋，黄色蛋盒，用右手，并把鸡蛋放进托盘"。**
1.00：如果机器人拿取了正确的物体并放置它在袋中; 0.50：如果机器人抓取了错误物体，但正确地将其放入袋中; 0.50：如果机器人抓取了正确物体，但未将其放入袋中; 0.00：如果机器人做了其他任何动作。	1.00：如果机器人抓取了正确的物体并放置它在袋子中; 0.50：如果机器人抓取了错误的物体但正确地将其放入袋子中; 0.50：如果机器人抓取了正确的物体但没有将其放入袋子中; 0.00：如果机器人做了任何其他动作。	1.00：如果机器人用右手拿起鸡蛋并将其放入托盘中; 0.67：如果机器人用右手拿起鸡蛋; 0.33：如果机器人用右手触碰了鸡蛋; 0.00：如果机器人做了任何其他动作。
"从右下槽位拾起鸡蛋黄色蛋盒，用左手将鸡蛋放入托盘"。	**"从桌上拿起最下面的绿色生菜，用你的右手"。**	**"从桌上拿起右侧的橙色交通锥用你的右手"。**
1.00：如果机器人用左手捡起鸡蛋并将其放入托盘; 0.67：如果机器人用左手捡起鸡蛋; 0.33：如果机器人用左手触碰了鸡蛋; 0.00：如果机器人做了任何其他事情。	1.00：如果机器人拿起了最下面的绿色生菜，用的是它的右手; 0.75：如果机器人触碰了最下面的绿色生菜，用的是它的右手; 0.50：如果机器人用它的右手拿起了生菜，但不是最下面的绿色生菜; 0.25：如果机器人用右手碰了生菜，但没有碰底部的那棵绿色生菜; 0.00：如果机器人做了其他任何事。	1.00：如果机器人拿起了右侧的橙色交通锥用它的右手; 0.75：如果机器人碰了右侧的橙色交通锥用其右手; 0.50：如果机器人拿起了一个交通锥，但不是正确的橙色交通锥; 0.25：如果机器人碰到了一个交通锥，但不是正确的橙色交通锥; 0.00：如果机器人做了其他任何事情。
"用你的右手拿起蓝色玩具车并把它放进白色碗里"	**"拿起浅粉色毛绒玩具，用你的左手，并把它放进棕色碗里"**	
1.00：如果机器人拿起了蓝色玩具车，用其右手，并把它放进白色碗里; 0.67：如果机器人拿起了蓝色玩具车，用它的右手; 0.33：如果机器人触碰了蓝色玩具车，用它的右手; 0.00：如果机器人做了任何其他事情。	1.00：如果机器人拿起了浅粉色毛绒玩具用左手将其放入棕色碗中; 0.67：如果机器人拿起了浅粉色的毛绒玩具用左手; 0.33：如果机器人触碰了浅粉色的毛绒玩具用左手; 0.00：如果机器人做了其他任何事情。	

表 **11** | 进度得分：Apollo 人形机器人（语义泛化）。

基准：Apollo 人形机器人 - 语义泛化。		
"捡起减压球，把它放进袋子里白色"。	**"捡起减压球，把它放进白色袋子里"。**	**"从……的右下方槽中拿起鸡蛋用左手拿着黄色鸡蛋盒并把鸡蛋放到托盘上"。**
1.00：如果机器人拿取了正确的物体并将其放置在袋子中; 0.50：如果机器人拿错了物体但正确地将其放入袋子中; 0.50：如果机器人抓取了正确物体但没有将其放入袋子中; 0.00：如果机器人做了其他任何事情。	1.00：如果机器人抓取了正确物体并放置将其放入袋子中; 0.50：如果机器人抓取了错误物体但正确地将其放入袋子中; 0.50：如果机器人拿起了正确的物体，但没有将其放入袋子中; 0.00：如果机器人做了其他任何动作。	1.00：如果机器人用左手拿起了鸡蛋并将其放入托盘中; 0.50：如果机器人用左手拿起鸡蛋; 0.00：如果机器人做了其他任何动作。
"从右下角的槽位拿起鸡蛋，用你的左手从黄色鸡蛋盒中，并把鸡蛋放进托盘"。	**"用你的右手拿起那棵绿色生菜，它在桌子尽头，靠近你"。**	**"从桌上拿起最下面的绿色生菜，用你的右手"。**
1.00：如果机器人用左手拿起鸡蛋并将其放入托盘中; 0.50：如果机器人用左手拿起鸡蛋; 0.00：如果机器人做了其他任何动作。	1.00：如果机器人拿起了最下面的绿色生菜，用的是它的右手; 0.75：如果机器人接触了最下面的绿色生菜，用的是它的右手; 0.50：如果机器人用它的右手拿起了生菜，但不是底部的绿色生菜; 0.25：如果机器人用右手碰到了生菜，但不是底部的绿色生菜; 0.00：如果机器人做了其他任何动作。	1.00：如果机器人拿起了底部的绿色生菜用右手; 0.75：如果机器人触碰了底部的绿色生菜，并用它的右手; 0.50：如果机器人用右手拿起生菜，但不是底部的绿色生菜; 0.25：如果机器人用右手触碰了生菜，但不是底部的绿色生菜; 0.00：如果机器人做了任何其他事情。
"拿起蓝色玩具车，把它放到白色容器里"。	**"拿起蓝色玩具车，把它放进白色的碗里"。**	
1.00：如果机器人拿起了蓝色玩具车并把它放进了白色碗里; 0.50：如果机器人拿起了蓝色玩具车; 0.00：如果机器人做了其他任何事情。	1.00：如果机器人拿起了蓝色玩具车并把它放进了白色碗里; 0.50：如果机器人拿起了蓝色玩具车; 0.00: 如果机器人做了其他任何事。	

Table 12 | Progress Scores: Apollo humanoid (Action Generalization).

Benchmark: Apollo humanoid - Action Generalization.		
"Pick up the black flashlight and place it in the white bag".	**"Pick up the black and brown snack bag and place it in the white bag".**	**"Pick up the orange mentos container and place it in the white bag".**
1.00: if the robot picked the correct object and placed it in the bag; 0.50: if the robot picked the wrong object but correctly placed it in the bag; 0.50: if the robot picked the correct object but didn't place it in the bag; 0.00: if the robot did anything else.	1.00: if the robot picked the correct object and placed it in the bag; 0.50: if the robot picked the wrong object but correctly placed it in the bag; 0.50: if the robot picked the correct object but didn't place it in the bag; 0.00: if the robot did anything else.	1.00: if the robot picked the correct object and placed it in the bag; 0.50: if the robot picked the wrong object but correctly placed it in the bag; 0.50: if the robot picked the correct object but didn't place it in the bag; 0.00: if the robot did anything else.
"Pick up the orange and place it in the white bag".	**"Pick up the purple die and place it in the white bag".**	**"Pick up the blue cereal box with your right hand".**
1.00: if the robot picked the correct object and placed it in the bag; 0.50: if the robot picked the wrong object but correctly placed it in the bag; 0.50: if the robot picked the correct object but didn't place it in the bag; 0.00: if the robot did anything else.	1.00: if the robot picked the correct object and placed it in the bag; 0.50: if the robot picked the wrong object but correctly placed it in the bag; 0.50: if the robot picked the correct object but didn't place it in the bag; 0.00: if the robot did anything else.	1.00: if the robot picked up the blue cereal box with its right hand; 0.67: if the robot picked up the blue cereal box with its left hand; 0.33: if the robot touched the blue cereal box with its right hand; 0.00: if the robot did anything else.
"Pick up the leftmost green pringles can from the top shelf of the black rack with your left hand".	**"Pick up the red cereal box with your left hand".**	
1.00: if the robot picked up the green pringles can with its left hand; 0.67: if the robot picked up the green pringles can with its right hand; 0.33: if the robot touched the green pringles can with its left hand; 0.00: if the robot did anything else.	1.00: if the robot picked up the red cereal box with its left hand; 0.67: if the robot picked up the red cereal box with its right hand; 0.33: if the robot touched the red cereal box with its left hand; 0.00: if the robot did anything else.	

表 **12** | 进展得分：Apollo 人形机器人（动作泛化）。

基准测试：Apollo 人形机器人——动作泛化。		
"拿起黑色手电筒，并将其放入白色包中"。	**"拿起黑色和棕色零食袋并将其放置在白色袋子中"。**	**"拿起橙色曼妥思容器并将其放置在白色袋子中"。**
1.00：如果机器人拿起了正确物体并把它放置在袋子中； 0.50：如果机器人拿错了物体但正确将其放入袋子中； 0.50：如果机器人拿对了物体但未将其放入袋子中； 0.00：如果机器人做了任何其他事情。	1.00：如果机器人拿对了物体并它在袋中； 0.50：如果机器人抓取了错误物体，但正确地将其放入袋中； 0.50：如果机器人抓取了正确物体，但未将其放入袋中； 0.00：如果机器人做了其他任何动作。	1.00：如果机器人抓取了正确的物体并放置它在袋子中； 0.50：如果机器人抓取了错误的物体但正确地将其放入袋子中； 0.50：如果机器人抓取了正确的物体但没有将其放入袋子中； 0.00：如果机器人做了其他任何事。
"拿起橙子并把它放进白色袋子中"。	**"拿起紫色骰子并把它放进白色袋子"。**	**"用右手拿起蓝色麦片盒"。**
1.00：如果机器人拿起了正确的物体并放置它在袋中； 0.50：如果机器人抓取了错误物体，但正确地将其放入袋中； 0.50：如果机器人抓取了正确物体，但未将其放入袋中； 0.00：如果机器人做了其他任何动作。	1.00：如果机器人抓取了正确的物体并放置它在袋子中； 0.50：如果机器人抓取了错误的物体但正确地将其放入袋子中； 0.50：如果机器人抓取了正确的物体但没有将其放入袋子中； 0.00：如果机器人做了其他任何事。	1.00：如果机器人捡起了蓝色麦片盒，用它的右手； 0.67：如果机器人捡起了蓝色麦片盒，用它的左手； 0.33：如果机器人用其触碰了蓝色麦片盒右手； 0.00：如果机器人做了任何其他事情。
"拿起最左边的绿色品客薯片罐，从黑色架子的最上层，用左手"。	**"用左手拿起红色麦片盒"。**	
1.00：如果机器人拿起了绿色品客薯片罐，用它的左手； 0.67：如果机器人拿起绿色品客薯片罐时用的是它的右手； 0.33：如果机器人触碰绿色品客薯片罐时用的是它的左手； 0.00：如果机器人做了任何其他事情。	1.00：如果机器人用其拿起红色麦片盒左手； 0.67：如果机器人用其拿起红色麦片盒右手； 0.33：如果机器人用其触碰红色麦片盒左手； 0.00: 如果机器人做了其他任何事。	

B.2.3.2 Qualitative Results

Controlling a humanoid is much more challenging than the Aloha or the bi-arm Franka robots, due to higher degree-of-freedom whole-body control, multi-finger dexterity, and limited visibility of the workspace that requires active perception. Fig. 27 shows qualitative examples which demonstrate remarkable degrees of generalization of Gemini Robotics 1.5 on the humanoid platform. We observe that our model can control Apollo to grasp novel objects and place them into receptacles that were not seen in the training data. Depending on the objects to manipulate, different grasping strategies emerge automatically. Furthermore, we also observe successful object manipulation on surfaces with different heights that were not seen during training as well.

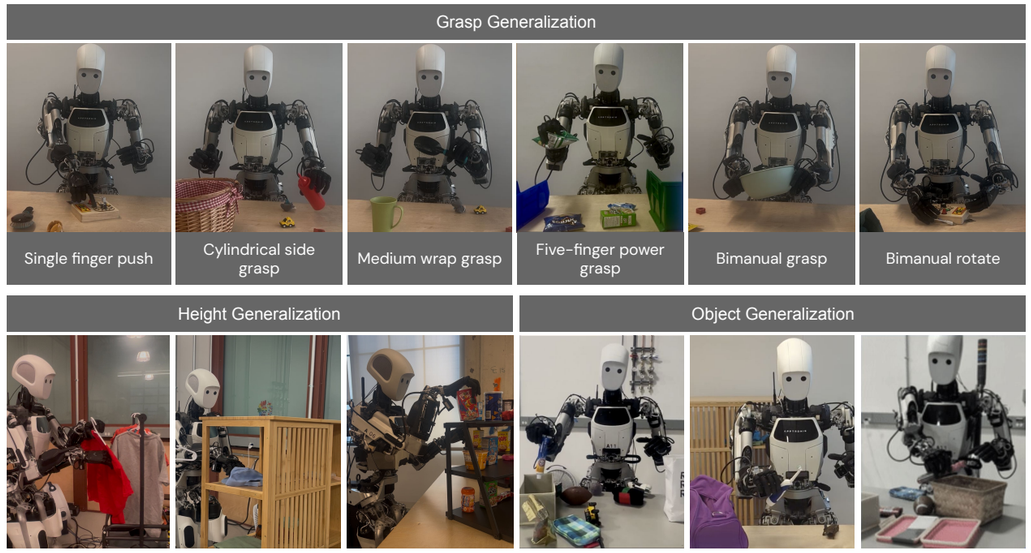


Figure 27 | Qualitative examples of generalization on the humanoid robot: Gemini Robotics 1.5 learns multiple grasp strategies, can manipulate objects at different heights, and grasp novel objects and place them into receptacles that were never seen during training.

B.2.3.2 定性结果

由于需要更高自由度的全身控制、多指灵巧操作，以及工作空间可见性有限而必须进行主动感知，控制人形机器人比控制 Aloha 或双臂 Franka 机器人要困难得多。图 27 展示了定性示例，这些示例证明了 Gemini Robotics 1.5 在人形机器人平台上具有显著的泛化能力。我们观察到，我们的模型能够控制 Apollo 抓取新物体，并将它们放入训练数据中未见过的容器中。根据要操作的物体不同，不同的抓取策略会自动涌现。此外，我们还观察到，在训练期间未见过的不同高度的表面上也能成功进行物体操作。

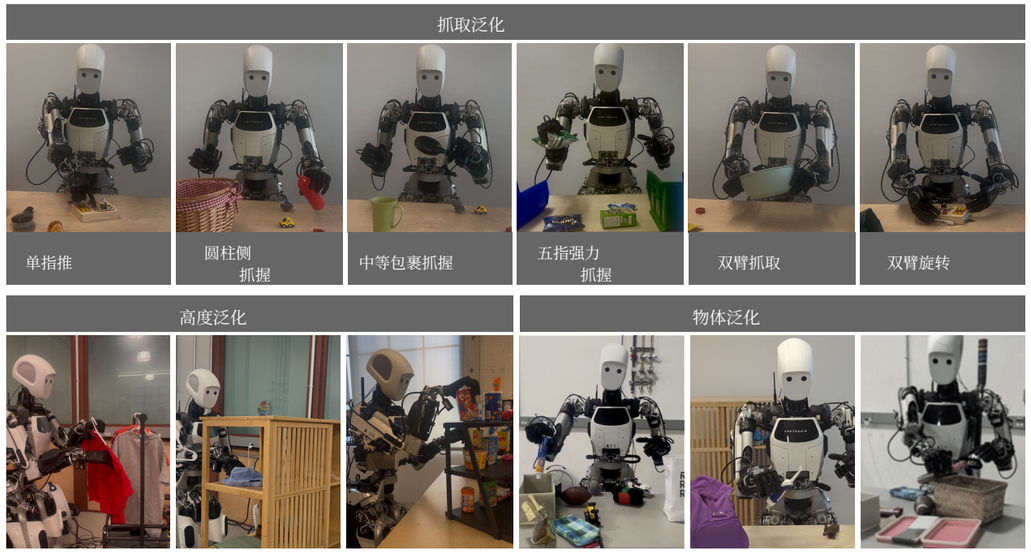


图 **27** | 人形机器人上泛化能力的定性示例：Gemini Robotics 1.5 学习多种抓取策略，能够在不同高度操作物体，并抓取新物体将其放入训练期间从未见过的容器中。

B.3. Cross-embodiment benchmark

In order to measure Motion Transfer across our three robots we define benchmarks for testing tasks on a Robot A for which data was only collected on Robot B and vice-versa. We focus on the following scenarios.

B.3.1. Bi-arm Franka → ALOHA benchmark

The ALOHA robot data is diverse and enables the execution of a multitude of tasks as demonstrated in our prior work (Gemini-Robotics-Team et al., 2025). However, the vast majority of this data focuses on table top tasks with limited interaction on the vertical axis. Meanwhile for Bi-arm Franka robot we have collected data involving interacting with a vertically mounted back-panel that goes beyond the typical motion range for ALOHA. As such, tasks in this scenario (e.g. hanging/unhanging tools) provide an ideal benchmark for motion transfer as a model trained with solely ALOHA data cannot solve them. Following this strategy, we design a set of 10 tasks in this benchmark. Figure 28 provides a visual overview of the tasks and Table 13 provides the progress score definition for each task.

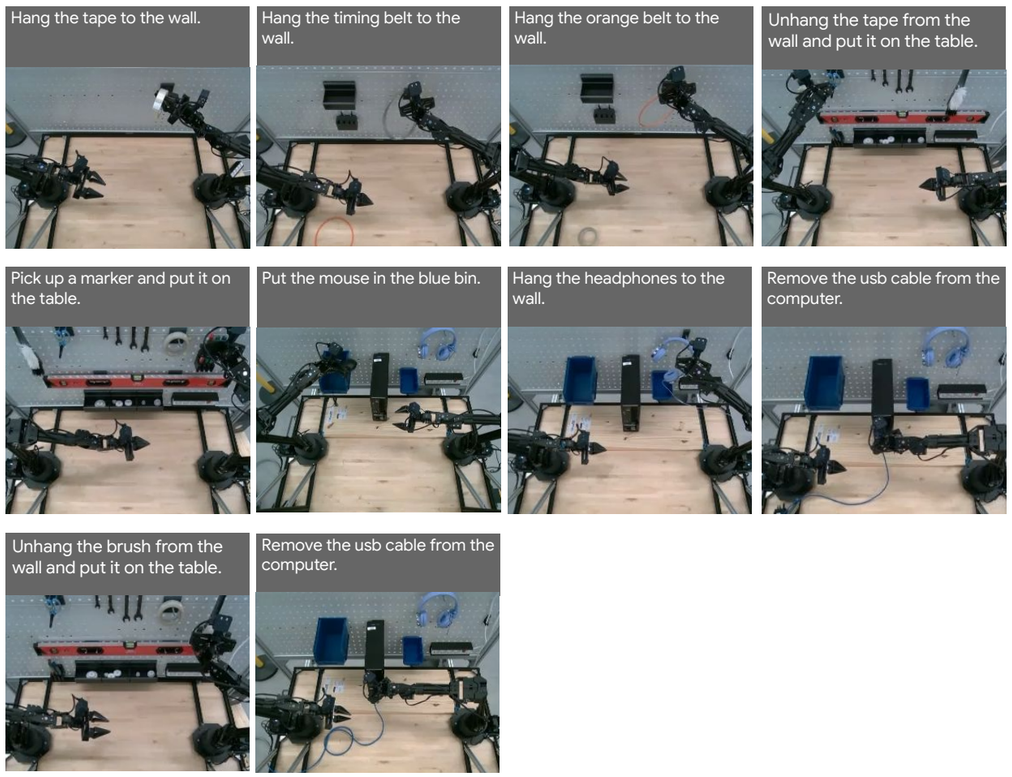


Figure 28 | Example of execution of cross-embodiment tasks for Bi-arm Franka → ALOHA benchmark.

B.3.2. ALOHA benchmark → Bi-arm Franka and Apollo humanoid robot → Bi-arm Franka

We identified in the ALOHA data tasks that require specific motions such as open drawers or closing a pear-shaped organizer (task that requires precise control) which are not available in the Bi-arm

B.3. 跨具身基准测试

为了衡量我们三款机器人之间的运动迁移，我们定义了若干基准，用于测试在机器人 A 上执行、但数据仅在机器人 B 上采集（反之亦然）的任务。我们重点关注以下场景。

B.3.1. 双臂 *Franka →ALOHA* 基准测试

ALOHA 机器人数据具有多样性，能够支持大量任务的执行，这一点已在我们先前的工作中得到展示（Gemini-Robotics-Team et al., 2025）。然而，这些数据中的绝大多数集中在桌面任务上，在垂直方向上的交互有限。与此同时，针对双臂 Franka 机器人，我们采集了涉及与垂直安装的背板进行交互的数据，此类交互超出了 ALOHA 的典型运动范围。因此，该场景中的任务（例如悬挂/取下工具）为运动迁移提供了理想的基准测试，因为仅使用 ALOHA 数据训练的模型无法解决这些任务。遵循这一策略，我们在该基准测试中设计了一组 10 个任务。图 28 以可视化方式概述了这些任务，表 13 给出了每个任务的进度得分定义。

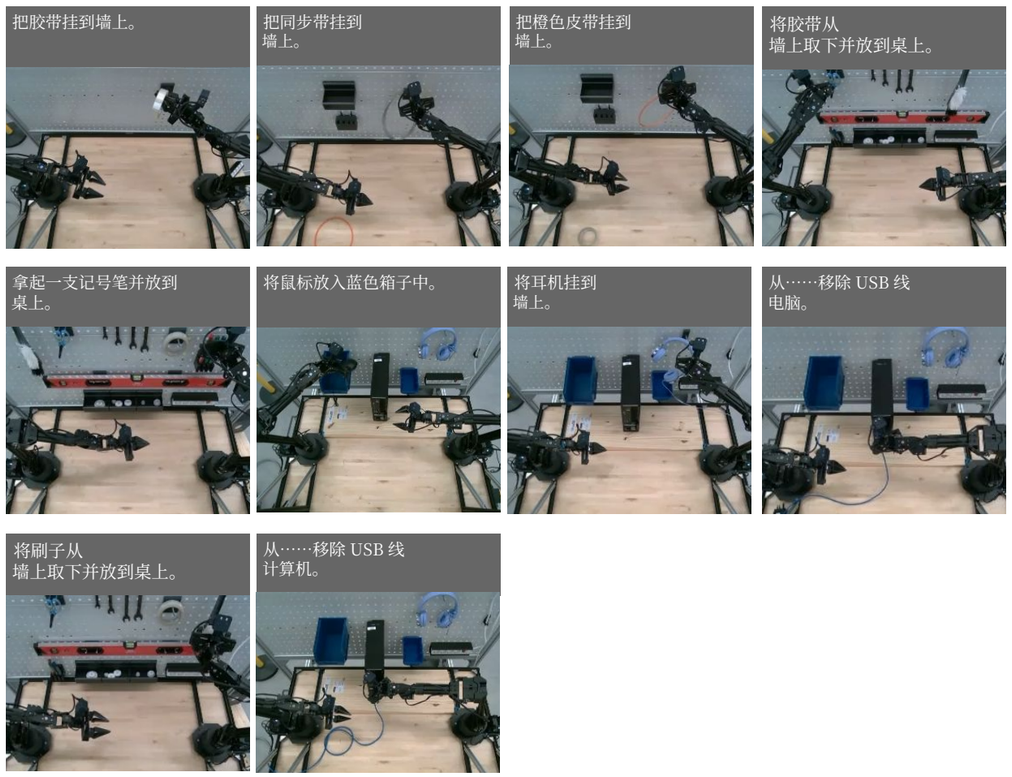


图 28 | 双臂 Franka 执行跨具身任务的示例 →ALOHA 基准 .

B.3.2. ALOHA 基准 → 双臂 *Franka* 与 *Apollo* 人形机器人 → 双臂 *Franka*

我们在 ALOHA 数据中确定了需要特定动作的任务，例如打开抽屉或合上梨形收纳盒（需要精确控制的任务），这些动作在双臂

Table 13 | Progress Scores: Bi-arm Franka → ALOHA Benchmark.

Benchmark: Bi-arm Franka → ALOHA.		
"Hang the orange belt on the small metal hook on the wall".	**"Hang the timing belt on the small metal hook on the wall".**	**"Hang the tape to the wall".**
1.00: if the robot hung the orange belt on the hook; 0.90: if the robot tried to hang the orange belt on the hook; 0.80: if the robot moved towards the hook after grasping the orange belt; 0.20: if the robot grasped the orange belt; 0.10: if the robot reached the orange belt; 0.00: if the robot didn't reach for the orange belt.	1.00: if the robot hung the timing belt on the hook; 0.90: if the robot tried to hang the timing belt on the hook; 0.80: if the robot moved towards the hook after grasping the timing belt; 0.20: if the robot grasped the timing belt; 0.10: if the robot reached the timing belt; 0.00: if the robot didn't reach for the timing belt.	1.00: if the robot hung the tape; 0.90: if the robot tried to hang the tape on the hook; 0.80: if the robot moved towards the hook after grasping the tape; 0.20: if the robot grasped the tape; 0.10: if the robot reached the tape; 0.00: if the robot didn't reach for the tape.
"Unhang the brush and put it on the table".	**"Pick up a marker and put it on the table".**	**"Unhang the tape and put it on the table".**
1.00: if the robot unhooked the brush and it ends up anywhere on the table; 0.70: if the robot grasped the brush but fails to unhook it; 0.30: if the robot attempted to grasp the brush on the backpanel but couldn't succeed in doing so; 0.00: if the robot has no success.	1.00: if the robot managed to pick up a marker and put it on the desk; 0.90: if the robot managed to pick up a marker; 0.50: if the robot tried to pick up a marker; 0.10: if the robot reached to a marker but didn't try to pick it up; 0.05: if the robot reached toward the markers but didn't get close enough; 0.00: if the robot didn't reach toward the markers.	1.00: if the robot unhooked the tape and it ends up anywhere on the table; 0.70: if the robot grasped the tape but fails to unhook it; 0.30: if the robot attempted to grasp the tape on the backpanel but couldn't succeed in doing so; 0.00: if the robot has no success.
"Unplug the usb cable".	**"Unplug the power cable".**	**"Hang the headphones on the small metal hook on the wall".**
1.00: if the robot unplugs the cable; 0.70: if the robot tried to unplug the cable; 0.40: if the robot grasped the cable; 0.20: if the robot reached toward the cable but didn't get close enough; 0.00: if the robot didn't reach toward the cable.	1.00: if the robot unplugs the cable; 0.70: if the robot tried to unplug the cable; 0.40: if the robot grasped the cable; 0.20: if the robot reached toward the cable but didn't get close enough; 0.00: if the robot didn't reach toward the cable.	1.00: if the robot managed to hang the headphones on the hook; 0.70: if the robot tried to hang the headphones on the hook; 0.50: if the robot moved the headphones toward the hook but didn't get close enough; 0.30: if the robot lifted the headphones; 0.20: if the robot grasped the headphones but didn't lift them; 0.10: if the robot reached toward the headphones but didn't get close enough; 0.00: if the robot didn't reach toward the headphones.
"Put the mouse in one of the blue bins".		
1.00: if the robot managed to put the mouse in the bin; 0.70: if the robot tried to put the mouse in a bin; 0.50: if the robot moved the mouse towards a bin but didn't get close enough; 0.30: if the robot lifted the mouse; 0.10: if the robot grasped the mouse but didn't lift it; 0.05: if the robot reached toward the mouse but didn't get close enough; 0.00: if the robot didn't reach toward the mouse.		

Franka data. In addition we added some easier packing tasks to see whether motion transfer could not only transfer new skills but also improves performance on easier tasks. We followed a similar procedure to define the tasks to measure motion transfer from the Humanoid robot to the Bi-arm Franka. Figure 29 and 30 provides a visual overview of the tasks (11 in total) and Table 14 provides the progress score definition for each task.

B.3.3. ALOHA benchmark → Humanoid robot

We followed a similar approach for the Humanoid robot: we looked into skills in the ALOHA dataset that are not covered by the action data for the Humanoid. One example to highlight this is the task "open the wardrobe", which is a motor skill completely outside the covergae of the Humanoid dataset. Fig 31 shows visuals of the 7 tasks in this benchmark and Table 15 reports the definition of the progress score for each task.

表 **13** | 进度得分：双臂 Franka →ALOHA 基准测试。

基准测试：双臂 Franka → ALOHA。		
"把橙色腰带挂在小金属挂钩上，在墙上"。	**"把同步带挂到小金属钩上，钩子在墙上"。**	**"把胶带挂到墙上"。**
1.00：如果机器人把橙色带子挂到了钩子上； 0.90：如果机器人试图把橙色带子挂到钩子上； 0.80：如果机器人在抓取后朝挂钩移动橙色皮带； 0.20：如果机器人抓住了橙色皮带； 0.10：如果机器人够到了橙色皮带； 0.00：如果机器人没有去够橙色皮带。	1.00：如果机器人将正时皮带挂到挂钩上； 0.90：如果机器人尝试将正时皮带挂到挂钩； 0.80：如果机器人在抓取后朝挂钩移动正时皮带； 0.20：如果机器人抓取了正时皮带； 0.10：如果机器人够到了正时皮带； 0.00：如果机器人没有伸手去拿同步带。	1.00：如果机器人挂上了胶带； 0.90：如果机器人尝试把胶带挂在挂钩上； 0.80：如果机器人在抓取后朝挂钩移动胶带； 0.20：如果机器人抓住了胶带； 0.10：如果机器人够到了胶带； 0.00：如果机器人没有去够胶带。
"把刷子从挂钩上取下来，放到桌子上"。	**"拿起一支记号笔，放到桌子上"。**	**"把胶带从挂钩上取下来，放到桌子上"。**
1.00：如果机器人把刷子从挂钩上取下来，并且它最终桌子上的任意位置； 0.70：如果机器人抓住了刷子，但未能将其从钩上取下it; 0.30：如果机器人尝试抓住刷子，而刷子位于背板，但未能成功做到； 0.00：如果机器人没有成功。	1.00：如果机器人成功拿起一支记号笔并放置将其放在桌上； 0.90：如果机器人成功拿起了一支记号笔； 0.50：如果机器人尝试拿起一支记号笔； 0.10：如果机器人伸手够向一支记号笔，但没有尝试拿起它； 0.05：如果机器人朝记号笔伸出手，但没有足够接近； 0.00：如果机器人没有朝标记物伸手。	1.00：如果机器人解开了胶带，并且胶带最终落到桌面上的任意位置； 0.70：如果机器人抓住了胶带但未能解开it; 0.30：如果机器人试图抓住位于其上的胶带背板，但未能成功做到这一点； 0.00：如果机器人没有成功。
"拔下 USB 线缆"。	**"拔下电源线"。**	**"将耳机挂在那个小金属挂钩上，挂钩位于墙上"。**
1.00：如果机器人拔下线缆； 0.70：如果机器人试图拔下线缆； 0.40：如果机器人抓住了线缆； 0.20：如果机器人朝线缆伸手，但没有足够接近； 0.00：如果机器人没有朝线缆伸手。	1.00：如果机器人拔下线缆； 0.70：如果机器人试图拔下线缆； 0.40：如果机器人抓住了线缆； 0.20：如果机器人朝线缆伸手，但没有足够接近； 0.00：如果机器人没有朝线缆伸手。	1.00：如果机器人成功将耳机挂到钩子上； 0.70：如果机器人尝试将耳机挂到钩子上； 0.50：如果机器人将耳机移向钩子，但没有靠得足够近； 0.30：如果机器人拿起了耳机； 0.20：如果机器人抓住了耳机但没有拿起它们； 0.10：如果机器人伸手够向耳机，但没有靠得足够近； 0.00：如果机器人没有伸手够向耳机。
"把鼠标放进其中一个蓝色箱子里"。		
1.00：如果机器人成功将鼠标放进箱子里； 0.70：如果机器人尝试将鼠标放进一个箱子里； 0.50：如果机器人将鼠标朝一个箱子移动，但没有靠得足够近； 0.30：如果机器人抬起了鼠标； 0.10：如果机器人抓住了鼠标但没有将其抬起； 0.05：如果机器人朝鼠标伸去，但未足够接近； 0.00：如果机器人没有朝鼠标伸去。		

Franka 数据中并不具备。此外，我们添加了一些更简单的装箱任务，以检验动作迁移是否不仅能迁移新技能，还能提升较简单任务上的表现。我们采用了类似的流程来定义任务，以衡量从人形机器人到双臂 Franka 的动作迁移。图 29 和 30 提供了任务（共 11 项）的视觉概览，表 14 提供了每项任务的进度分数定义。

B.3.3. ALOHA 基准测试 → 人形机器人

我们针对人形机器人采用了类似的方法：我们研究了 ALOHA 数据集中那些人形机器人动作数据未涵盖的技能。一个能突显这一点的例子是“打开衣柜”任务，这是一项完全超出人形机器人数据集覆盖范围的运动技能。图 31 展示了该基准中 7 项任务的可视化结果，表 15 报告了每项任务进度分数的定义。

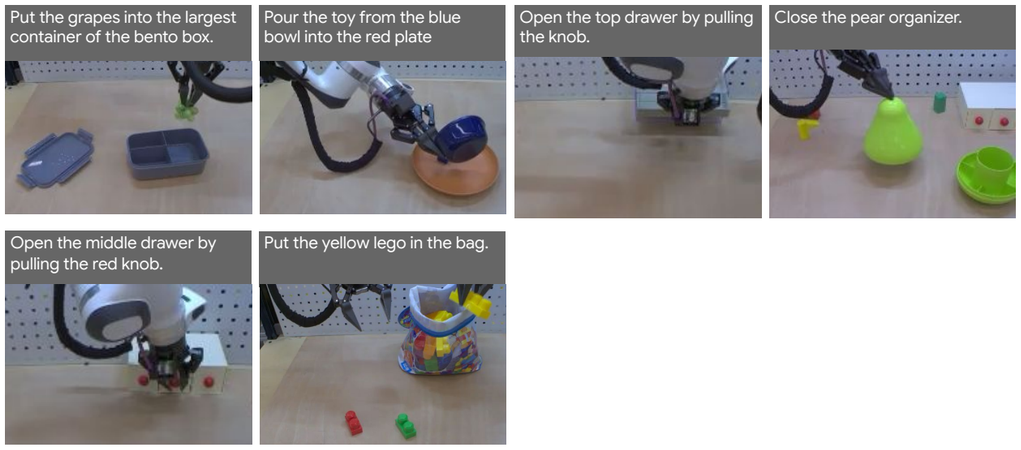


Figure 29 | Example of execution of cross-embodiment tasks for ALOHA → Bi-arm Franka benchmark.

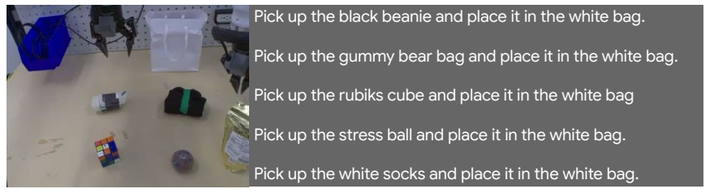


Figure 30 | Cross-embodiment tasks for Apollo humanoid → Bi-arm Franka benchmark.

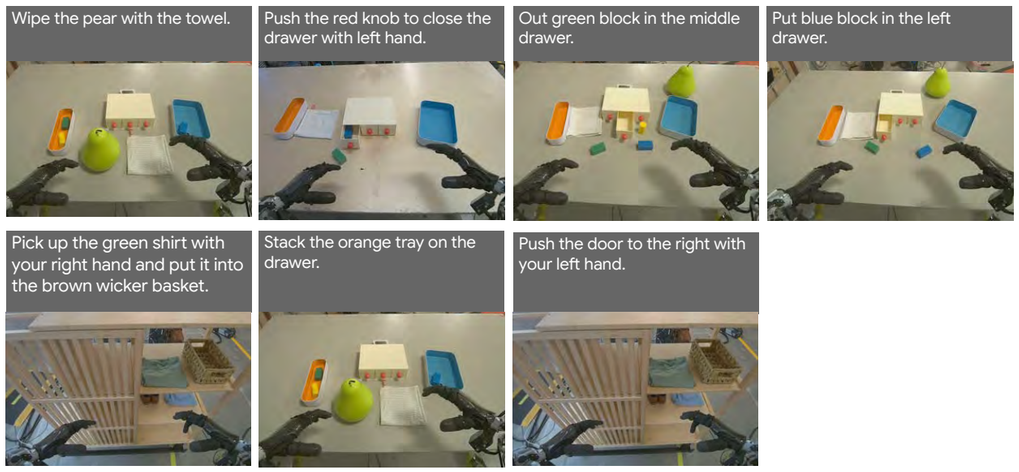


Figure 31 | Cross-embodiment tasks for ALOHA → Humanoid benchmark.

图29 | ALOHA 跨具身任务执行示例 →双臂 Franka 基准 .

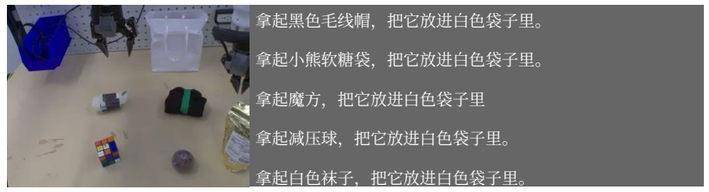


图 **30** | Apollo 人形机器人的跨具身任务 → 双臂 Franka 基准。

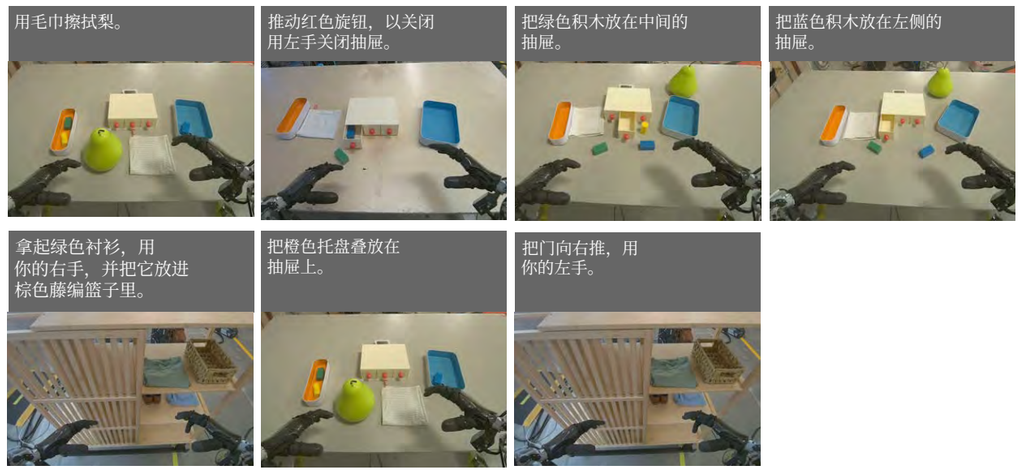


图 **31** | ALOHA 的跨具身任务 → 人形机器人基准。

Table 14 | Progress Scores: ALOHA/Humanoid → Bi-arm Franka.

Benchmark: ALOHA/Humanoid → Bi-arm Franka.		
"Close the pear organizer".	**"Put the grapes in the largest compartment of the bento box"**.	**"Pour the toy from the blue bowl into the red plate"**.
1.00: if the robot placed the pear top onto the bottom so it covers the compartment; 0.50: if the robot held the pear top above the center compartment of the base; 0.20: if the robot grasped the pear top; 0.05: if the robot reached toward the pear top; 0.00: if the robot did anything else.	1.00: if the robot put the grapes in the largest compartment; 0.50: if the robot put the grapes anywhere on the bento box; 0.20: if the robot grasped the grapes; 0.10: if the robot reached for the grapes; 0.00: if the robot did anything else.	1.00: if the robot poured at least one toy onto the plate; 0.70: if the robot tried to pour the toys onto the plate; 0.30: if the robot held the bowl above the plate; 0.20: if the robot grasped the bowl; 0.05: if the robot reached for the bowl; 0.00: if the robot did anything else.
"Open the top drawer by pulling the knob".	**"Put the yellow lego in the bag"**.	**"Open the middle drawer by pulling the red knob"**.
1.00: if the robot opened the top drawer by pulling the knob; 0.90: if the robot opened the top drawer; 0.80: if the robot opened any drawer; 0.50: if the robot grasped any knob; 0.10: if the robot reached for any knob; 0.05: if the robot reached toward the drawer; 0.00: if the robot did anything else.	1.00: if the robot put the yellow block in the bag; 0.70: if the robot put any block in the bag; 0.50: if the robot held the yellow block above the bag; 0.20: if the robot grasped the yellow block; 0.10: if the robot reached for the yellow block; 0.00: if the robot did anything else.	1.00: if the robot opened the correct drawer; 0.60: if the robot opened any drawer; 0.30: if the robot grasped any knob; 0.10: if the robot reached for any knob; 0.05: if the robot reached toward the drawer; 0.00: if the robot did anything else.
"Pick up the stress ball and place it in the white bag".	**"Pick up the white socks and place it in the white bag"**.	**"Pick up the rubiks cube and place it in the white bag"**.
1.00: if the robot placed the object in the white bag; 0.50: if the robot grasped the object and brings it over the white bag; 0.25: if the robot grasped the object; 0.00: if the robot did not grasp the object.	1.00: if the robot placed the object in the white bag; 0.50: if the robot grasped the object and brings it over the white bag; 0.25: if the robot grasped the object; 0.00: if the robot did not grasp the object.	1.00: if the robot placed the object in the white bag; 0.50: if the robot grasped the object and brings it over the white bag; 0.25: if the robot grasped the object; 0.00: if the robot did not grasp the object.
"Pick up the black beanie and place it in the white bag".	**"Pick up the gummy bear bag and place it in the white bag"**.	
1.00: if the robot placed the object in the white bag; 0.50: if the robot grasped the object and brings it over the white bag; 0.25: if the robot grasped the object; 0.00: if the robot did not grasp the object.	1.00: if the robot placed the object in the white bag; 0.50: if the robot grasped the object and brings it over the white bag; 0.25: if the robot grasped the object; 0.00: if the robot did not grasp the object.	

Table 15 | Progress Scores: ALOHA → Humanoid Benchmark.

Benchmark: ALOHA → Humanoid robot.		
"Pick up the green shirt with its right hand and put it into the brown wicker basket".	**"Push the door to the right with its left hand"**.	**"Push the red knob to close the drawer with its left hand"**.
1.00: if the robot picked up the green shirt with its right hand and put it into the brown wicker basket; 0.50: if the robot picked up the green shirt with its right hand; 0.00: if the robot did anything else.	1.00: if the robot pushed the door to the right with its left hand (door should be at least halfway through); 0.50: if the robot's left hand touched the door; 0.00: if the robot did anything else.	1.00: if the robot closes the drawer at least halfway through; 0.50: if the robot's hand touched the red knob; 0.00: if the robot did anything else.
"Stack the orange tray on the drawer".	**"Put blue block in the left drawer"**.	**"Put green block in the middle drawer"**.
1.00: if the robot picked up the orange tray and stacked it on the drawer; 0.50: if the robot picked up the orange tray; 0.00: if the robot did anything else.	1.00: if the robot picked up the blue block and put it in the left drawer; 0.50: if the robot picked up the blue block; 0.00: if the robot did anything else.	1.00: if the robot picked up the green block and put it in the middle drawer; 0.50: if the robot picked up the green block; 0.00: if the robot did anything else.
"Wipe the pear with towel".		
1.00: if the robot wiped the pear with the towel; 0.50: if the robot grabbed the towel; 0.00: if the robot did anything else.		

表 **14** | 进展得分：ALOHA/人形 → 双臂 Franka。

基准：ALOHA/人形 → 双臂 Franka。		
"合上梨形收纳盒"。	**"把葡萄放入……的最大隔间便当盒"**。	**"把玩具从蓝色碗里倒入红色盘子"**。
1.00：如果机器人将梨形顶部放到底部上使其覆盖隔间; 0.50：如果机器人将梨形顶部保持在中心上方基座的隔间; 0.20：如果机器人抓住了梨的顶部; 0.05：如果机器人伸向梨的顶部; 0.00：如果机器人做了其他任何事情。	1.00：如果机器人将葡萄放入最大的隔-间; 0.50：如果机器人将葡萄放在便当盒上的任何位置box; 0.20：如果机器人抓住了葡萄; 0.10：如果机器人伸手去抓葡萄; 0.00：如果机器人做了其他任何事情。	1.00：如果机器人至少将一个玩具倒到盘子上; 0.70：如果机器人试图将玩具倒到盘子上; 0.30：如果机器人将碗举在盘子上方; 0.20：如果机器人抓住了碗; 0.05：如果机器人伸手去够碗; 0.00：如果机器人做了其他任何事情。
"通过拉动旋钮打开最上面的抽屉"。	**"把黄色乐高积木放进袋子里"**。	**"通过拉动红色旋钮打开中间的抽屉"**。
1.00：如果机器人打开了顶层抽屉，方式为拉动该旋钮; 0.90：如果机器人打开了顶层抽屉; 0.80：如果机器人打开了任意抽屉; 0.50：如果机器人抓住了任意旋钮; 0.10：如果机器人伸手去够任意旋钮; 0.05：如果机器人朝抽屉伸出了手; 0.00：如果机器人做了其他任何事情。	1.00：如果机器人将黄色积木放进了袋子里; 0.70：如果机器人将任意积木放进了袋子里; 0.50：如果机器人将黄色积木举在袋子上方; 0.20：如果机器人抓住了黄色积木; 0.10：如果机器人伸手去够黄色积木; 0.00：如果机器人做了其他任何动作。	1.00：如果机器人打开了正确的抽屉; 0.60：如果机器人打开了任意一个抽屉; 0.30：如果机器人抓住了任意一个旋钮; 0.10：如果机器人伸手去够任意一个旋钮; 0.05：如果机器人朝抽屉伸手; 0.00：如果机器人做了任何其他事情。
"拿起压力球，把它放进白色袋子里"。	**"拿起白色袜子，把它放进白色袋子"**。	**"拿起魔方，把它放进白色袋子"**。
1.00：如果机器人将物体放入白色袋子中; 0.50：如果机器人抓取了物体并将其带到白色袋子; 0.25：如果机器人抓取了物体; 0.00：如果机器人没有抓取物体。	1.00：如果机器人将物体放入白色袋子中; 0.50：如果机器人抓住物体并将其带过来白色袋子; 0.25：如果机器人抓住物体; 0.00：如果机器人未抓住物体。	1.00：如果机器人将物体放入白色袋子中; 0.50：如果机器人抓取了物体并将其带过来白色袋子; 0.25：如果机器人抓取了物体; 0.00：如果机器人没有抓取物体。
"拿起黑色毛线帽，把它放进白色袋子。"	**"拿起小熊软糖袋，把它放进白色袋子"**。	
1.00：如果机器人将物体放入白色袋子; 0.50：如果机器人抓取物体并将其移到白色袋子上方; 0.25：如果机器人抓取物体; 0.00：如果机器人未抓取物体。	1.00：如果机器人将物体放入白色袋子中; 0.50：如果机器人抓取了物体并将其带过来白色袋子; 0.25：如果机器人抓取了物体; 0.00：如果机器人未抓取物体。	

表 **15** | 进度得分：ALOHA → 人形机器人基准。

基准：ALOHA → 人形机器人。		
"用右手拿起绿色衬衫，并将其放入棕色藤编篮子"。	**"用左手将门向右推"**。	**"用左手推动红色旋钮以关闭抽屉手"**。
1.00：如果机器人用其右捡起绿色衬衫手并把它放进棕色柳条篮; 0.50：如果机器人用其右捡起绿色衬衫手; 0.00：如果机器人做了其他任何事。	1.00：如果机器人用它的左手（门应至少过了一半）; 0.50：如果机器人的左手碰到了门; 0.00：如果机器人做了其他任何动作。	1.00：如果机器人将抽屉至少关闭一半以上; 0.50：如果机器人的手触碰了红色旋钮; 0.00：如果机器人做了其他任何动作。
"将橙色托盘叠放在抽屉上"。	**"将蓝色积木放进左侧抽屉"**。	**"将绿色积木放进中间抽屉"**。
1.00：如果机器人拿起橙色托盘并进行了堆叠将其放在抽屉上; 0.50：如果机器人拿起了橙色托盘; 0.00：如果机器人做了其他任何事情。	1.00：如果机器人拿起蓝色方块并将其放入左侧抽屉; 0.50：如果机器人拿起了蓝色方块; 0.00：如果机器人做了其他任何动作。	1.00：如果机器人拿起绿色积木并将其放入中间抽屉; 0.50：如果机器人拿起绿色积木; 0.00：如果机器人做了其他任何动作。
"用毛巾擦拭梨子"。		
1.00：如果机器人用毛巾擦拭了梨; 0.50：如果机器人抓住了毛巾; 0.00：如果机器人做了其他任何事情。		

B.4. Multi-step benchmark

The multi-step benchmarks combine individual tasks into compound instructions (i.e., A then B then C) or abstract, goal-oriented instructions. For the compound instructions, we typically require a particular order for the individual tasks in order to achieve a full score of 1.0. For example, for the Apollo humanoid, several of our multi-step tasks combine individual gift packing tasks into compound instructions such as "put the stress ball in the white bag, then put the gummy bear bag in the white bag, then put the white socks in the white bag" and require this exact order for a perfect score. We also include abstract, goal-oriented instructions such as “pack all the gifts” or “sort the snacks into their matching containers by color”, though these are less common across the benchmarks. The next paragraphs show the tasks and their progress score system for each embodiment.

B.4.1. ALOHA robot

Figure 32 shows visuals of the tasks and Table 16 defines the progress score for each task.

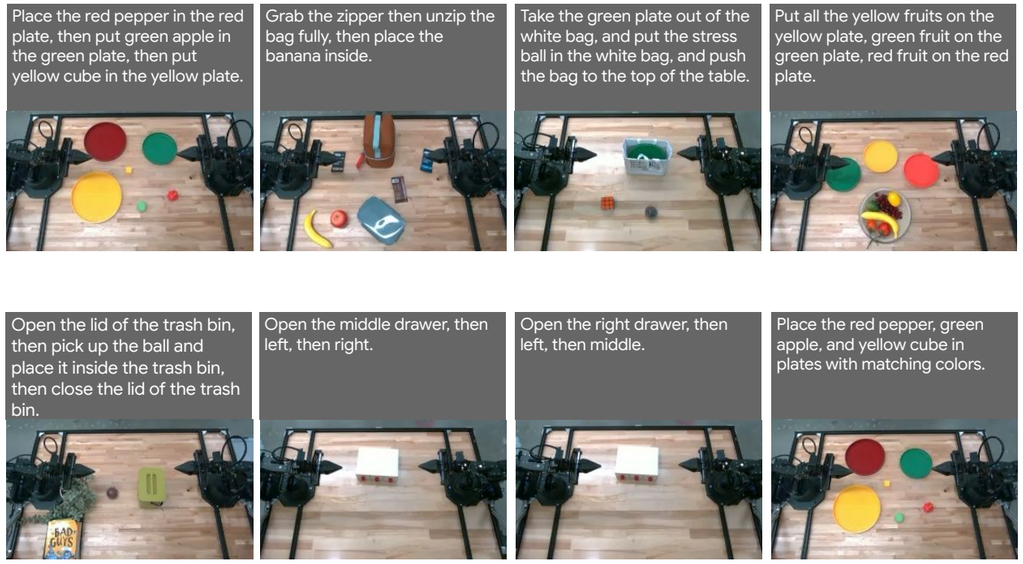


Figure 32 | Tasks in the multi-step benchmark for the ALOHA.

B.4.2. Bi-arm Franka robot

Figure 32 shows visuals of the tasks and Table 17 defines the progress score for each task.

B.4.3. Humanoid robot

Figure 34 shows visuals of the tasks and Table 18 defines the progress score for each task.

B.4. 多步骤基准测试

多步骤基准测试将各个单独任务组合成复合指令（即 A 然后 B 然后 C）或抽象的、目标导向的指令。对于复合指令，我们通常要求各个单独任务按特定顺序执行，才能获得 1.0 的满分。例如，对于 Apollo 人形机器人，我们的若干多步骤任务会将各个单独的礼品包装任务组合成复合指令，例如“把减压球放进白色袋子，然后把小熊软糖袋放进白色袋子，然后把白色袜子放进白色袋子”，并要求严格按此顺序执行才能获得满分。我们还包含抽象的、目标导向的指令，例如“打包所有礼物”或“按颜色将零食分拣到与之匹配的容器中”，不过这些指令在各基准测试中不太常见。接下来的段落将展示每种具身形态的任务及其进度评分系统。

B.4.1. ALOHA 机器人

图 32 展示了这些任务的可视化画面，表 16 定义了每个任务的进度得分。

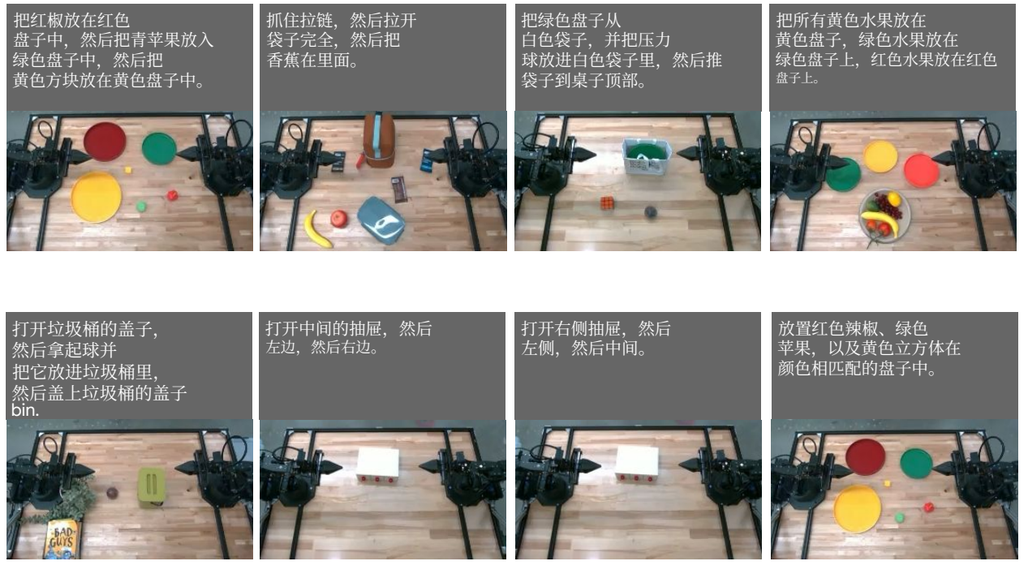


图 **32** | 用于 ALOHA 的多步骤基准测试中的任务。

B.4.2. 双臂 *Franka* 机器人

图 32 展示了这些任务的可视化画面，表 17 定义了每个任务的进度得分。

B.4.3. 人形机器人

图 34 展示了各任务的可视化结果，表 18 定义了每个任务的进度分数。

Table 16 | Progress Scores: ALOHA Robot (Multi-step benchmark).

Benchmark: ALOHA Robot - Multi-step.		
"Unzip the lunchbag and then place the banana in the lunchbag".	**"Place the red pepper, green apple, and yellow cube in plates with matching colors"**.	**"Place the red pepper in the red plate, then put green apple in the green plate, then put yellow cube in the yellow plate"**.
1.00: if the robot fully unzipped the lunch bag and placed the banana in the lunchbag; 0.50: if the robot fully unzipped the lunch bag; 0.00: if the robot did anything else.	1.00: if the robot placed all three objects correctly in the matching plates; 0.70: if the robot placed two objects on the correct matching plates, but failed for third object; 0.40: if if the robot picked one object and successfully placed it on the correct plate, but failed to do that for second object; 0.20: if the robot picked one object but failed to put it on the right plate; 0.00: if the robot didn't approach any of the objects or approached plates directly.	1.00: if the robot placed all three objects correctly in the matching plates; 0.70: if the robot placed two objects on the correct matching plates, but failed for third object; 0.40: if the robot picked one object and successfully placed it on the correct plate, but failed to do that for second object; 0.20: if the robot picked one object but failed to put it on the right plate; 0.00: if the robot didn't approach any of the objects or approached plates directly.
"Open the middle drawer, then left, then right".	**"Open the right drawer, then left, then middle"**.	**"Open the lid of the trash bin, then pick up the ball and place it inside the trash bin, then close the lid of the trash bin"**.
1.00: if the robot opened all three drawers in correct order; 0.60: if the robot opened the middle then left drawer but failed to open the right drawer; 0.30: if the robot opened the middle drawer but failed to open the left drawer next; 0.00: if the robot failed to open the middle drawer first.	1.00: if the robot opened all three drawers in correct order; 0.60: if the robot opened the right then left drawer but failed to open the middle drawer; 0.30: if the robot opened the right drawer but failed to open the left drawer next; 0.00: if the robot failed to open the right drawer first.	1.00: if the robot successfully opened the lid, put the ball in the bin, then closes the lid; 0.75: if the robot failed to close the lid after putting the ball in the trash bin; 0.50: if the robot opened the trash bin lid, picked up the ball, but failed to put the ball in the trash bin; 0.25: if the robot opened the trash bin lid but failed to pick up the ball next; 0.00: if the robot didn't open the trash bin lid.
"Take the green plate out of the white bag, and put the stress ball in the white bag, and push the bag to the top of the table".	**"Put all the yellow fruits on the yellow plate, green fruit on the green plate, red fruit on the red plate"**.	
1.00: if the robot completes the entire task successfully; 0.70: if the robot took the green plate out of the bag, put the stress ball in the bag, but failed to push the bag to the top of the table; 0.30: if the robot took the green plate out of the bag but failed to put the stress ball into the bag; 0.00: if the robot failed to take the green plate out of the bag.	1.00: if the robot put all fruits in the correct plate; 0.80: if the robot put 4 fruits in the correct plate; 0.60: if the robot put 3 fruits in the correct plate; 0.20: if the robot put 2 fruits in the correct plate; 0.20: if the robot put 1 fruit in the correct plate; 0.00: if the robot put none of the fruits in the correct plate.	

表 **16** | 进度得分：ALOHA 机器人（多步基准）。

基准：ALOHA 机器人 - 多步。		
"拉开午餐袋的拉链，然后把香蕉放入午餐袋"。	**"把红椒、绿苹果和黄色方块放进颜色相匹配的盘子里"**。	**"把红椒放进红色盘子，然后把绿苹果放进绿色盘子，然后把黄色方块放进黄色盘子"**。
1.00：如果机器人完全拉开了午餐袋的拉链，并且将香蕉放入午餐袋中； 0.50：如果机器人完全拉开了午餐袋的拉链； 0.00：如果机器人做了其他任何事情。	1.00：如果机器人将三个物体全部正确地放入对应的盘子； 0.70：如果机器人将两个物体放在正确的匹配盘子，但在第三个物体上失败了； 0.40：如果机器人拾取了一个物体并成功将其放在正确的盘子上，但未能对第二个物体； 0.20：如果机器人拾取了一个物体但未能将其放到正确的盘子上； 0.00：如果机器人没有接近任何物体，或直接接近了盘子。	1.00：如果机器人将全部三个物体正确地放入相匹配的盘子中； 0.70：如果机器人将两个物体放在正确的相匹配的盘子上，但第三个物体未成功放置； 0.40：如果机器人抓取了一个物体并成功将其放在正确的盘子上，但未能为以下对象做到这一点：第二个物体； 0.20：如果机器人抓取了一个物体，但未能将其放置在正确的盘子上； 0.00：如果机器人没有靠近任何一个物体，或者直接靠近盘子。
"打开中间的抽屉，然后打开左边的，再打开右边的"。	**"打开右边的抽屉，然后打开左边的，再打开中间的"**。	**"打开垃圾桶的盖子，然后捡起球并把它放进垃圾桶里，然后盖上盖子垃圾桶的"**。
1.00：如果机器人打开了全部三个抽屉且按正确顺序； 0.60：如果机器人先打开中间抽屉，再打开左侧抽屉，但未能打开右侧抽屉； 0.30：如果机器人打开了中间抽屉，但未能接着打开左侧抽屉； 0.00：如果机器人未能先打开中间抽屉。	1.00：如果机器人打开了全部三个抽屉且按照正确顺序； 0.60：如果机器人先打开右侧抽屉再打开左侧抽屉，但未能打开中间抽屉； 0.30：如果机器人打开了右侧抽屉但未能接下来打开左侧抽屉； 0.00：如果机器人未能先打开右侧抽屉。	1.00：如果机器人成功打开了盖子，则将球放入箱子，然后关闭盖子； 0.75：如果机器人在放入后未能关闭盖子球放入垃圾桶； 0.50：如果机器人打开了垃圾桶盖、捡起了球，但未能将球放入垃圾桶； 0.25：如果机器人打开了垃圾桶盖但未能接着捡起球； 0.00：如果机器人没有打开垃圾桶盖。
"把绿色盘子从白色袋子里拿出来，并把白色袋子里的减压球，然后把袋子推到桌子顶部"。	**"把所有黄色水果放到黄色盘子上，绿色水果放到绿色盘子上，红色水果放到红色盘子上"**。	
1.00：如果机器人成功完成整个任务； 0.70：如果机器人把绿色盘子从袋子里拿出来，把减压球放进袋子里，但未能推动袋子到桌子顶部； 0.30：如果机器人将绿色盘子从袋子里取出但未能将减压球放入袋子里； 0.00：如果机器人未能将绿色盘子从袋子里。	1.00：如果机器人将所有水果放入正确的盘子； 0.80：如果机器人将4个水果放入正确的盘子； 0.60：如果机器人将3个水果放入正确的盘子； 0.20：如果机器人将2个水果放入正确的盘子； 0.20：如果机器人将1个水果放入正确的盘子； 0.00：如果机器人没有将任何水果放入正确的盘子板。	

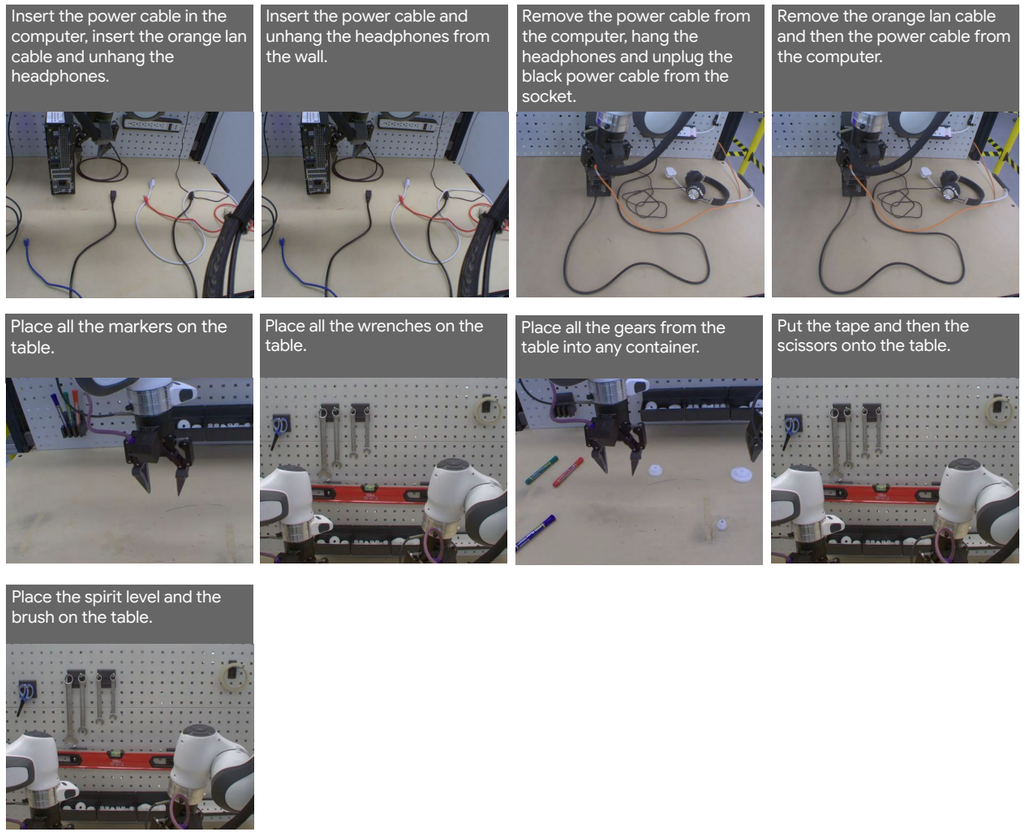


Figure 33 | Tasks in the multi-step benchmark for the Bi-arm Franka.

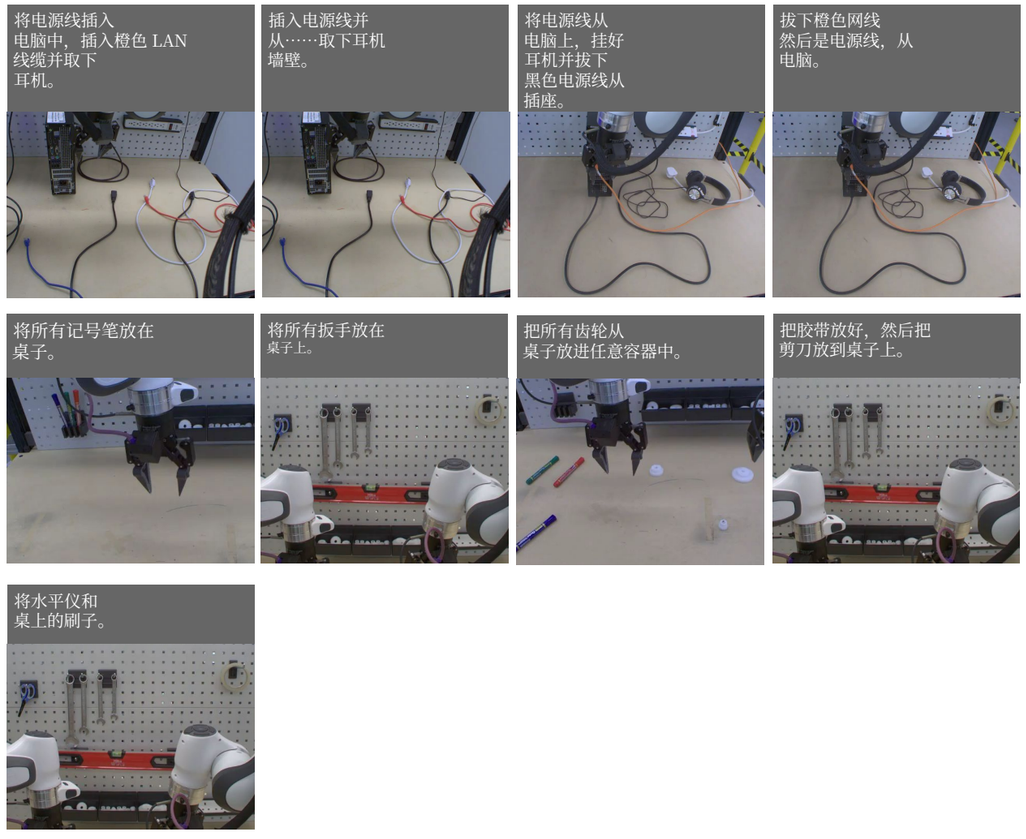


图 **33** | 双臂 Franka 多步基准测试中的任务。

Table 17 | Progress Scores: Bi-arm Franka (Multi-step benchmark).

Benchmark: Bi-arm Franka - Multi-step.		
"Insert the power cable in the computer, insert the orange lan cable and unhang the headphones".	**"Insert the power cable and unhang the headphones from the wall"**.	**"Remove the power cable from the computer, hang the headphones and unplug the black power cable from the socket"**.
1.00: if the robot successfully completed the third task; 0.75: if the robot successfully completed the remaining task; 0.60: if the robot successfully completed the second task; 0.45: if the robot grasped and attempted one of the remaining two tasks; 0.30: if the robot successfully inserted the power cable or the lan cable or unhung the headphones; 0.15: if the robot grasped and attempted to insert the power cable or the lan cable or unhang the headphones; 0.00: if the robot didn't attempt to insert the power cable or insert the lan cable or unhang the headphones.	1.00: if the robot successfully completed both tasks; 0.75: if the robot attempted to complete the remaining task; 0.50: if the robot successfully inserted the cable or unhung the headphones; 0.25: if the robot attempted to insert the cable or unhang the headphones; 0.00: if the robot didn't do anything.	1.00: if the robot successfully completed the remaining task; 0.75: if the robot grasped and attempted the remaining task; 0.60: if the robot is successful in completing one of the remaining two tasks; 0.45: if the robot grasped and attempted one of the remaining two tasks; 0.30: if the robot is successful in removing the power cable, unplugging the plug from the socket or hanging the headphones; 0.15: if the robot grasped and attempted to either remove the power cable, unplug the plug from the socket or to hang the headphones; 0.00: if the robot didn't attempt to remove the power cable, unplug the plug from the socket or to hang the headphones.
"Remove the orange lan cable and then the power cable from the computer".	**"Place all the markers on the table"**.	**"Place all the wrenches on the table"**.
1.00: if the robot successfully removed both cables; 0.75: if the robot grasped and attempted to remove the remaining cable; 0.50: if the robot successfully removed either the lan cable or the power cable from the computer; 0.25: if the robot grasped and attempted to remove either the lan cable or the power cable from the computer; 0.00: if the robot didn't attempt to remove either cable.	1.00: if the robot placed the remaining marker on the table; 0.80: if the robot grasped the remaining marker; 0.70: if the robot reached for the remaining marker of any colour; 0.67: if the robot placed the second marker on the table; 0.50: if the robot grasped the second marker of any colour; 0.40: if the robot reached for the second marker of any colour; 0.33: if the robot placed the first marker on the table; 0.20: if the robot grasped the first marker of any colour; 0.10: if the robot reached for a first marker of any colour; 0.00: if the robot placed no markers on the table.	1.00: if the robot successfully unhung and placed the fourth wrench on the table; 0.90: if the robot grasped the fourth wrench; 0.85: if the robot reached for a fourth wrench; 0.75: if the robot successfully unhung and placed the third wrench on the table; 0.65: if the robot grasped the third wrench; 0.60: if the robot reached for a third wrench; 0.50: if the robot successfully unhung and placed the second wrench on the table; 0.40: if the robot grasped the second wrench; 0.35: if the robot reached for a second wrench; 0.25: if the robot successfully unhung and placed the first wrench on the table; 0.15: if the robot grasped the first wrench; 0.10: if the robot reached for a first wrench; 0.00: if the robot placed no wrenches on the table.
"Place all the gears from the table into any container".	**"Put the tape and then the scissors onto the table"**.	**"Place the spirit level and the brush on the table"**.
1.00: if the robot placed the remaining gear in any container; 0.80: if the robot grasped the remaining third gear on the table; 0.70: if the robot reached for the remaining third gear on the table; 0.66: if the robot placed the second gear in any container; 0.50: if the robot grasped the second gear; 0.40: if the robot reached for a second gear on the table; 0.33: if the robot placed the first gear in any container; 0.20: if the robot grasped the first gear; 0.10: if the robot reached for a first gear on the table; 0.00: if the robot didn't place or attempt to put back any gears back in the container.	1.00: if the robot placed the scissors on the table; 0.75: if the robot grasped the scissors; 0.60: if the robot reached for the scissors; 0.50: if the robot successfully placed the tape on the table; 0.25: if the robot grasped the tape; 0.10: if the robot reached for the tape; 0.00: if the robot didn't place any item on the table.	1.00: if the robot placed the remaining item on the table; 0.75: if the robot grasped the remaining item; 0.60: if the robot reached for the remaining item; 0.50: if the robot successfully placed the spirit level or the brush on the table; 0.25: if the robot grasped the spirit level or the brush; 0.10: if the robot reached for either the spirit level or the brush; 0.00: if the robot didn't place any item on the table.

表 **17** | 进度分数：双臂 Franka（多步基准）。

基准：双臂 Franka - 多步。		
"将电源线插入电脑，插入橙色网线，并取下耳机"。	**"插入电源线并取下耳机从墙上"**。	**"从电脑上拔下电源线，挂好耳机，并拔下黑色电源线从插座上"**。
1.00：如果机器人成功完成了第三项任务； 0.75：如果机器人成功完成了剩余的任务； 0.60：如果机器人成功完成了第二项任务； 0.45：如果机器人抓取并尝试了其中一项剩余两项任务； 0.30：如果机器人成功插入电源线或 LAN 线，或取下了耳机； 0.15：如果机器人抓住并尝试插入电源线或 LAN 线，或取下耳机； 0.00：如果机器人未尝试插入电源线缆，或插入 LAN 线缆，或取下耳机。	1.00：如果机器人成功完成了两项任务； 0.75：如果机器人尝试完成剩余的任务； 0.50：如果机器人成功插入电缆或摘下耳机； 0.25：如果机器人尝试插入电缆或摘把耳机挂起来； 0.00：如果机器人没有做任何事。	1.00：如果机器人成功完成了剩余的任务； 0.75：如果机器人抓取并尝试了剩余的任务； 0.60：如果机器人成功完成以下任务之一剩余的两项任务； 0.45：如果机器人抓取并尝试了以下任务之一剩余的两项任务； 0.30：如果机器人成功移除了电源线缆，将插头从插座中拔出或悬挂耳机； 0.15：如果机器人抓取并试图要么重新-移开电源线，将插头从插座中拔出或悬挂耳机； 0.00：如果机器人没有尝试移除电源线缆，将插头从插座中拔出，或悬挂耳机。
"拔掉橙色网线，然后将电源线从计算机上拔下"。	**"将所有记号笔放在桌子上"**。	**"将所有扳手放在桌子上"**。
1.00：如果机器人成功移除了两根线缆； 0.75：如果机器人抓取并试图移除剩余的电缆； 0.50：如果机器人成功移除了 LAN 电缆或计算机的电源电缆； 0.25：如果机器人抓取并试图移除任一 LAN 电缆或计算机的电源电缆； 0.00：如果机器人没有尝试移除两根线缆中的任何一根。	1.00：如果机器人将剩余的记号笔放在桌子上； 0.80：如果机器人抓取了剩余的记号笔； 0.70：如果机器人伸手去够剩余记号笔，任意颜色； 0.67：如果机器人将第二个标记物放在桌上； 0.50：如果机器人抓取了第二个标记物的任意颜色； 0.40：如果机器人伸手去够第二个标记物的任意颜色； 0.33：如果机器人将第一个标记物放在桌上； 0.20：如果机器人抓取了任意颜色的第一个标记； 0.10：如果机器人伸手去够第一个标记，任意颜色； 0.00：如果机器人没有在桌上放置任何标记。	1.00：如果机器人成功取下并放置了第四个扳手在桌上； 0.90：如果机器人抓住了第四把扳手； 0.85：如果机器人伸手去拿第四把扳手； 0.75：如果机器人成功地从悬挂处取下并放置了第三把扳手放在桌子上； 0.65：如果机器人抓住了第三把扳手； 0.60：如果机器人伸手去拿第三把扳手； 0.50：如果机器人成功取下并放置了第二把扳手在桌子上； 0.40：如果机器人抓住了第二把扳手； 0.35：如果机器人伸手去拿第二把扳手； 0.25：如果机器人成功取下并放置了第一把扳手在桌子上； 0.15：如果机器人抓住了第一把扳手； 0.10：如果机器人伸手去拿第一把扳手； 0.00：如果机器人没有把任何扳手放在桌上。
"把桌上的所有齿轮放入任意容器。"	**"把胶带放到桌上，然后再把剪刀放到桌上。"**	**"将水平仪和刷子放在桌子上"**。
1.00：如果机器人将剩余齿轮放入任意容器； 0.80：如果机器人抓取了剩余的第三个齿轮在桌子上； 0.70：如果机器人伸手去抓剩余的第三个齿轮在桌子上； 0.66：如果机器人将第二个齿轮放入任意容-器； 0.50：如果机器人抓住了第二个齿轮； 0.40：如果机器人伸手去拿桌子上的第二个齿轮； 0.33：如果机器人将第一个齿轮放入任意容器； 0.20：如果机器人抓取了第一个齿轮； 0.10：如果机器人伸手去拿桌上的第一个齿轮； 0.00：如果机器人没有放置或尝试放回任何齿轮放回容器中。	1.00：如果机器人将剪刀放在桌子上； 0.75：如果机器人抓取了剪刀； 0.60：如果机器人伸手去拿剪刀； 0.50：如果机器人成功地将胶带放在桌子上； 0.25：如果机器人抓取了胶带； 0.10：如果机器人伸手去拿胶带； 0.00：如果机器人没有将任何物品放在桌子上。	1.00：如果机器人将剩余物品放置在桌子上； 0.75：如果机器人抓住了剩余物品； 0.60：如果机器人伸手去拿剩余物品； 0.50：如果机器人成功将水平仪或刷子放在桌子上； 0.25：如果机器人抓取了水平仪或刷子； 0.10：如果机器人伸手去拿水平仪或刷子； 0.00：如果机器人没有将任何物品放在桌上。

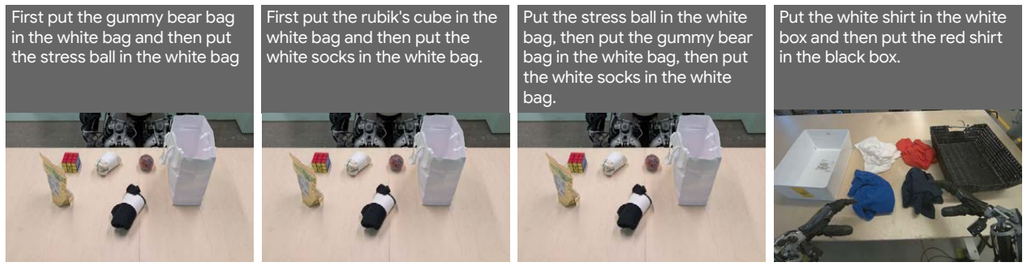


Figure 34 | Tasks in the multi-step benchmark for the Apollo humanoid.

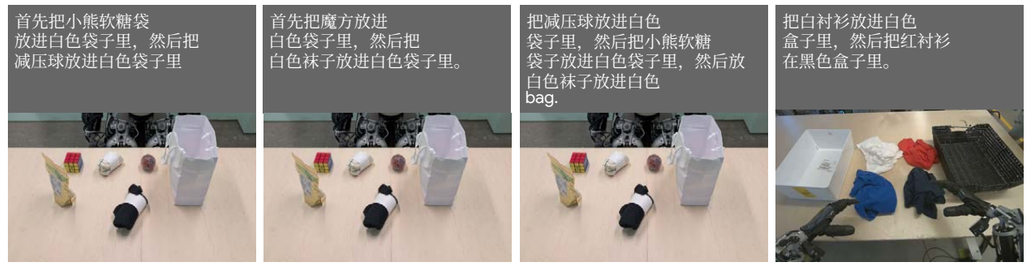


图 **34** | Apollo 人形机器人在多步基准测试中的任务。

Table 18 | Progress Scores: Humanoid Robot (Multi-step benchmark).

Benchmark: Humanoid Robot - Multi-step.		
"First put the gummy bear bag in the white bag and then put the stress ball in the white bag".	**"First put the rubik's cube in the white bag and then put the white socks in the white bag".**	**"Put the stress ball in the white bag, then put the gummy bear bag in the white bag, then put the white socks in the white bag".**
1.00: if the robot successfully put gummy bear bag into the white bag and then put stress ball into the white bag and only those 2 items; 0.90: if the robot put the gummy bear bag in the white bag and then put the stress ball in the white bag; 0.75: if the robot put the gummy bear bag in the white bag and then picked up the stress ball; 0.60: if the robot put the gummy bear bag in the white bag; 0.45: if the robot picked up the gummy bear bag and did not place it in the white bag; 0.30: if the robot put the stress ball and the gummy bear bag in the white bag but in the wrong order; 0.20: if the robot put the stress ball in the white bag; 0.10: if the robot put something in the white bag but itś not the stress ball or the gummy bear bag; 0.00: if the robot did anything else.	1.00: if the robot successfully put the rubikś cube in the white bag and then put the white socks into the white bag and only those 2 items; 0.90: if the robot put the rubikś cube in the white bag and then put the white socks in the white bag – order matters; 0.75: if the robot put the rubikś cube in the white bag and then picked up the white socks – order matters; 0.60: if the robot put the rubikś cube in the white bag; 0.45: if the robot picked up the rubikś cube and did not place it in the white bag; 0.30: if the robot put the white socks in the bag and then put the rubikś cube in the white bag; 0.20: if the robot put the white socks in the white bag; 0.10: if the robot put something in the white bag but itś *not* the rubikś cube or white socks; 0.00: if the robot did anything else.	1.00: if the robot put the stress ball in the white bag and then put the gummy bears in the white bag and then put the white socks in the white bag; 0.90: if the robot put the stress ball in the white bag and then put the gummy bears in the white bag and then put the white socks in the white bag; 0.75: if the robot put the stress ball in the white bag and then put the gummy bears in the white bag and then picked up the white socks; 0.60: if the robot put the stress ball in the white bag and then put the gummy bears in the white bag; 0.45: if the robot put the stress ball in the white bag and then picked up the gummy bears; 0.30: if the robot put the stress ball in the white bag; 0.15: if the robot picked up the stress ball; 0.00: if the robot did anything else.
"Stack the right orange cone on top of the left orange cone and then put the green lettuce in the brown bowl".	**"Put the dark colored clothes in the black box and then put the white clothes in the white box".**	**"Put the white shirt in the white box and then put the red shirt in the black box".**
1.00: if the robot stacked the right orange cone on the left orange cone and then put the green lettuce in the brown bowl; 0.75: if the robot stacked the right orange cone on the left orange cone and then touched the green lettuce; 0.50: if the robot stacked the right orange cone on the left orange cone; 0.25: if the robot touched the right orange cone; 0.00: if the robot did anything else.	1.00: if the robot put 3 colored shirts in the black box, *then* put the white shirt in the white box; 0.75: if the robot put 3 colored shirts in the black box; 0.50: if the robot put 2 colored shirts in the black box; 0.25: if the robot put 1 colored shirt in the black box; 0.00: if the robot did anything else.	1.00: if the robot put the white shirt in the white box, then put the red shirt in the black box; 0.75: if the robot put the white shirt in the white box, then touched the red shirt; 0.50: if the robot put the white shirt in the white box; 0.25: if the robot touched the white shirt; 0.00: if the robot did anything else.
"Place all the green lettuces on the table into the brown bowl and then picked up the orange cone".	**"Put the leftmost light blue chips bag in the blue bin and then put the bottom green chips in the green bin".**	**"Put the white clothes in the black box and then put the dark colored clothes in the white box".**
1.00: if the robot put 3 lettuces in the brown bowl, then picked up the orange cone; 0.75: if the robot put 3 lettuce in the brown bowl; 0.50: if the robot put 2 lettuce in the brown bowl; 0.25: if the robot put 1 lettuce in the brown bowl; 0.00: if the robot did anything else.	1.00: if the robot put the leftmost blue chips bag in the blue bin and then put the bottom green chips in the green bin; 0.75: if the robot put the leftmost blue chips bag in the blue bin and then touched the bottom green chips; 0.50: if the robot put the leftmost blue chips bag in the blue bin; 0.25: if the robot put any blue snack in the blue bin; 0.00: if the robot did anything else.	1.00: if the robot put the white shirt in the black box, then put 3 colored shirts in the white box; 0.75: if the robot put the white shirt in the black box, then put 2 colored shirts in the white box; 0.50: if the robot put the white shirt in the black box, then put 1 colored shirt in the white box; 0.25: if the robot put the white shirt in the black box; 0.00: if the robot did anything else.
"First put all the blue snacks in the blue bin and then put all the green snacks in the green bin".	**"Sort the snacks into their matching containers by color".**	**"Pack all the gifts".**
1.00: if the robot put 3 blue snacks in the blue bin, then put 2 green snacks in the green bin; 0.80: if the robot put 3 blue snacks in the blue bin, then put 1 green snack in the green bin; 0.60: if the robot put 3 blue snacks in the blue bin; 0.40: if the robot put 2 blue snacks in the blue bin; 0.20: if the robot put 1 blue snack in the blue bin; 0.00: if the robot did anything else.	1.00: if the robot put 5 snacks into their matching color bins; 0.80: if the robot put 4 snacks into their matching color bins; 0.60: if the robot put 3 snacks into their matching color bins; 0.40: if the robot put 2 snacks into their matching color bins; 0.20: if the robot put 1 snack into its matching color bin; 0.00: if the robot did anything else.	1.00: if the robot packed all 5 gifts into the white bag; 0.80: if the robot packed 4 gifts into the white bag; 0.60: if the robot packed 3 gifts into the white bag; 0.40: if the robot packed 2 gifts into the white bag; 0.20: if the robot packed 1 gift into the white bag; 0.00: if the robot did anything else.
"Place the stress ball and the gummy bears in the white bag".	**"First put the stress ball in the white bag and then put the gummy bear bag in the white bag".**	
1.00: if the robot successfully put stress ball and gummy bear bag into the white bag and only those 2 items; 0.80: if the robot successfully put stress ball and gummy bear bag into the white bag but also put other items in the bag; 0.60: if the robot put either the stress ball or the gummy bear bag in the white bag, but not both, and successfully picked up the other item but did not put it in the white bag; 0.40: if the robot put either the stress ball or the gummy bear bag in the white bag, but not both; 0.20: if the robot put something in the white bag but itś *not* the stress ball or the gummy bear bag; 0.00: if the robot did anything else.	1.00: if the robot successfully put stress ball into the white bag, then put gummy bear bag into the white bag and only those 2 items – order matters; 0.90: if the robot put the stress ball in the white bag and then put the gummy bear bag in the white bag; 0.75: if the robot put the stress ball in the white bag and then picked up the gummy bear bag; 0.60: if the robot put the stress ball in the white bag; 0.45: if the robot picked up the stress ball and did not place it in the white bag; 0.30: if the robot put the gummy bear bag and the stress ball in the white bag but in the wrong order; 0.20: if the robot put the gummy bear bag in the white bag; 0.10: if the robot put something in the white bag but itś not the stress ball or the gummy bear bag; 0.00: if the robot did anything else.	

表 **18** | 进展分数：人形机器人（多步基准）。

基准测试：人形机器人 - 多步。		
"首先把小熊软糖袋放进白色袋子里，然后把压力球放进白色袋子里"。	**"首先把魔方放进白色袋子里，然后把白色袜子放进白色袋子里"。**	**"把减压球放进白色袋子里，然后把把小熊软糖袋放进白色袋子里，然后把白色袜子放进白色袋子里"。**
1.00：如果机器人成功将小熊软糖袋放入白色袋子，并且只放那2件物品； 0.90：如果机器人将小熊软糖袋放入白色袋子，然后把压力球放入白色袋子； 0.75：如果机器人将小熊软糖袋放入白色袋子，然后拿起压力球； 0.60：如果机器人把小熊软糖袋放进白色袋子bag; 0.45：如果机器人拿起小熊软糖袋并没有把它放进白色袋子； 0.30：如果机器人把减压球和软糖小熊袋放进白色袋子，但顺序错误； 0.20：如果机器人把减压球放进白色袋子； 0.10：如果机器人把某样东西放进白色袋子，但它不是减压球，也不是小熊软糖袋； 0.00：如果机器人做了任何其他事情。	1.00：如果机器人成功将魔方放入白色袋子，然后把白色袜子放入白色袋子中，并且仅限这两样物品； 0.90：如果机器人把魔方放进白色袋子然后把白袜子放进白色袋子——顺序很重要； 0.75：如果机器人把魔方放进白色袋子然后拿起白袜子——顺序很重要； 0.60：如果机器人把魔方放进白色袋子； 0.45：如果机器人拿起了魔方而没有把它放进白色袋子里； 0.30：如果机器人把白袜子放进袋子里并且然后把魔方放进白色袋子里； 0.20：如果机器人把白袜子放进白色袋子里； 0.10：如果机器人把某样东西放进白色袋子里但这*不是*魔方，也不是白色袜子； 0.00：如果机器人做了任何其他事情。	1.00：如果机器人把减压球放进了白色袋子里然后把小熊软糖放进白色袋子里，并且然后把白色袜子放进白色袋子里； 0.90：如果机器人把减压球放进了白色袋子里然后把小熊软糖放进白色袋子里，并且然后把白色袜子放进白色袋子里； 0.75：如果机器人把压力球放进白色袋子里然后把小熊软糖放进白色袋子里，并且然后捡起了白色袜子； 0.60：如果机器人把压力球放进白色袋子里然后把小熊软糖放进白色袋子里； 0.45：如果机器人把减压球放进白色袋子里然后拿起小熊软糖； 0.30：如果机器人把减压球放进白色袋子里； 0.15：如果机器人拿起减压球； 0.00：如果机器人做了其他任何事情。
"把右边的橙色锥筒叠放到左边的橙色锥筒上，然后把绿色生菜放进棕色的碗里"。	**"把深色衣服放进黑色盒子里，再把白色衣服放进白色盒子里"。**	**"把白色衬衫放进白色盒子里，然后把黑色盒子里的红色衬衫"。**
1.00：如果机器人将右侧的橙色锥筒叠放在左侧的橙色锥筒上，然后将绿色生菜放进棕色碗里； 0.75：如果机器人将右侧的橙色锥筒叠放在左侧的橙色锥筒上，然后触碰了绿色生菜； 0.50：如果机器人将右侧橙色锥桶叠放在左侧橙色锥桶上； 0.25：如果机器人触碰了右侧橙色锥桶； 0.00：如果机器人做了其他任何动作。	1.00：如果机器人将3件彩色衬衫放入黑色箱子，*然后*将白色衬衫放入白色箱子； 0.75：如果机器人把3件彩色衬衫放进黑色盒子里； 0.50：如果机器人把2件彩色衬衫放进黑色盒子里； 0.25：如果机器人把1件彩色衬衫放进黑色盒子里； 0.00：如果机器人做了任何其他事情。	1.00：如果机器人把白色衬衫放进白色盒子里，然后把红色衬衫放进黑色盒子里； 0.75：如果机器人将白色衬衫放入白色盒子中，然后触摸了红色衬衫； 0.50：如果机器人将白色衬衫放入白色盒子中； 0.25：如果机器人触摸了白色衬衫； 0.00：如果机器人做了任何其他事情。
"将桌子上的所有绿色生菜放入棕色碗，然后捡起了橙色锥体"。	**"把最左边的浅蓝色薯片袋放进蓝色垃圾桶然后把最下面的绿色薯片放进绿色垃圾桶"。**	**"把白色衣物放进黑色盒子，然后把深色衣物放进白色盒子"。**
1.00：如果机器人把3颗生菜放进棕色碗里，那么捡起了橙色锥桶； 0.75：如果机器人把3颗生菜放进棕色碗里； 0.50：如果机器人把2颗生菜放进棕色碗里； 0.25：如果机器人把1颗生菜放进棕色碗里； 0.00：如果机器人做了其他任何事。	1.00：如果机器人将最左侧的蓝色薯片袋放入蓝色箱中，然后将底部的绿色薯片放入绿色箱中； 0.75：如果机器人将最左侧的蓝色薯片袋放入蓝色箱中，然后触摸了底部的绿色薯片； 0.50：如果机器人将最左侧的蓝色薯片袋放入蓝色箱子； 0.25：如果机器人将任何蓝色零食放入蓝色箱子； 0.00：如果机器人做了其他任何事情。	1.00：如果机器人将白色衬衫放入黑色盒子，然后将3件彩色衬衫放入白色盒子； 0.75：如果机器人将白色衬衫放入黑色盒子，然后把2件彩色衬衫放进白色盒子里； 0.50：如果机器人把白衬衫放进了黑盒子，然后把1件彩色衬衫放进白色盒子里； 0.25：如果机器人把白衬衫放进了黑盒子； 0.00：如果机器人做了其他任何事。
"首先把所有蓝色零食放进蓝色箱子里，然后把所有绿色零食放进绿色箱子里"。	**"将零食分类放入对应的容器中，按颜色"。**	**"打包所有礼物"。**
1.00：如果机器人将3个蓝色零食放入蓝色箱子，那么将2个绿色零食放入绿色箱子； 0.80：如果机器人把3个蓝色零食放进蓝色箱子，那么把1个绿色零食放进绿色箱子； 0.60：如果机器人把3个蓝色零食放进蓝色箱子； 0.40：如果机器人把2个蓝色零食放进蓝色箱子； 0.20：如果机器人把1个蓝色零食放进蓝色箱子； 0.00：如果机器人做了任何其他事情。	1.00：如果机器人将5个零食放入与其颜色匹配的箱子； 0.80：如果机器人将4个零食放入与其颜色匹配的箱子； 0.60：如果机器人将3个零食放入与其颜色匹配的箱子； 0.40：如果机器人将 2 份零食放入与其颜色匹配的收纳盒; 0.20：如果机器人将 1 份零食放入与其颜色匹配的bin; 0.00：如果机器人做了其他任何事。	1.00：如果机器人将所有 5 件礼物装入白色袋子； 0.80：如果机器人将 4 件礼物装入白色袋子； 0.60：如果机器人将 3 件礼物装进了白色袋子； 0.40：如果机器人将 2 件礼物装进了白色袋子； 0.20：如果机器人将 1 件礼物装进了白色袋子； 0.00：如果机器人做了其他任何事情。
"把减压球和小熊软糖放进白色袋子"。	**"首先把减压球放进白色袋子，然后把小熊软糖袋放进白色袋子"。**	
1.00：如果机器人成功将减压球和小熊软糖袋放入白色袋子，且仅放入这两件物品； 0.80：如果机器人成功将减压球和小熊软糖袋放入白色袋子，但也放入其他物品袋子； 0.60：如果机器人将压力球或小熊软糖小熊软糖袋放入白色袋子中，但不能两者都放，并且成功拿起了另一件物品，但没有把它放入白色bag; 0.40：如果机器人将压力球或小熊软糖小熊软糖袋放入白色袋子中，但不能两者都放; 0.20：如果机器人把某样东西放进了白色袋子，但它 *不是* 减压球或小熊软糖袋； 0.00：如果机器人做了任何其他事情。	1.00：如果机器人成功将减压球放入白色袋子，然后将小熊软糖袋放入白色袋子并且只放入这 2 件物品——顺序很重要； 0.90：如果机器人将压力球放入白色袋子然后将小熊软糖袋放入白色袋子； 0.75：如果机器人将压力球放入白色袋子然后拿起了小熊软糖袋； 0.60：如果机器人将压力球放入白色袋子； 0.45：如果机器人拿起了压力球并且没有把它放进白色袋子； 0.30：如果机器人把小熊软糖袋和减压球放进白色袋子，但顺序错误； 0.20：如果机器人把小熊软糖袋放进白色袋子bag; 0.10：如果机器人把某样东西放进白色袋子，但它不是减压球或小熊软糖袋； 0.00：如果机器人做了其他任何事情。	

B.5. Success rate

In this section we report the *success rate* for all our results in Section 3.

B.5.1. Success rate for generalization performance

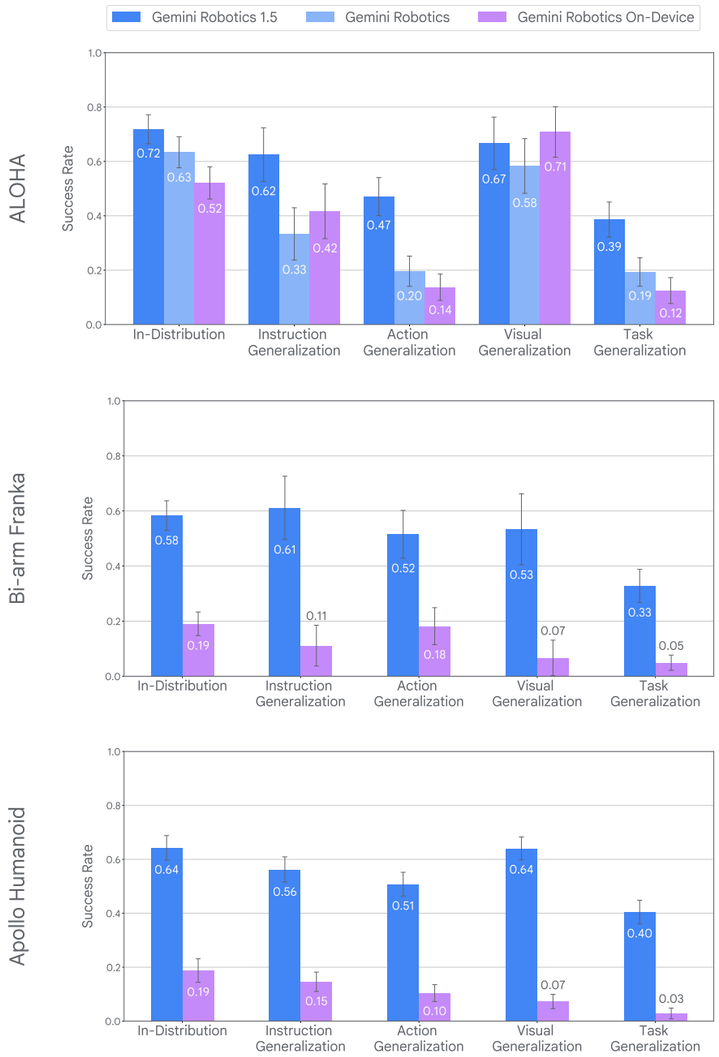


Figure 35 | Breakdown of GR 1.5 generalization capabilities across our robots. GR 1.5 consistently outperforms the baselines and handles all four types of variations more effectively.

B.5. 成功率

在本节中，我们报告第 3 节中所有结果的成功率。

B.5.1. 泛化性能的成功率

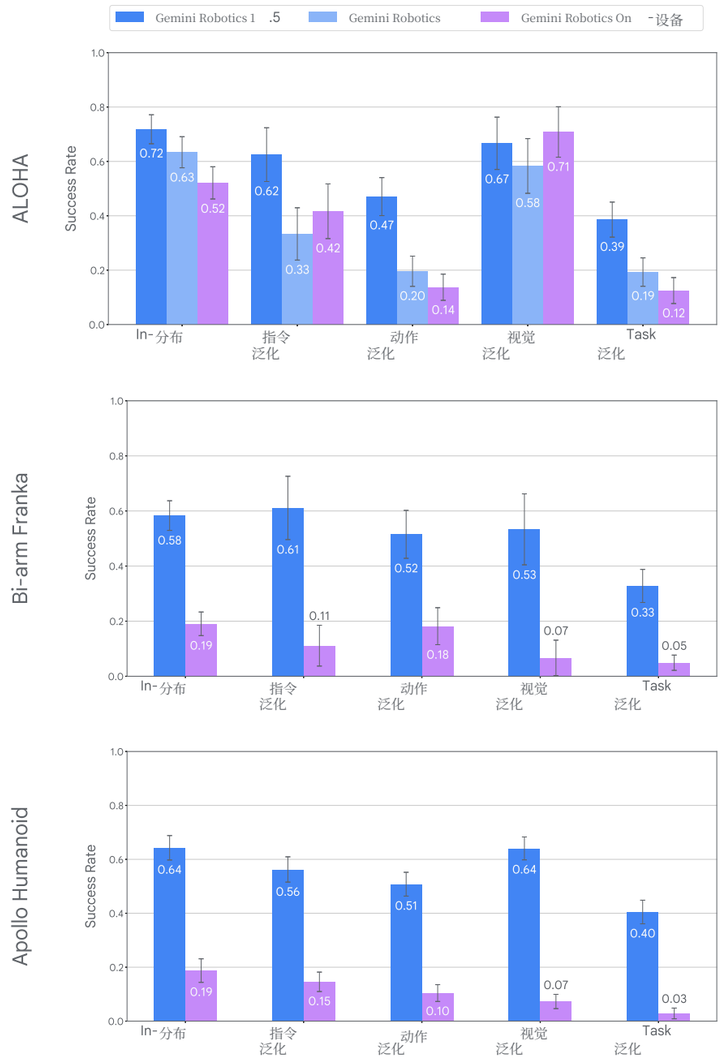


图 **35** |： GR 1.5 在各机器人上的泛化能力分解。GR 1.5 始终优于基线，并更有效地处理全部四类变化。

B.5.2. Success rate for data and model ablation

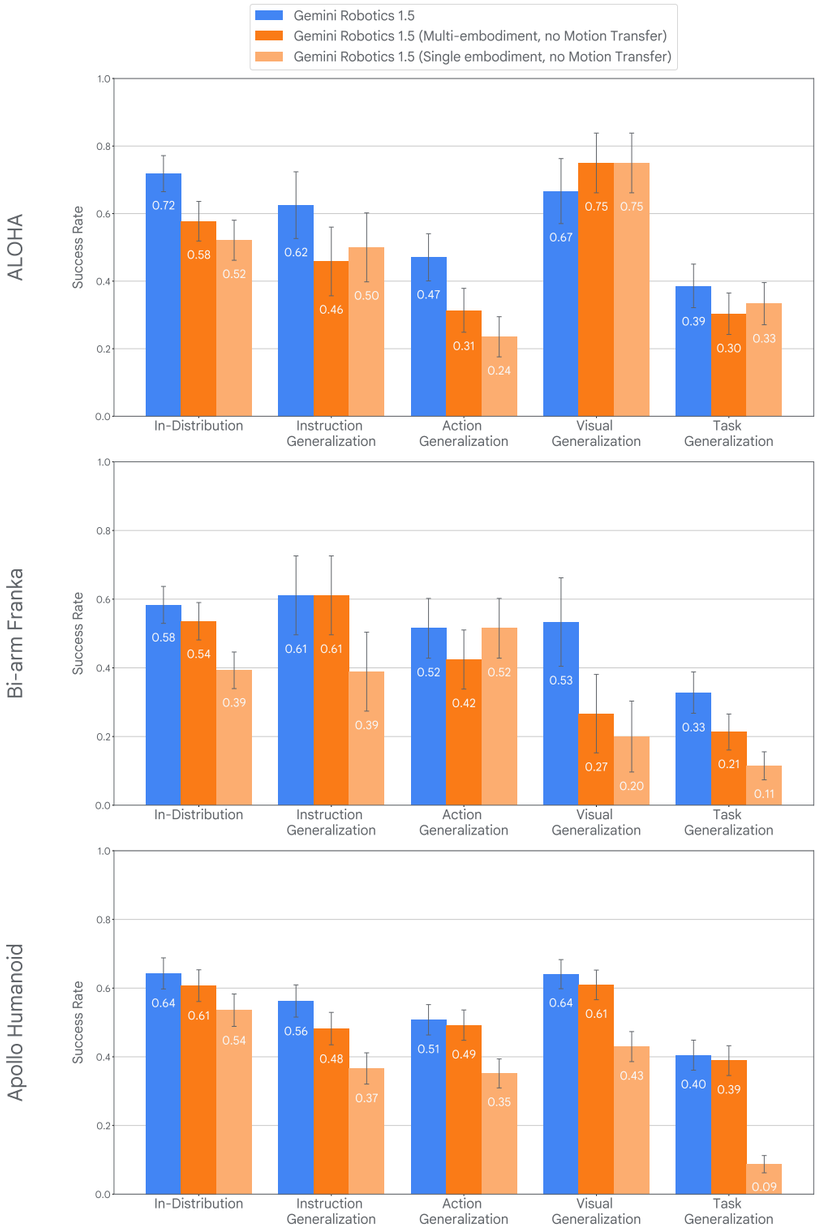


Figure 36 | Ablation on datasets and training recipes on our robots: GR 1.5 consistently outperforms our baselines: GR 1.5 trained on single or multi-robot data without the MT recipe.

B.5.2. 数据和模型消融的成功率

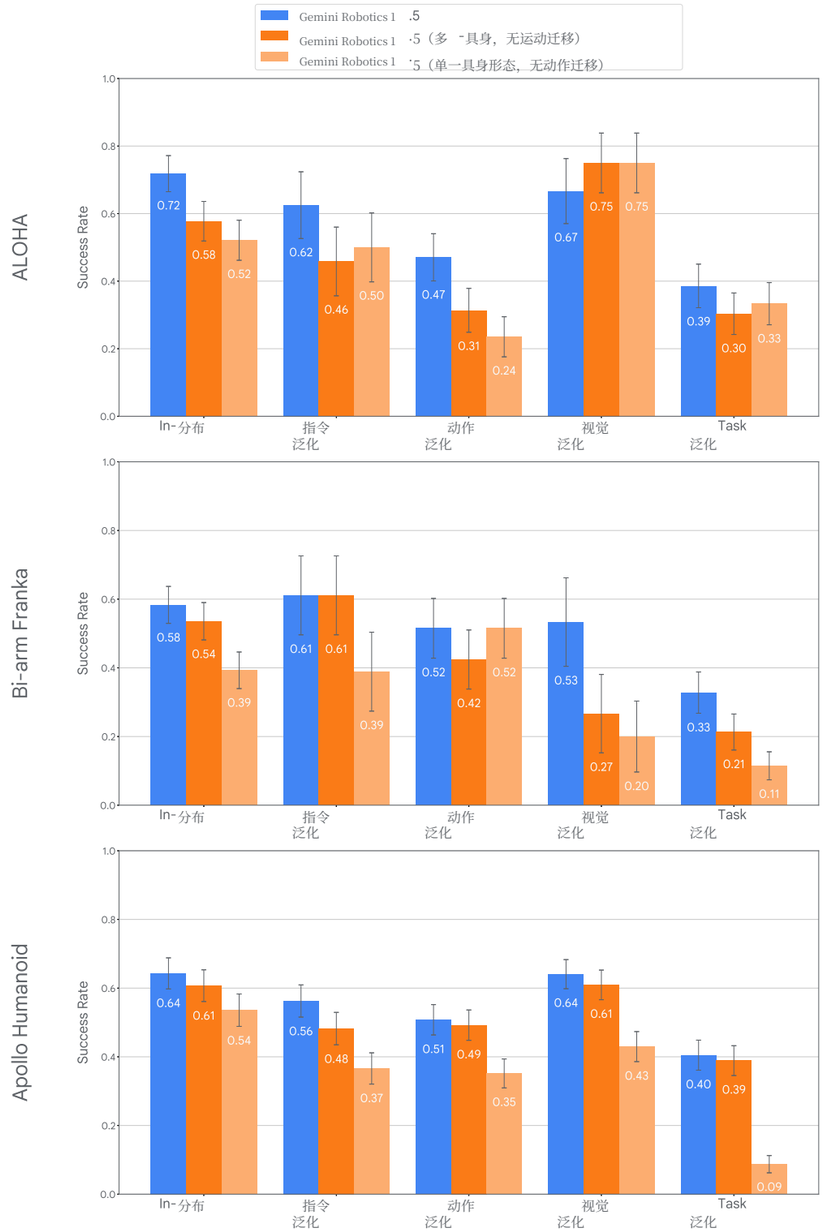


图 **36** | 在我们的机器人上针对数据集和训练配方进行的消融：GR 1.5 始终优于我们的基线：在单机器人或多机器人数据上训练且未采用 MT 配方的 GR 1.5。

B.5.3. Success rate for thinking ablation

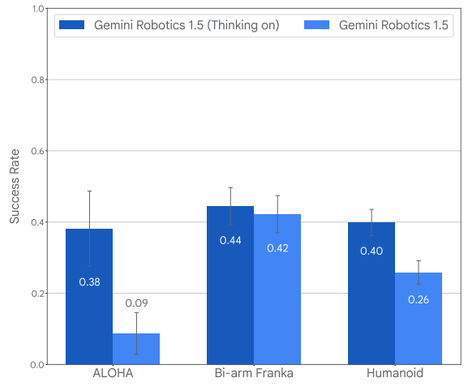


Figure 37 | Ablation of thinking: success rate in the multi-step benchmark with and without enabling thinking during inference.

B.5.3. 思考消融的成功率

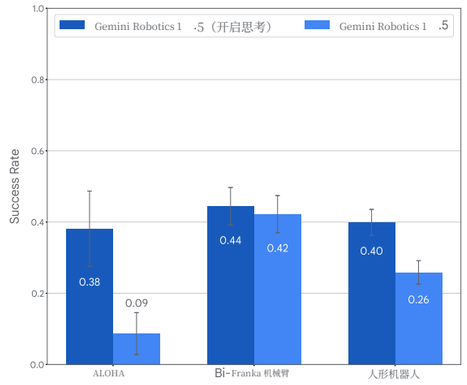


图 **37** | 思考消融：在推理过程中启用和不启用思考时，多步基准测试中的成功率。

C. Gemini Robotics-ER 1.5 is a generalist embodied reasoning model

C.1. Evaluation Details: Generality

To assess an overall approximation of model embodied reasoning performance, we evaluate Gemini Robotics-ER 1.5 and other multimodal models on a mix of 15 academic benchmarks. The aggregated results are reported in Fig. 8, and the individual benchmark performance results are shown in Table 19 and Table 20. For text-based VQA evaluation benchmarks, we used Gemini 2.5 Flash to grade response accuracy for both multiple-choice and freeform question formats.

The Gemini 2.5 and GPT-5 models were accessed in between September 1, 2025 and September 20, 2025, using default thinking budgets and without tool use.

Model	GR-ER 1.5	GR-ER 1.5	GR-ER	Gemini 2.5 Pro	Gemini 2.5 Flash	GPT-5	GPT-5-mini
Thinking	Yes	No	No	Yes	Yes	Yes	Yes
Point-Bench	71.6	73.3	**75.7**	62.7	61.7	43.6	39.5
RefSpatial	48.5	41.8	**49.3**	33.6	41.2	23.5	23.0
RoboSpatial-Pointing	**31.1**	25.3	30.3	8.3	7.9	19.0	12.5
Where2Place	**59.0**	48.0	41.0	37.0	48.0	37.0	33.5
Spatial average	**52.6**	47.1	49.1	35.4	39.7	30.8	27.1
BLINK	57.8	65.2	60.1	69.2	46.1	**71.3**	66.4
CV-Bench	84.3	83.6	83.2	**85.9**	85.5	**86.1**	**85.9**
ERQA	54.8	47.0	45.3	56.0	47.5	**59.0**	57.3
EmbSpatial	78.4	73.4	56.4	78.0	76.2	**81.5**	78.8
MindCube	54.7	47.7	47.4	**59.2**	55.4	58.0	55.6
RoboSpatial-VQA	**79.3**	57.7	66.2	71.3	73.4	69.3	70.7
SAT	76.7	62.0	64.7	74.7	73.3	**86.7**	81.3
Cosmos-Reason1	72.2	68.3	62.0	73.8	72.1	**79.4**	76.3
Min Video Pairs	72.5	67.1	59.5	72.8	69.2	**77.0**	73.0
OpenEQA	55.0	50.5	38.3	55.7	45.3	**64.4**	59.2
VSI-Bench	45.8	39.9	34.1	51.1	45.3	**52.9**	46.2
QA average	66.5	60.2	56.1	68.0	62.7	**71.4**	68.2
ER Score	**59.6**	53.7	52.6	51.7	51.2	51.1	47.7
Overall Average	**62.8**	56.7	54.2	59.3	56.5	60.6	57.3

Table 19 | Model performance on a mix of 15 academic embodied reasoning benchmarks. GPT-5 and GPT-5-mini results obtained via API in September 2025.

C.2. Evaluation Details: Pointing

Table 21 shows detailed breakdown of the evaluation for complex pointing.

C.3. Additional Examples

Fig. 38 illustrates sampled thoughts from GR-ER 1.5 on embodied reasoning tasks.

C. Gemini Robotics-ER 1.5 是一种通用型具身推理模型

C.1. 评估详情：通用性

为了总体评估模型具身推理性能的近似水平，我们在 15 个学术基准的混合集上评估了 Gemini Robotics-ER 1.5 及其他多模态模型。汇总结果见图 8，各基准的性能结果见表 19 和表 20。对于基于文本的 VQA 评估基准，我们使用 Gemini 2.5 Flash 对多项选择题和自由形式问题格式的回答准确性进行评分。

Gemini 2.5 和 GPT-5 模型的访问时间为 2025 年 9 月 1 日至 2025 年 9 月 20 日，使用默认思考预算，且未使用工具。

模型	GR-ER 1.5	GR-ER 1.5	GR-ER	Gemini 2.5 Pro	Gemini 2.5 Flash	GPT-5	GPT-5-mini
思考	Yes	No	No	Yes	Yes	Yes	Yes
Point-Bench	71.6	73.3	**75.7**	62.7	61.7	43.6	39.5
RefSpatial	48.5	41.8	**49.3**	33.6	41.2	23.5	23.0
RoboSpatial 指向	**31.1**	25.3	30.3	8.3	7.9	19.0	12.5
Where2Place	**59.0**	48.0	41.0	37.0	48.0	37.0	33.5
空间平均	**52.6**	47.1	49.1	35.4	39.7	30.8	27.1
BLINK	57.8	65.2	60.1	69.2	46.1	**71.3**	66.4
CV-Bench	84.3	83.6	83.2	**85.9**	85.5	**86.1**	**85.9**
ERQA	54.8	47.0	45.3	56.0	47.5	**59.0**	57.3
EmbSpatial	78.4	73.4	56.4	78.0	76.2	**81.5**	78.8
MindCube	54.7	47.7	47.4	**59.2**	55.4	58.0	55.6
RoboSpatial-VQA	**79.3**	57.7	66.2	71.3	73.4	69.3	70.7
SAT	76.7	62.0	64.7	74.7	73.3	**86.7**	81.3
Cosmos-Reason1	72.2	68.3	62.0	73.8	72.1	**79.4**	76.3
最少视频对	72.5	67.1	59.5	72.8	69.2	**77.0**	73.0
OpenEQA	55.0	50.5	38.3	55.7	45.3	**64.4**	59.2
VSI-Bench	45.8	39.9	34.1	51.1	45.3	**52.9**	46.2
QA 平均值	66.5	60.2	56.1	68.0	62.7	**71.4**	68.2
ER 得分	**59.6**	53.7	52.6	51.7	51.2	51.1	47.7
总体平均值	**62.8**	56.7	54.2	59.3	56.5	60.6	57.3

表 19 | **模式** l 在 15 个学术具身推理基准的混合集上的表现。GPT-5 和 GPT-5-min i 所得结果 于 2025 年 9 月通过 API。

C.2. 评估细节：指向

表 21 展示了复杂指向评估的详细细分。

C.3. 补充示例

图 38 展示了 GR-ER 1.5 在具身推理任务中采样得到的思考内容。

Model	GR-ER 1.5	GR-ER 1.5	GR-ER	Pro 2.5	Flash 2.5	GPT-5	GPT-5-mini
Thinking	Yes	No	No	Yes	Yes	Yes	Yes
MMMU	80.7	79.3	67.0	82.0	79.7	82.0	78.0
GPQA	83.3	81.3	59.6	86.4	82.8	88.4	78.3
Aider Polyglot	57.3	44.4	16.0	82.2	56.7	81.3	66.7
Average	73.8	68.3	47.5	83.5	73.1	83.9	74.3

Table 20 | Model performance on MMMU, GPQA and Aider Polyglot benchmarks. Results for GPT-5 and GPT-5-mini obtained via API with default thinking settings and no tool use in September 2025.

模型	GR-ER 1.5	GR-ER 1.5	GR-ER	Pro 2.5	Flash 2.5	GPT-5	GPT-5-mini
思考	Yes	No	No	Yes	Yes	Yes	Yes
MMMU	80.7	79.3	67.0	82.0	79.7	82.0	78.0
GPQA	83.3	81.3	59.6	86.4	82.8	88.4	78.3
Aider Polyglot	57.3	44.4	16.0	82.2	56.7	81.3	66.7
平均值	73.8	68.3	47.5	83.5	73.1	83.9	74.3

表 **20** | 模型在 MMMU、GPQA 和 Aider Polyglot 基准测试上的表现。GPT-5 和 GPT-5-mini 的结果于 2025 年 9 月通过 API、在默认思考设置且不使用工具的情况下获得。

Table 21 | Model performance on complex pointing benchmarks, broken down by subtask.

Model	GR-ER 1.5	GR-ER 1.5	GR-ER	Gemini 2.5 Pro	Gemini 2.5 Flash	GPT-5	GPT-5-mini
Thinking	Yes	No	No	Yes	Yes	Yes	Yes
Standard Pointing							
Point-Bench-Affordance	70.9	76.5	**87.9**	65.3	67.8	58.1	50.0
Point-Bench-Counting	86.8	86.8	**88.4**	77.5	73.1	53.7	56.8
Point-Bench-Reasoning	61.7	**69.0**	64.8	55.4	49.4	33.0	28.3
Steerable Pointing							
Point-Bench-Steerable	**67.8**	61.8	65.8	53.4	61.3	38.0	32.0
Spatial Pointing							
Point-Bench-Spatial	71.0	**72.6**	71.9	62.7	57.2	35.4	30.3
RefSpatial	48.5	41.8	**49.2**	33.6	41.1	23.5	23.0
RoboSpatial	**31.1**	25.3	30.3	8.3	7.9	19.0	12.5
Where2Place	**59.0**	48.0	41.0	37.0	48.0	37.0	33.5
Point-to-Count							
PixMo Count	**80.0**	65.0	60.0	76.0	64.0	73.0	77.0
Average	**52.6**	47.1	49.1	35.4	39.7	30.8	27.1

表 **21** | 模型在复杂指向基准测试上的表现，按子任务细分 k.

模型	GR-ER 1.5	GR-ER 1.5	GR-ER	Gemini 2.5 Pro	Gemini 2.5 Flash	GPT-5	GPT-5-mini
思考	Yes	No	No	Yes	Yes	Yes	Yes
标准指向							
Point-Bench-可供性	70.9	76.5	**87.9**	65.3	67.8	58.1	50.0
Point-Bench-计数	86.8	86.8	**88.4**	77.5	73.1	53.7	56.8
Point-Bench-推理	61.7	**69.0**	64.8	55.4	49.4	33.0	28.3
可操控指向							
Point-Bench-可引导	**67.8**	61.8	65.8	53.4	61.3	38.0	32.0
空间指向							
Point-Bench-空间	71.0	**72.6**	71.9	62.7	57.2	35.4	30.3
指代空间	48.5	41.8	**49.2**	33.6	41.1	23.5	23.0
机器人空间	**31.1**	25.3	30.3	8.3	7.9	19.0	12.5
放置位置	**59.0**	48.0	41.0	37.0	48.0	37.0	33.5
指向计数							
PixMo 计数	**80.0**	65.0	60.0	76.0	64.0	73.0	77.0
平均	**52.6**	47.1	49.1	35.4	39.7	30.8	27.1

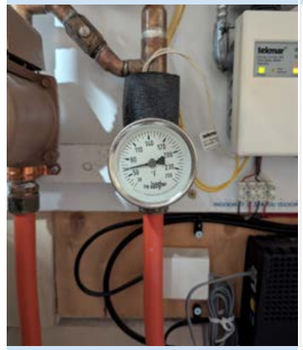
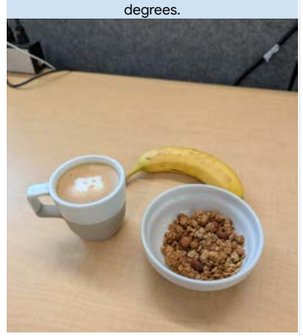
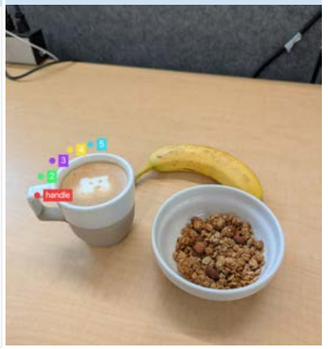
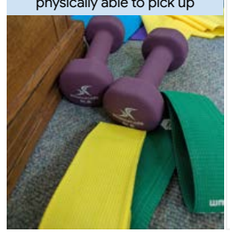
What does the gauge read?	Select model thoughts	Model response
	***Analyze the gauge's features:*** ** **Scale:** The gauge measures temperature in degrees Fahrenheit (°F). This is indicated by "°F" written on the lower part of the dial.* ** **Numbers:** The major markings on the scale are 30, 50, 80, 110, 140, 170, 200, 230, 250* ***Read the pointer position:*** **The pointer is between the numbers 50 and 80.* ** If the interval between 50 and 80 is 30 degrees, and there are two tick marks dividing it into three equal sections, each section would be 10 degrees. The first tick mark after 50 would represent 60. The second tick mark would represent 70.* ** The pointer is pointing to the unnumbered tick mark that is immediately after the 50 mark.* ** Therefore, the value is 50 + 10 = 60 degrees Fahrenheit.*	Based on the image, the gauge reads approximately **60°F.** Here's how to interpret the reading: 1. The large numbers on the dial are in increments, such as 30, 50, 80, 110, etc. The unit is marked as °F (degrees Fahrenheit). 2. The pointer is located between 50°F and 80°F. 3. Between 50°F and 80°F, there are two unlabeled tick marks. This divides the 30-degree interval into three sections, with each section representing 10 degrees (50 + 10 = 60; 60 + 10 = 70; 70 + 10 = 80). 4. The pointer is pointing directly at the first tick mark after 50°F, which represents 60°F.
Match all of the socks by pointing to them and labeling them based on their pattern. Make sure matching socks have the same label.	Select model thoughts	Model response
	Identify individual socks and their patterns: ** **Sock 3 (center-top):** Light blue sock with small white dots. It's partially obscured. Pattern: small white dots. Let's look for a match.* ** **Sock 4 (bottom-center):** Light blue sock with small white dots. This matches Sock 3.* ***Create pairs based on matching patterns:*** ** **Pair 1:** Cat face socks. Sock 1 ([y: 250, x: 380]) and Sock 2 ([y: 350, x: 700]). Label: "cat face".* ** **Pair 2:** Small dot socks. Sock 3 ([y: 350, x: 580]) and Sock 4 ([y: 800, x: 400]). Label: "small dots".*	
Point to the handle of the mug. Plot a trajectory that traces the position of the handle as it is rotated in place by 90 degrees.	Select model thoughts	Model response
	***Determine the center of rotation:*** *For "rotated in place," the center of rotation is likely the center of the mug itself. The mug's center is roughly around [500, 250]. The handle starts at approximately [558, 100].* ***Visualize the rotation:*** *The handle is currently at the 9 o'clock position relative to the mug's center. A 90-degree rotation clockwise would move it to the 6 o'clock position (bottom of the mug).* ***Refine trajectory points:*** *The trajectory should be an arc. Let's make it more visually representative of a rotation around the mug's center.* ** Start: [558, 100] (initial handle location)* ** Point 1: [500, 120] (moving upwards and inwards)* ** Point 2: [450, 180] (approaching the top-left corner)* ** Point 3: [400, 250] (arriving at the top of the mug)*	
I am a robot with a payload of 10LBs. Point to all the objects in the image I am physically able to pick up	Select model thoughts	Model response
	***Dumbbells:*** *There are two purple dumbbells. The label on the dumbbells clearly reads "Prosource Fit 5LBS". This means each dumbbell weighs 5 pounds.* ***Resistance Bands/Loop Bands:*** *There are several resistance bands on the floor. Resistance bands typically weigh very little, usually a few ounces to less than a pound each. A 10 lb payload limit is more than sufficient to lift a single resistance band, or even all visible bands together.* ***Furniture/Chair:*** *There is a wooden piece of furniture (desk or cabinet) on the left and a chair base on the right. These objects are clearly much heavier than 10lbs. A robot with a 10lb payload cannot lift them.* ***Dumbbells:*** *Each dumbbell weighs 5lbs. The robot can lift one dumbbell (5lbs <= 10lbs). Can it lift both? 5lbs + 5lbs = 10lbs. The payload limit is exactly 10lbs. The question asks "pick up", which could mean one or both.*	

Figure 38 | Sample thoughts from GR-ER 1.5 performing embodied reasoning tasks.

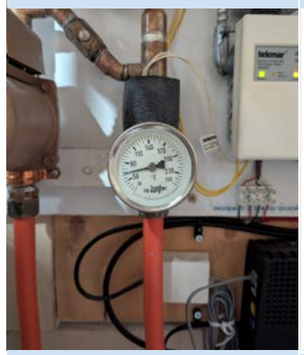
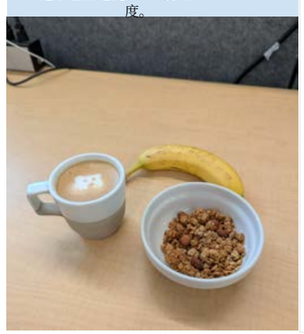
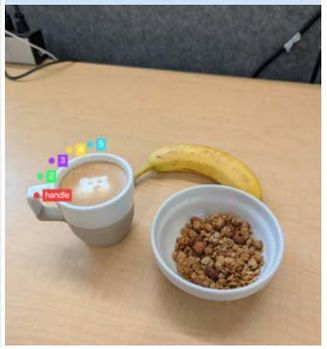
仪表读数是多少？	选择模型思考	模型响应
	分析仪表的特征： *** 刻度：该仪表以度为单位测量温度华氏度（*°F*）。这由*"°F"*表明，其写在刻度盘的下部。 *** 数字：刻度上的主要标记为 *30*、*50*、*80*、*110*、*140*、*170*、*200*、*230*、*250* 读取指针位置： *** 指针位于数字 *50* 和 *80* 之间。 *** 如果 *50* 和 *80* 之间的间隔为 *30* 度，并且有两条刻度线将其分成三个相等的部分，每个部分将是 *10* 度。第一个刻度 *50* 后的刻度线将表示 *60*。第二个刻度线将表示 *70*。 *** 指针正指向未编号的刻度线该刻度线紧接在 *50* 刻度之后。 *** 因此，数值为 *50 + 10 = 60* 华氏度。	根据图像，仪表读数为约 60°F。 以下是解读读数的方法： 1. 表盘上的大数字以增量表示，例如 30、50、80、110 等。该单位标记为 °F（华氏度）。 2. 指针位于 50°F 与以下数值之间：80°F。 3. 在 50°F 到 80°F 之间，有两个未标记的刻度线。这会将30 度区间分成三个部分，其中每个部分代表 10 度（50 + 10 = 60；60 + 10 = 70；70 + 10 = 80）。 4. 指针直接指向第一个50°F 之后的勾号，表示 60°F。
通过指向来配对所有的袜子它们，并根据它们的图案。确保配对的袜子具有相同的标签。	选择模型思考	模型响应
	识别单只袜子及其图案： *** 袜子 *3*（顶部中央）：浅蓝色袜子，带细小白色圆点。它部分被遮挡。图案：细小白色圆点。 我们来找找配对的袜子吧。 *** 袜子 *4*（底部中央）：浅蓝色袜子，带细小白色圆点。这与袜子 *3* 相匹配。 根据匹配模式创建配对： *** 配对 *1*：猫脸袜子。袜子 *1*（*[y: 250, x: 380]*）和袜子 *2*（*[y: 350, x: 700]*）。标签："猫脸"。 *** 配对 *2*：小圆点袜子。袜子 *3*（*[y: 350, x: 580]*）和袜子 *4*（*[y: 800, x: 400]*）。标签："小圆点"。	
指向马克杯的把手。绘制一条轨迹，以追踪以下对象的位置：把手在原地旋转 90 度时度。	选择模型思考	模型响应
	确定旋转中心：对于*"原地旋转"*，旋转中心很可能是马克杯本身的中心。马克杯的中心大致在 *[500, 250]* 附近。杯柄大约从 *[558, 100]* 处开始。 设想旋转：杯柄当前位于 *9* 相对于杯子中心的钟点位置。一个 *90* 度顺时针旋转会将其移动到 *6* 点钟位置（杯子底部）。 细化轨迹点：轨迹应为一条弧线。让我们使其更能从视觉上代表一次旋转围绕杯子中心。 *** 开始：*[558, 100]*（初始把手位置） *** 点 *1*：*[500, 120]*（向上并向内移动） *** 点 *2*：*[450, 180]*（接近左上角） *** 点 *3*：*[400, 250]*（到达马克杯顶部）	
我是一台机器人，有效载荷为 10 磅。请指向所有图像中的物体，我身体上能够拿起的	选择模型思考	模型响应
	哑铃：有两个紫色哑铃。哑铃上的标签清楚地写着*"Prosource Fit 5LBS"*。这意味着每个哑铃重*5*磅。 阻力带/环形带：地板上有几条阻力带。 阻力带通常重量很轻，每条一般只有几盎司到不足一磅。*10*磅的负载上限足以抬起单条阻力带，甚至可一次抬起所有可见的带子。 家具/椅子：左侧有一件木制家具（桌子或柜子）右侧有一个椅子底座。这些物体显然比*10*磅重得多。一台有效载荷为*10*磅的机器人无法举起它们。 哑铃：每个哑铃重*5*磅。机器人可以举起一个哑铃（*5*磅 <= *10*磅）。它能同时举起两个吗？*5*磅 + *5*磅 = *10*磅。有效载荷上限正好是*10*磅。这个问题问的是"拿起"，这可能指一个或两个。	

图 **38** | GR-ER 1.5 执行具身推理任务时的思考示例。

D. Gemini Robotics 1.5: A Physical Agent

D.1. Long-horizon benchmarks

Our long-horizon benchmarks evaluate the combination of the GR-ER 1.5 model with the VLA as an autonomous agent. Fig. 39 shows visuals of the 4 tasks in the ALOHA long-horizon benchmark. The progress is scored as the sum of points scored along each subtask (Table 22). Fig. 40 shows visuals of the 4 tasks on the Bi-arm Franka long-horizon benchmark. The progress is scored as the sum of points scored along each subtask (Table 23).

Table 22 | Progress Scores: ALOHA Robot (Long-horizon Benchmark).

Benchmark: ALOHA Robot - Long-horizon.			
Trash Sorting: "Put the compostables into the green bin, the recyclables into the blue bin, and the waste into the black bin".	**Desk Organization: "What is the state of the objects in the table? Return them to their original locations".**	**Packing Suitcase: "Put the hat and socks into the suitcase then pack the colorful shirt that's on the hanger".**	**Blocks in drawer: "Open each drawer, and put one block in each drawer".**
0.2 is added per item in the correct bin: – red grapes in the green bin; – lettuce leaf in the green bin; – aluminum can in the blue bin; – plastic cup in the blue bin; – energy bar wrapper in the black bin.	0.2 is added per item in the correct state: – red pen in the pen holder; – blue pen in the pen holder; – green marker in the cork tray; – glasses case closed; – laptop closed.	0.25 is added per: – white beanie in the suitcase; – blue socks in the suitcase; – shirt taken off the hanger; – shirt in the suitcase.	0.11 is added per: – left drawer was opened; – any block in left drawer; – left drawer closed; – middle drawer was opened; – any block in middle drawer; – middle drawer closed; – right drawer was opened; – any block in right drawer; – right drawer closed.

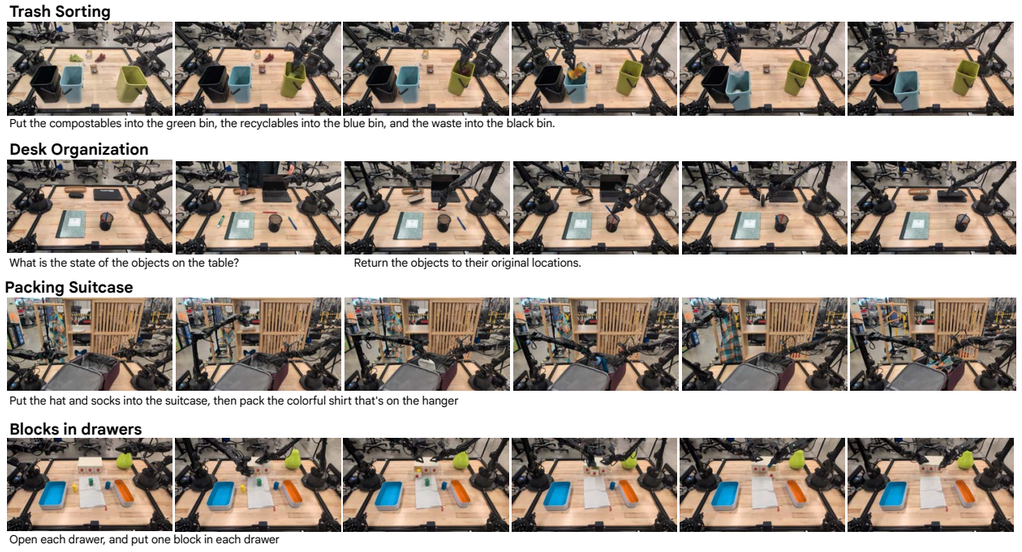


Figure 39 | ALOHA long-horizon benchmark.

D. Gemini Robotics 1.5：物理智能体

D.1. 长时程基准

我们的长时程基准评估 GR-ER 1.5 模型与 VLA 作为自主智能体的组合。图 39 展示了 ALOHA 长时程基准中 4 项任务的可视化结果。进度按沿每个子任务所得分数之和计分（表 22）。图 40 展示了双臂 Franka 长时程基准中 4 项任务的可视化结果。进度按沿每个子任务所得分数之和计分（表 23 ）。

表 **22** | 进度得分：ALOHA 机器人（长时程基准）。

基准：ALOHA 机器人 - 长时程。			
垃圾分类："将可堆肥物放入绿色垃圾桶，可回收物放入蓝色垃圾桶，并将废弃物放入黑色垃圾桶"。	**桌面整理："状态是什么桌子上的物体？将它们放回到它们原来的位置"。**	**打包行李箱："把帽子和把袜子放进行李箱，然后打包挂在衣架上的彩色衬衫"。**	**抽屉里的积木："打开每个抽屉，并在每个抽屉里放一块积木"。**
每将一件物品放入正确的箱子，加 0.2 分： – 红葡萄放入绿色箱子； – 生菜叶在绿色垃圾桶中； – 铝罐在蓝色垃圾桶中； – 塑料杯在蓝色垃圾桶中； – 能量棒包装纸在黑色垃圾桶中。	每件处于正确状态的物品加 0.2 分： – 红笔在笔筒中； – 笔筒里的蓝色钢笔； – 软木托盘里的绿色马克笔； – 眼镜盒已合上； – 笔记本电脑已合上。	每项增加 0.25： – 行李箱里的白色毛线帽； – 行李箱里的蓝色袜子； – 衬衫从衣架上取下来； – 行李箱里的衬衫。	每项增加 0.11： – 左侧抽屉被打开了； – 左侧抽屉里的任意积木； – 左侧抽屉已关闭； – 中间抽屉已打开； – 中间抽屉中的任意方块； – 中间抽屉已关闭； – 右侧抽屉已打开； – 右侧抽屉中的任意方块； – 右侧抽屉已关闭。

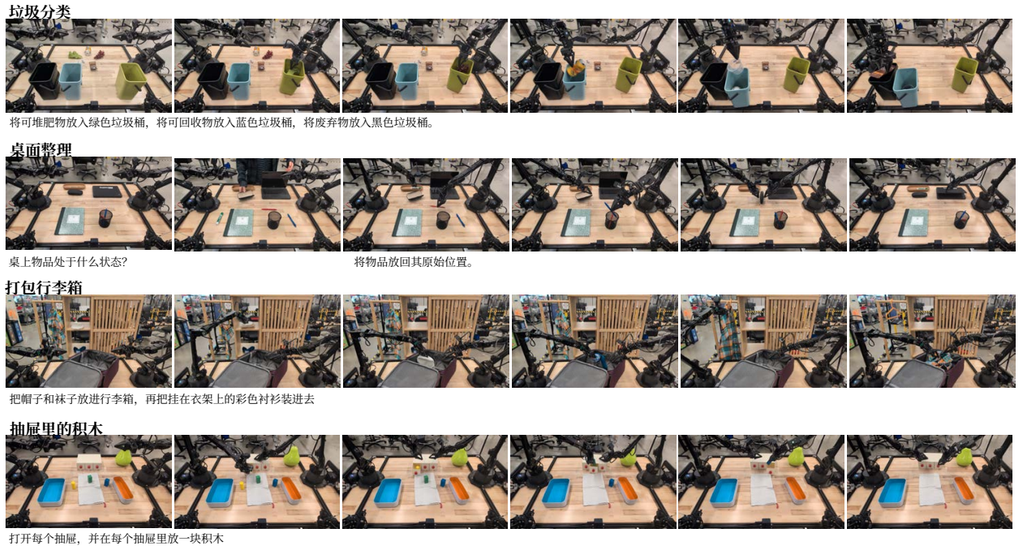


图 **39** | ALOHA 长时程基准。

Table 23 | Progress Scores: Bi-arm Franka (Long-horizon Benchmark).

Benchmark: Bi-arm Franka - Long-horizon.			
Swap: "Swap the sardines and the yellow bottle".	**Top shelf to the table: "Put all the objects from the top right shelf onto the table".**	**Mushroom risotto: "Pack all ingredients for a mushroom risotto into the basket".**	**Vegetarian with nut allergy: "I am vegetarian and allergic to nuts. Can you put all the food I can't eat into the basket".**
0.33 is added per each subtask: – lemon juice is in the correct location; – can of sardines is in the correct location; – no unrelated task done.	0.25 per each subtask: – rice is on the table; – corn is on the table; – lemon juice is on table; – no unrelated task done.	0.25 per each subtask: – mushrooms are in the basket; – rice is in the basket; – stock cubes are in the basket; – no unrelated task done.	0.33 per each subtask: – the can of sardines is into the basket; – the granola is into the basket; – no unrelated task done.

Figure 40 | Bi-arm Franka long-horizon benchmark.

表 **23** | 进度得分：双臂 Franka（长时程基准）。

基准测试：双臂 Franka - 长时程。			
交换：“交换沙丁鱼和黄色瓶子”。	**从顶层架子到桌子：“把所有物品从右上方的架子放到桌子上”。**	**蘑菇烩饭：“把所有配料用于蘑菇烩饭，装进篮子”。**	**坚果过敏的素食者：“我是素食者，并且对坚果过敏。你能把所有我不能吃的食物放进篮子”。**
每个子任务加 0.33 分： – 柠檬汁位于正确位置； – 沙丁鱼罐头位于正确的位置； – 未执行无关任务。	每个子任务加 0.25 分： 米饭在桌子上； – 玉米在桌子上； – 柠檬汁在桌子上； – 名词相关任务已完成。	每个子任务 0.25： – 蘑菇在篮子里； – 大米在篮子里； – 汤料块在篮子里； – 未执行无关任务。	每个子任务 0.33： – 沙丁鱼罐头在篮子里； – 格兰诺拉麦片在篮子里； – 未执行任何无关任务。

图 **40** | 双臂 Franka 长时程基准测试。

D.2. Success rate

Fig. 41 shows the *success rate* for the results in Section 5.

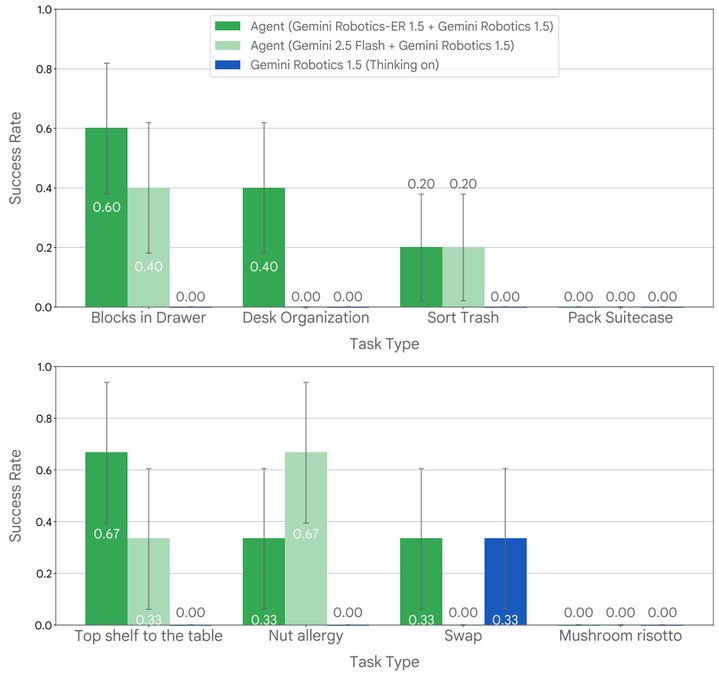


Figure 41 | Long-horizon evaluations for the GR 1.5 Agent on ALOHA (top) and Bi-arm Franka (bottom), consisting of tasks that require advanced real-world understanding, tool use, long-horizon task planning, execution and error recovery to successfully complete the complex long-horizon tasks.

D.2. 成功率

图 41 展示了第 5 节中结果的成功率。

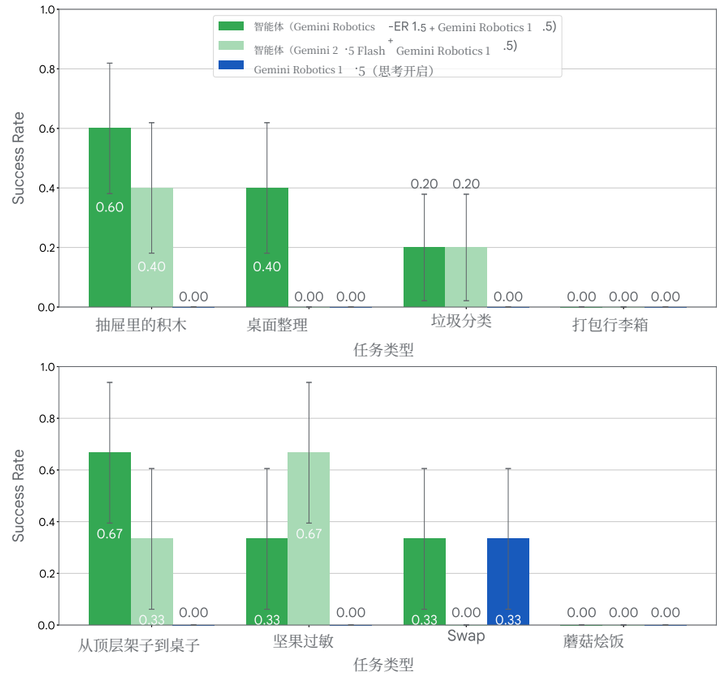


图 **41** | GR 1.5 Agent 在 ALOHA（上）和双臂 Franka（下）上的长时程评估，包含需要高级真实世界理解、工具使用、长时程任务规划、执行和错误恢复才能成功完成复杂长时程任务的任务。